wealth-tensor: a point-in-time capture

Jason C. Braatz
Independent researcher
jason@braatzresearch.com

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Supply and demand are not independent equations: price formation from a single distribution of reservation prices

Jason C. Braatz
Independent researcher
jason@braatzresearch.com

Draft — not yet submitted. Version 0.1, 2026-08-10.

⚠ SUPERSEDED — read `REVIEW-002-internal-referee.md` before this file

This draft was rejected by its own internal referee on the day it was written, and the rejection was substantially correct. It is retained as written, rather than revised or deleted, because `docs/` is a working lab notebook (ADR-001 §Consequences) and a paper that was wrong in instructive ways teaches more than its own corrected successor.

What is wrong with it, in short:

1. **Contribution 2 is Böhm-Bawerk (1889).** The term *marginal pairs* is his, for this exact object; his horse-market numbers reproduce this paper's formula to the shilling. The modern formalisation is Shapley & Shubik (1971), whose §4 is titled *"The Horse Market of Böhm-Bawerk"*. Neither is cited below.
2. **Contribution 5 is Theocharis (1960) and Fisher (1961)**, and the bound $d < 4/(n+1)$ is eq. (2.26) of a 2010 Springer monograph, presented there as routine. Not new.
3. **The central negative claim is false in this paper's own model.** *"The schedules cannot be perturbed independently"* does not follow from the invariance, and the referee shifted one schedule while holding the other pointwise fixed, using this repository's unmodified code.

4. **§3.4 breaks §3.1**, because its behavioural transform makes $c(m)$ a function of the allocation. At $\lambda = 1.3$ the invariance gives 25 distinct excess-demand schedules, not 1.
5. **§3.3 is circular**: `marshallian_cross` is computed *from* excess demand, so the reduction result substitutes §3.1 into itself. And "exactly recovered" is a coincidence of two resolutions — at $N = 4000$ it is 0/25.
6. **§4.2's gain expression was wrong** (the Jacobian has a second eigenvalue, $1 - d/2$). The stability condition happens to survive it. Fixed in the code and the tests; **left standing below** so the review has something to point at.

What may survive, and it is stated at the strength the audit supports and no further: no source checked states the *invariance* of $z(p)$ to which agents hold the units — but the auditor's own warning stands, that this is one short step from Coase, Gorman, Böhm-Bawerk and Shapley–Shubik, and is plausibly folklore in a literature not yet searched. **Searching Gul–Stacchetti, Demange–Gale–Sotomayor and the law-and-economics treatment of Coase in indivisible-goods markets is a precondition of any successor draft, not an optional extra.**

Abstract

With indivisible units and no income effects, the supply and demand schedules are not two equations. They are two readings of one distribution of reservation prices, and the reading is a bookkeeping choice rather than an economic fact. This paper demonstrates that as an identity rather than an inference. Across 25 allocations of the same 400 reservation prices and the same stock of 150 units there are **25 distinct demand schedules, 25 distinct supply schedules, and exactly one excess-demand schedule**, equal to $\#\{i : m_i > p\} - S$ at every price that is not itself a reservation price: the allocation cancels from the difference algebraically, because the holders partition at any price into those above it and those below it. Three consequences follow. The market-clearing interval is the **marginal pair**, the S-th and $(S{+}1)$-th highest reservation prices. The Marshallian cross is **exactly**

recovered for any fixed allocation, making it a valid snapshot and an invalid comparative static — the schedules cannot be perturbed independently because their difference does not move. And behaviour enters as a stated transform of the distribution rather than a free coefficient, reproducing the endowment effect's documented volume decline (93 ⟶ 49 units) as a consequence. The same exclusion boundary appears in Cournot competition, where the damping that stabilises tâtonnement is shown to vanish like $4/n$, so the repair for Cournot's instability needs more information than his own assumption grants. Every number regenerates from one command against a public repository.

Keywords: reservation prices · excess demand · Marshallian cross · comparative statics · Cournot competition · tâtonnement stability · Sonnenschein–Mantel–Debreu · endowment effect

JEL classification: D40, D41, D51, L13, C62

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1 · Introduction

The supply-and-demand cross is the first diagram in economics and the last one most readers ever see. Two schedules, drawn as independent objects, crossing at a price. The construction is old enough that its status is rarely examined: it is treated as a *model*, in which the two curves are primitives that can be shifted one at a time, and comparative statics is the exercise of doing so.

This paper's claim is that in the simplest exchange setting the two schedules are not primitives at all, and that the demonstration is an identity rather than an argument. Given a population of reservation prices and a stock of indivisible units, an *allocation* records who currently holds one. Demand at price p counts the non-holders who value a unit above p; supply counts the holders who value theirs below it. Change the allocation and both schedules change. Their difference does not change at all.

That is not a near-invariance or a statistical regularity. The S holders partition at any price into those above it and those below it, so the allocation enters the two counts with opposite signs and cancels. The excess-demand function is $\#\{i : m_i > p\} - S$: a function of the reservation-price distribution and the stock, and of nothing else. **The split of that single function into a supply half and a demand half carries no economic content.** The textbook draws a bookkeeping convention as though it were data, and then perturbs it.

This is an instance of a proposition stated elsewhere in this programme, and it is cited rather than restated. P3

— *measured aggregates are folds over units; no aggregate is more fundamental than its constituents* — is stated with its domain in the companion paper on the dual tensor [III, §2.2]. Supply and demand are exactly such folds: two different ways of summing over the same units, presented as two objects. The general proposition and this instance were developed together and it would be circular for each to support the other, so no support is claimed in either direction. What is claimed is that the price system is where P3 is cheapest to check, because here the fold can be written down and the cancellation performed in one line.

Contributions. This paper is short and its claims are specific.

1. An **identity**: with indivisible units and no income effects, aggregate excess demand is invariant to the allocation *at every price*, not merely at its zero, and equals $\#\{i : m_i > p\} - S$ (§3.1). This is stronger than the usual observation that the schedules co-move, and it is what licenses the rest.
2. The identification of the market-clearing interval with the **marginal pair**, the S-th and $(S{+}1)$-th highest reservation prices, together with the demonstration that excess demand steps $+1 \longrightarrow 0 \longrightarrow -1$ across it for every allocation (§3.2).
3. A **reduction result**: for any fixed allocation the Marshallian construction recovers the structural clearing interval exactly (§3.3). The cross is therefore a correct *snapshot* and an incorrect *comparative static*, and the failure is located precisely — not in the diagram's accuracy but in the independence its perturbation assumes.
4. A demonstration that **behaviour enters as a shape transform of the distribution**, not as a fitted coefficient: raising holders' reservation prices reproduces the endowment effect's documented volume decline monotonically, 93 $\longrightarrow$ 49 units (§3.4).
5. The **same exclusion boundary in Cournot competition** (§4): a firm whose marginal cost exceeds the price the survivors set is excluded rather than loss-making, which is the marginal pair arriving from the quantity side. And a sharpening of the classical instability result — the damping that rescues tâtonnement is not a constant but **vanishes like $\boldsymbol{4/n}$**, so the repair requires each firm to know how many rivals it has (§4.2).
6. A **stated limit on the claim**: this construction produces a monotone, single-crossing excess demand and therefore does *not* exhibit Sonnenschein–Mantel–Debreu pathology (§3.5). The limit is enforced by a test rather than by a promise.

A boundary, stated once. Everything here concerns a model class. The setting is one good, unit demand, indivisible units and no income effects; §5 records what that costs and §6 records which of the results survive its relaxation and which do not. No claim is made about any market.

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2 · The construction

2.1 · Reservation prices, and the two readings

N agents. Agent i holds a reservation price m_i, drawn from a distribution $c(m)$. There are S indivisible units of a single good, at most one per agent, and an **allocation** is the subset of agents currently holding one. At price p:

a holder **sells** if $m_i < p$ — a non-holder **buys** if $m_i > p$

Nothing here partitions the population into buyers and sellers. An agent's side of the market is not a property of the agent; it is a property of the pair (reservation price, current holding) evaluated at p. This is the standard content of excess demand as Walras (1874) formulated it, where $z_i(p) = x_i(p) - e_i$ is positive for a net buyer and negative for a net seller — the same agent, resolved by price — and it is worth saying plainly that the observation is *not* a critique of general equilibrium. It **is** general equilibrium. The object under examination is the cross of Marshall (1890), a pedagogical device that the general-equilibrium tradition does not itself use: Arrow and Debreu (1954) prove existence without it. §5 records what happened when that distinction was not maintained.

2.2 · What is measured

Demand $D(p)$ = #{non-holders with $m_i > p$}. **Supply** $S(p)$ = #{holders with $m_i < p$}. **Excess demand** $z(p) = D(p) - S(p)$. The **marginal pair** is the S-th and $(S+1)$-th highest reservation prices. **Volume** is the number of units changing hands on the way to the efficient allocation at the clearing price.

Throughout: $N = 400$, reservation prices lognormal($\mu = 3.0$, $\sigma = 0.6$) under NumPy's `default_rng(7)`, $S = 150$, and 25 allocations drawn under `default_rng(0…24)`. Every figure in §3 is produced by one command against the committed modules; §7 gives it.

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3 · Results

3.1 · The allocation cancels from the difference, identically

Take the same $c(m)$, the same S, and 25 different allocations. Measured at 399 interior grid points, ties excluded:

object	distinct values across 25 allocations
demand schedule $D(p)$	**25**
supply schedule $S(p)$	**25**
excess demand $\boldsymbol{z(p) = D(p) - S(p)}$	**1**

Every allocation gives a different demand curve. Every allocation gives a different supply curve. All 25 give the same difference, and that difference equals $\#\{i : m_i > p\} - S$ at every grid point.

The reason is a partition and it takes one line. At any price p that is not itself a reservation price, the S holders divide into those with $m_i > p$ and those with $m_i < p$. So

$$S(p) = \#\{\text{holders}, m_i < p\} = S - \#\{\text{holders}, m_i > p\}$$

and therefore

$$z(p) = \#\{\text{non-holders}, m_i > p\} - S + \#\{\text{holders}, m_i > p\}$$
$$= \boldsymbol{\#\{m_i > p\} - S.}$$

The allocation appears in both counts with opposite signs and leaves.

Why the identity is worth more than the co-movement. The usual way to state this result is that the schedules move while the crossing does not, from which a reader infers that two things which both shift under a perturbation leaving the equilibrium fixed cannot be independent equations. The inference is sound and unnecessary. The identity says something strictly stronger and says it without inference: **the decomposition of z into a supply half and a demand half is a bookkeeping choice.** It records which side of the ledger each agent is written on, and the economy is invariant to that choice at every price, not only where the two halves happen to be equal.

An economist may reasonably respond that this is obvious once written down. It is — that is the point of writing it down. What is not obvious is that a construction with this property is routinely perturbed one curve at a time.

3.2 · The clearing interval is the marginal pair

The invariant object has a zero, and the zero is structural. For the population above:

quantity	value
(S+1)-th highest reservation price (first excluded)	21.461883
S-th highest reservation price (last included)	21.498548
interval width	0.036665
distinct clearing intervals across 25 allocations	**1**

Any price strictly inside the interval clears the market, for every allocation. Excess demand steps **+1 ⟶ 0 ⟶ −1** across it, and the triple (z just below, z at the midpoint, z just above) takes exactly one value — (1, 0, −1) — across all 25 allocations.

The interval is the manuscript's **marginal pair**: the last agent who can hold a unit and the first who cannot. It is a property of $c(m)$ and S. It has no dependence on who currently holds what, which is the same statement as §3.1 evaluated at one price.

3.3 · The textbook construction is recovered exactly

The reduction result. For each of the 25 allocations, read $D(p)$ and $S(p)$ off the schedules and find where they cross, exactly as the diagram instructs:

quantity	value
allocations tested	25
crosses landing inside the structural interval	**25 / 25**
distinct cross values across allocations	1
cross value	21.469835
structural interval	[21.461883, 21.498548]

The Marshallian cross is not wrong. It is a correct instantaneous description, and this paper contains it rather than contradicting it — which is the only form of concession that survives review, because it is a theorem rather than a courtesy.

What the cross cannot do is comparative statics that move the allocation. The diagram's method is to shift one schedule and read off the new intersection, and that method presumes the schedules can be shifted separately. They cannot: §3.1 says their difference is fixed, so any perturbation of the allocation moves both in lockstep and moves the crossing not at all. **The failure is not in the picture. It is in the operation performed on the picture** — and locating it there is what makes this a result rather than a complaint.

3.4 · Behaviour is a shape transform, not a coefficient

The endowment effect — named by Thaler (1980) — is the finding that holders value what they hold more highly than non-holders value acquiring it. In this construction that is not an additional mechanism. It is a statement about where reservation prices sit, and reservation prices are the model's only primitive. Multiply holders' m_i by a factor $\lambda \geq 1$ and re-read the same two schedules:

λ	volume	change
1.00	93	—
1.05	88	−5
1.15	82	−6
1.30	74	−8
1.60	62	−12
2.00	49	−13

Volume falls monotonically, by 47.3 % of baseline across the range. **Reduced trading volume is the documented experimental finding**, and it is the right one to match: Kahneman, Knetsch and Thaler (1990) report that observed volume in mug-exchange experiments is persistently below the Coase prediction, and that markets in induced-value tokens run by the same subjects hit the predicted volume — which is what rules out transaction costs as the explanation. The undertrading, not the willingness-to-accept/willingness-to-pay gap alone, is the result reproduced here, and here it is a **consequence of a stated transform** rather than a coefficient chosen to produce it. The distinction is the whole of the methodological claim: a transform of $c(m)$ is measurable and therefore falsifiable, and it makes a prediction the modeller does not control. A free parameter fitted to the same data forbids nothing.

Note also what the transform does *not* do. It does not move the clearing interval away from the marginal pair; it moves the marginal pair, because it moves the distribution. The identity of §3.1 is untouched — this is a different $c(m)$, not a different reading of the same one.

3.5 · What this is not: the Sonnenschein–Mantel–Debreu boundary

Sonnenschein (1972, 1973), Mantel (1974) and Debreu (1974) established that aggregate excess demand inherits from individually rational agents only continuity, homogeneity of degree zero and Walras's Law — not downward slope, not uniqueness, not stability under tâtonnement. Aggregate excess demand can take essentially arbitrary shape. Two conditions of the construction are worth stating rather than eliding, because they are where the result's scope actually lies: the agents' preferences are *not* weakened — they remain continuous, convex and monotone, which is the entire force of the theorem — and the number of households is taken to be at least the number of commodities, Debreu's construction using n households for n goods. Within those conditions it is the strongest available statement that the aggregate schedules are not well-behaved primitives, it is fifty years old, and it was established inside the mainstream.

It is not what this paper demonstrates, and the two must not be conflated. Excess demand here is monotone and single-crossing: 0 monotonicity violations across 500 grid points, running from +249 to −150 with one sign change. It is perfectly well behaved.

The reason is worth stating precisely, because the loose version of it is wrong. It is tempting to say that SMD "requires at least two goods" and that this model has one. That is not the operative distinction — a single traded good priced against money is already a two-commodity partial equilibrium, so the count of goods is not what does the work. **What does the work is unit demand together with the absence of income effects.** With each agent demanding at most one unit and no wealth channel from the endowment back into demand, aggregate demand is a non-increasing step function of price by construction, and there is nothing for the SMD construction to get purchase on. This is the same restriction that §6 identifies as load-bearing for §3.1's identity, and it is the restriction known to tame the aggregate more generally. Mas-Colell, Whinston and Green (1995, §10.C) note that the quasi-linear partial-equilibrium price is uniquely defined and warn in the same breath that uniqueness "need not hold in more general settings in which wealth effects are present". The chain is theirs: absent wealth effects the aggregate excess demand is that of a single representative consumer, which satisfies the weak axiom (*ibid.*, §17.F) — and the weak axiom delivers both uniqueness of the normalised equilibrium and convergence of tâtonnement (*ibid.*, Propositions 17.F.2 and 17.H.1). **The two results are therefore not rivals but opposite ends of one axis**, and §7 states the relation. Note that this is *not* the gross-substitutes route to the same conclusion, which is a logically independent condition: the same source records that gross substitutes can fail in quasi-linear one-consumer economies and that neither property implies the other.

Selling this construction as a demonstration of SMD would be a genuine error and an easy one, since the two arguments are adjacent and point the same way. The monotonicity is therefore asserted in the test suite deliberately — `test_excess_demand_is_monotone_here_so_this_is_not_an_SMD_result` — as a standing limit on the claim rather than as a property being celebrated. **The relationship between the two results is complementarity, not support**, and §6 states it precisely.

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4 · The same boundary from the quantity side

ADR-001 allocates the Cournot material to this paper for a reason that is a result rather than a convenience: **the Cournot corner solution is the marginal pair, arrived at from the other direction.** The two chapters of the original manuscript described one object and did not cite each other.

4.1 · The exclusion boundary is an equilibrium object

Linear inverse demand $p = 100 - Q$, five firms with marginal costs [8, 9, 10, 11, 30]. The closed-form interior solution returns a **negative** output for the fifth firm, and the correct reading of that is not that the firm loses money:

firm	MC	q	profit	share
1	8	19.6	384.16	0.2707
2	9	18.6	345.96	0.2569
3	10	17.6	309.76	0.2431
4	11	16.6	275.56	0.2293
5	30	**0**	0	0

with $Q = 72.4$ and $p =$ **27.6**. The fifth firm is **excluded**, because its marginal cost exceeds the price the survivors set, and $27.6 < 30$ confirms the exclusion is self-consistent. The implementation refuses the interior closed form on this input rather than returning the negative quantity, and says why.

That boundary — the last participant inside and the first outside, determined by the price the market reaches rather than by any participant's own circumstances — is the marginal pair of §3.2. In the exchange setting it is the S-th and $(S{+}1)$-th reservation prices; here it is the highest marginal cost below p and the lowest above it. **Same object, two directions**, and this is the join that makes these two modules one paper.

4.2 · Tâtonnement fails where its own expectation is weakest — and the repair needs what the model denies

Cournot's adjustment process has each firm best-respond to its rivals' current output. Its linearised map has gain $(n-1)/2$, so undamped simultaneous adjustment is stable at $n = 2$, marginal at $n = 3$, and non-convergent beyond. Measured:

n	gain $(n{-}1)/2$	undamped, 5000 iterations
2	0.5	converged in 47
3	1.0	did not converge
4	1.5	did not converge
6	2.5	did not converge
10	4.5	did not converge
20	9.5	did not converge

Output is floored at zero, so the failure is bounded oscillation rather than divergence. The process rests on an expectation falsified every period out of equilibrium: each firm assumes its rivals hold output constant, and each period they all move. Bertrand (1883), reviewing Cournot four decades later, objected to the choice of strategic variable rather than to the adjustment — firms set prices, not quantities, and under price competition the equilibrium collapses to marginal cost. That objection is orthogonal to the one here and is not answered by it: the point below is that even granting Cournot his quantity-setting firms, his own adjustment process does not reach his own equilibrium without an assumption he did not make.

The standard repair is damping — firms adjust part of the way, $q \longleftarrow q + d(\mathrm{BR}(q) - q)$ — and the standard gloss is that damping restores convergence at the cost of an inertia assumption the original model does not contain. That is true and it undersells the point. The damped map has linearised gain $|1 - d(n{+}1)/2|$, so it is stable **iff $\boldsymbol{d < 4/(n{+}1)}$**:

n	$4/(n{+}1)$	largest d converging	smallest d failing
2	1.3333	1.32	1.34
3	1.0000	0.98	1.00
4	0.8000	0.78	0.80
6	0.5714	0.56	0.58
10	0.3636	0.36	0.38
20	0.1905	0.18	0.20

measured on a 0.02 grid, bracketing the prediction at every n tested.

The threshold is not a constant. It vanishes like $4/n$. So the damping that rescues Cournot's own adjustment process is not a single inertia parameter chosen once — it is a quantity each firm must condition on the number of rivals it has, slowing itself in proportion as the market gets more competitive. **Rescuing the dynamic requires every firm to know n and act on it, which is more information than the static expectation the dynamic is built on grants them.** The repair needs precisely what the model denies.

This is a claim about the structure of the adjustment process and it does not expire. It is also sharper than the usual objection: not *damping is an extra assumption*, but *which* assumption and *how much* of it, in a direction that worsens without bound as n grows.

4.3 · Identities checked rather than assumed

The markup condition of Lerner (1934), $(p - \mathrm{MC}_i)/p = s_i/|\varepsilon|$, holds to machine precision at equilibrium (maximum absolute residual 1.1×10^{-16} across the cost vectors tested) — a verification of the markup equation rather than an assumption baked into the solver. The Cournot limit theorem is exhibited rather than asserted: with symmetric MC = 10, $p - \mathrm{MC}$ falls 30.0 ⟶ 8.18 ⟶ 0.891 ⟶ 0.0899 for $n = 2, 10, 100, 1000$. HHI equals $1/n$ exactly for symmetric firms. Three independent solution routes — closed form, simultaneous first-order conditions, and damped tâtonnement — agree to 10^{-6}, so each checks the others.

5 · Abandoned approaches

This section is not a formality and it is not an appendix. A result reported without the routes that failed is a result the reader cannot calibrate — they are shown the one path that worked and left to assume it was the only one considered. The test applied to every entry below is: had this route worked, which sentence in this paper would be different?

Attacking the diagram rather than the theory. The original framing argued that partitioning agents into fixed buyers and sellers is mathematically invalid, and presented that as a critique of general equilibrium. It is not. Walrasian excess demand already has exactly this property, and Arrow–Debreu never used the Marshallian cross. The framing was abandoned because a referee's reply was available and fatal — *the author is attacking the introductory diagram rather than the theory* — and the section could not have recovered from it. **The cost was the paper's most rhetorically satisfying claim.** What replaced it is the reduction result of §3.3, which concedes that the cross is correct as a snapshot and locates the failure in the *operation* instead. That is a smaller claim and a true one, and §3.3 exists only because the larger one was given up.

A superposition framing for agent role. Proposed as a way to soften the critique for traditionalists: the agent is "in superposition" between buyer and seller until the price is observed. Rejected on technical grounds. Superposition denotes genuine indeterminacy prior to measurement, and these agents hold a definite reservation price at all times; their role is a deterministic threshold function of that price against p. It is a sign function, not a superposition. The cost of using it anyway would have been a reviewer observing that quantum mechanics was invoked to describe a piecewise-constant function, in a paper whose defensive strategy rests on dimensional rigour — and econophysics carries enough reputational damage from loose physics metaphor already. **The metaphor also undersells the claim**, which is the part worth keeping: "both schedules are readings of one distribution, therefore they are not independent equations, therefore perturbing them separately is invalid" is a precise mathematical statement, and fog makes it weaker. Had this route been taken, §1 and §2.1 would be written in a vocabulary the result does not need.

Reporting the invariance as partial. §3.1's first measurement used a 12-point grid spanning the full range of reservation prices and returned **4** distinct excess-demand schedules rather than 1. That reads as a partial invariance, it is the kind of finding that gets written up as one, and it was very nearly written up as one. Both grid endpoints coincide with data points — the minimum and maximum reservation prices — and the strict inequalities in the demand and supply counts then disagree about a single agent, whose holding status varies by allocation. Two endpoints × two holding states = 4. **It was a tie convention, not an economic effect.** Had it gone in, the paper's central claim would have been stated one full step weaker than it is true, and the identity in §3.1 would never have been looked for. The grid now excludes any point within 10^{-9} of a reservation price, in the regeneration script and in the test, with the reason recorded beside it.

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6 · Limitations

1. **The identity is a property of the no-income-effect case, and that is the load-bearing restriction.** In general, $z(p) = \Sigma_i \; d_i(p) - E$, and the allocation enters only through the aggregate endowment E precisely when each agent's demand d_i does not depend on their own endowment. Income effects break that, and income effects are exactly what SMD requires to generate pathology. So the two results sit at opposite ends of one axis: **at zero income effects the allocation cancels exactly and the schedules are one object; with income effects the aggregate inherits almost nothing and the schedules are not well-behaved objects at all.** The conclusion that they are not independent primitives survives at both ends, but *for different reasons*, and **only the first end is proved here.** The middle of that axis is not characterised by this paper and the general statement above is algebra, not one of this paper's demonstrated results. That the no-income-effect end is the tame one is not this paper's discovery — the absence of wealth effects is the standard route to a unique and tâtonnement-stable normalised equilibrium (§3.5) — which makes the identity in §3.1 a sharp instance of a known regularity rather than an isolated curiosity, and correspondingly less impressive than it would be if the general case behaved this way. It does not. One further caveat belongs here rather than in a footnote: the no-wealth-effect property itself holds only while numéraire consumption is interior, so even the tame end has an edge.
2. **Indivisible units, one per agent.** Multi-unit and divisible demand are not modelled. The partition argument in §3.1 is written for the unit-demand case and its generalisation is stated in Limitation 1 rather than demonstrated.
3. **One good.** There are no relative prices, no substitution and no budget constraint binding across markets. The construction is an exchange economy in the narrowest available sense.
4. **No production, no entry, no dynamics in §3.** The exchange results are static; the only dynamics in the paper are Cournot's adjustment process in §4.2, which belongs to a different model.
5. **The endowment-effect transform is exogenous and uncalibrated.** λ is swept, not estimated. The monotone volume decline is a qualitative match to a documented experimental finding; no claim is made that any particular λ

describes any particular population, and the magnitudes are not compared to experimental magnitudes.

6. **Finite *N*, one distributional family, one seed.** $N = 400$ and $c(m)$ is lognormal. The identity of §3.1 is exact and distribution-free by the partition argument, so it does not depend on these; the reported figures do.
7. **§4.2's threshold is a linearised result verified on a grid.** The bracket is 0.02 wide and the test asserts a bracket rather than an equality. The scaling is the claim; the third decimal is not defended.

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7 · Relation to existing work

The non-independence of the schedules is not itself new, and the paper is stronger for saying so. Walras's excess-demand formulation already resolves an agent's side of the market by price rather than by type, and the Sonnenschein–Mantel–Debreu theorems already deny that the aggregate inherits the properties the cross depends on. The contribution here is narrower and more elementary: the *identity* in §3.1 — the algebraic cancellation of the allocation from the difference in the unit-demand case — makes the point without invoking either, and it locates the failure of the textbook construction in one specific operation rather than in the construction as a whole.

The capital controversies are the nearest precedent for the shape of the argument, and the relation is a debt rather than an application. Sraffa (1926) on returns under competitive conditions, and then Sraffa (1960), where prices and distribution are derived from the technical conditions of production without an aggregate capital measure entering as a primitive; Robinson (1953), objecting that a scalar "capital" presupposes the price system it is used to derive; and Samuelson (1966, p. 568), conceding that reswitching and capital-reversing cannot be excluded once capital goods are heterogeneous, so that the neoclassical parables relating a falling interest rate to greater roundaboutness "cannot be universally valid" — the formal withdrawal of the non-switching theorem being Levhari and Samuelson (1966). **That concession is narrower than it is often reported to be and is stated here at its actual width:** it concerns aggregate-capital parables, not general-equilibrium or marginal-productivity theory, and it is a claim about what cannot be excluded rather than about what is empirically common. All the same, the move is one move: an aggregate treated as a primitive turns out to be a fold over units whose validity requires conditions on the units. This paper's object is smaller — a schedule rather than a capital stock — and its conditions are cheaper to state, but the structure of the objection is theirs.

The reservation-price distribution is a limit order book, and this is the empirical opening the construction offers rather than a metaphor: one population, bids below and asks above, and the "intersection" is where the book crosses. That places the formulation inside an existing microstructure literature with abundant data — Bouchaud, Farmer, Lillo and the surrounding work on order-book dynamics and market impact — rather than outside all literatures. **No empirical claim is made here.** The connection is stated because it names where a test would come from, and because a construction with an empirical object available is a different proposition from one without.

On accounting consistency, the cancellation in §3.1 is a stock–flow statement: the allocation is a stock, trade is a flow, and the identity holds because every unit held is a unit not demanded. The stock–flow consistent tradition, in the form developed by Godley and Lavoie, makes the general discipline explicit — that every flow originates somewhere and terminates somewhere, and that models which do not enforce this can produce results that are artefacts of the omission.

On the Cournot side, the instability of simultaneous best-response adjustment is classical and the $(n-1)/2$ gain is standard. The extension in §4.2 — that the stabilising damping shrinks like $4/n$, so the informational requirement of the repair grows with the size of the field — is stated here as a small result rather than a survey of the stability literature, which is large and which this paper does not attempt to cover.

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8 · Data and code availability

All results in this paper are produced by open code, and no proprietary or restricted data is used, because no empirical data is used at all — every number is generated by computation over a specified distribution.

- **Repository:** `https://github.com/jasoncbraatz/wealth-tensor` (public)
- **Modules:** `src/wealth_tensor/excess_demand.py`, `src/wealth_tensor/cournot.py`
- **Regenerate every number in §3 and §4:** `python3 scripts/wt018_report.py`

- **Test suite:** `python3 -m pytest tests/ -q` — 109 tests across the repository, of which 22 hold this paper's two modules in place (`tests/test_excess_demand.py`, `tests/test_cournot.py`)
- **Commit for the results reported here: 6492157** — the last commit touching `src/`, and therefore the state of the code that produced every number. *A head-of-repository SHA will additionally be pinned when this paper is posted.*

Pinning the last commit that touched `src/` rather than a bare placeholder is deliberate: it is non-circular (a paper cannot cite the commit that adds the paper), it is verifiable today, and it names the object a replicator actually needs — the state of the code, not the state of the prose.

Two of the tests exist specifically to make overclaiming fail loudly rather than quietly. `test_excess_demand_is_monotone_here_so_this_is_not_an_SMD_result` pins the limit stated in §3.5, so that a future draft cannot promote this construction into an SMD result. `test_excess_demand_is_identically_invariant_to_the_allocation` asserts all three counts of §3.1 (25 / 25 / 1) *and* the closed form, so that the central claim cannot be quietly weakened back to the crossing-only version it was stated as before. A test suite that constrains the author is a different object from one that flatters him.

The repository's `docs/` directory is deliberately public and contains the project's working notebook, including the ledger entries in which this paper's central claim was found to be one step weaker than the code supported, and in which §5's tie-convention near-miss is recorded in full.

References

The citation rule this list follows. *The edition cited is the edition* ***consulted*** *— the copy in the author's possession — not the earliest printing a catalogue happens to list. Where the original's date does argumentative work, because the entry is a translation or because a claim about priority rests on it, the entry is* ***dual-dated*** `original/consulted`*. A reprint that changes no pagination is a printing, not an edition, and is not dual-dated.*

Verified 2026-08-10 (session wealthTensor-06). ✓ — checked against a publisher page, a library-catalogue record, a Crossref record or the issuing body's own documentation, not recalled. ✓✎ — additionally checked against **the author's own copy**, by reading that copy's title page and colophon.

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'Notes' " reprinted from his 1898 Quarterly Journal of Economics *article. The French original is* Recherches sur les principes mathématiques de la théorie des richesses*, L. Hachette, Paris, 1838. The entry is dual-dated because §4 attributes the oligopoly model and its adjustment process to Cournot's priority, so the 1838 date does argumentative work.)*

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Accounting consistency

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This programme

[II] Braatz, J. C. (2026). *The base caps the region, the rate moves* *you within it: redistribution as a parameter space.* Manuscript, `docs/papers/paper-II-redistribution/paper-II.md` in the repository named in §8. *Not yet posted; this citation will carry a public identifier at submission.*

[III] Braatz, J. C. (2026). *A crisis is deferred information arriving at once: the dual tensor of wealth, and a pre-registered prediction it lost.* Manuscript, `docs/papers/paper-III-dual-tensor/paper-III.md` in the repository named in §8. *Not yet posted; this citation will carry a public identifier at submission.* §1 cites P3 from its §2.2.

How this list was checked, recorded because a reference section that silently improves teaches a reader nothing.

Two passes ran, in this order, and the second found what the first structurally could not.

1. **Bibliographic** — *does this work exist with these details?* Every entry checked against a publisher page, a library catalogue, a Crossref record or the issuing body's documentation. It found **one error**, Bertrand's volume number, and two entries stated more narrowly than the record supports (Marshall's 1890 imprint is Volume I; Walras's is the first instalment of a work completed in 1877). Otherwise clean.

2. **Cited-in-text** — *does this entry do any work in the body?* It found **two failure modes, and most of the list had one of them.** Several entries were listed and never mentioned in any form: Bertrand, Sraffa (1960), Thaler (1980), and a Kahneman–Knetsch–Thaler survey from 1991. More were invoked *eponymously* — "Walrasian excess demand", "the Marshallian cross", "the Lerner condition", "the documented experimental finding" — prose that names a person or a construction and cites nothing, so the body reads as attributed while the list reads as used, and neither document reveals the gap on its own. **That is the failure a bibliographic pass structurally cannot see**, because every entry it checks is real.

   Disposition: Bertrand, Sraffa (1960) and Thaler were given the work they had been listed for, and the eponymous invocations now carry years. The 1991 survey was **removed** — the 1990 paper does the same job better, and an entry retained for completeness is an entry doing no work.

A third pass — provenance, against the author's own library — was run and is partially complete. It located Cournot and corrected *that entry from the 1838 French original to the Kelley reprint of the Bacon translation actually consulted. It did* ***not*** *locate Marshall, Sraffa, Robinson, Samuelson, Mas-Colell et al., or the Kahneman–Knetsch–Thaler and Thaler papers.* ***That is a null result from an indexed subset, not an absence*** *— journal articles are not expected in a book archive, and the book collection is mid-reorganisation across several devices. No citation has been altered or removed on the strength of it. The outstanding question, which needs the author and not a session, is which edition of Marshall, Sraffa (1960) and Mas-Colell et al. he consulted.*

The base caps the region, the rate moves you within it: redistribution as a parameter space

Jason C. Braatz
Independent researcher
jason@braatzresearch.com

Draft — not yet submitted. Version 0.3, 2026-08-24.

Declaration of interest. The author is employed by a company building accounting software for very small businesses. This work was conducted independently, on personal time, and without company funding, data or direction.

Use of AI assistance. Anthropic Claude Opus 5, at high reasoning effort, was used throughout as a research and drafting assistant: literature retrieval, adversarial review, code review and prose drafting. All claims, results and final text are the author's, and every computational result is produced by committed code in the repository named in §7.

Revision note: ***v0.3****, this draft — independent review passes; results, numbers, claims and citations were corrected where those passes found them wrong.*

Abstract

In multiplicative wealth processes, unopposed growth condenses: log-wealth variance grows without bound and the Gini approaches unity. Any distribution that does not condense is being opposed by something. This paper classifies the opposing mechanisms not by institutional origin but by four coordinates — **base, rate, periodicity, threshold** — and asks which regions bound inequality below unity. First, the **base sets a ceiling the rate cannot cross**: at a matched rate the two bases differ by roughly an order of magnitude in κ, the levy's compressive budget, for which the flow base admits a closed form. The stronger prediction it was built to test — that a flow levy fails to oppose the multiplicative term *regardless of rate* — is **false**, and §3.1's rate sweep is what falsified it: the frontiers are **nested**, stock 0.000 against flow 0.125. Second, the surviving claim is narrower and better: the decisive quantity is **realisation** — the share of a period's gain the base can see. At zero realisation the flow base is uniform, so a **100 % levy on flow leaves wealth exactly unchanged** (Gini 0.994 and top decile 1.000 in both). Third, periodicity and threshold are trim, not structure: they modulate the effective rate without opening or closing a region; a threshold at a quarter of the mean is close to free. The claims are properties of a model class; no causal claim about any institution is made. All results reproduce from open code; 18 tests pin them.

Keywords: kinetic exchange models · wealth condensation · multiplicative growth · Gini coefficient · tax base · realisation · agent-based models · econophysics

JEL classification: D31, D63, H23, H24, C63

1 · Introduction

A multiplicative wealth process with a positive mean growth rate condenses. This is not a pathology of any particular model; it is what multiplicative processes do. The variance of log-wealth grows without bound, an additive wage becomes negligible in comparison, and the wealth share of the top holder tends to one. The kinetic-exchange literature has established this repeatedly and by several routes (Drăgulescu and Yakovenko, 2000; Patriarca, Chakraborti and Kaski, 2004; Yakovenko and Rosser, 2009).

The observation that motivates this paper is the contrapositive, and it is not often stated plainly: **every wealth distribution that is not condensed is being opposed by something.** Redistribution is not, in this framing, a policy question that arrives after the economics. It is a term in the process, and its absence is the special case, not its presence.

That reframing makes a question available which the usual framing does not. Redistributive mechanisms are ordinarily compared by institutional identity — a progressive income tax, a wealth tax, zakat, a land value tax — and any comparison drawn on that axis is immediately and correctly read as an argument about which tradition is preferable. But institutional identity is not what the process responds to. The process responds to five numbers — the levy's four coordinates and the realisation share of its base (§3.2) — and it cannot see where any of them came from.

Contributions. This paper is short and its claims are specific.

1. A **parameterisation** of any period-assessed levy by four structural coordinates — base, rate, periodicity, threshold — plus the realisation share of the base, and a demonstration that the process's behaviour is a function of these alone (§2).
2. The result that the **base caps the reachable region and the rate only moves you within it**, with κ — the levy's

compressive *budget*, not its mechanism — separating the bases by an order of magnitude, and a closed form for the flow base's κ that the simulation reproduces to within 7 % at every rate tabulated (§3.1). The stock-versus-flow contrast this result sharpens is prior and is credited in §3.1 and §6; what is new is κ itself — the levy's budget, separated from its mechanism — and the closed form for it.

3. The identification of **realisation as the decisive quantity**, including the limiting result that a confiscatory levy on flow, at zero realisation, leaves the wealth vector exactly unchanged — its base is uniform, not absent (§3.2). This is a statement about what a base can *observe*, not about how hard it squeezes.
4. A methodological result of independent interest: **a summary statistic with a hard ceiling cannot serve as a convergence criterion.** The Gini is capped at (N−1)/N, so a fully condensed economy also stops rising, and a drift test scores total condensation as bounded (§3.4).
5. A **reproducible artefact**: every number below is regenerated from a public repository by the two commands §7 names — save the six quantities §7 enumerates, which no command prints — and the claims are held in place by the 18 tests in `tests/test_redistribution.py`, one of which exists specifically to make overclaiming fail loudly — alongside a second, in a companion module of the same suite, that does the same office for the superseded price-formation manuscript (§7).

A boundary. Everything here is positive. The claims are properties of a model class. Where a historical institution is mentioned it is mentioned as a *coordinate* — an existence proof that a region of the parameter space is implementable — and never as a recommendation. Zakat is named in this paper for exactly one reason: it is assessed on stock held above a threshold across a full year rather than on income received, which places it somewhere specific on the base axis. That is a measurement, not an endorsement.

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2 · The model

2.1 · The process

N agents. Each period, every agent's wealth is multiplied by an idiosyncratic growth factor and receives a common additive wage:

$$\mathbf{w\_i(t+1) = w\_i(t)\cdot(1 + \eta\_i(t)) + a}, \eta\_i \sim \mathcal{N}(\mu, \sigma^2)$$

The multiplicative term is the engine of condensation; the additive wage *a* is the only force opposing it in the absence of a levy, and it is not enough. Throughout: $N = 800$, $\mu = 0.05$, $\sigma = 0.20$, $a = 0.05$, $T = 1200$ periods, with reported statistics averaged over the final quarter of the path.

2.2 · The levy, as four numbers

A period-assessed levy is described by:

coordinate	meaning
base	what is assessed — the **stock** held, or the **flow** received
rate *r*	the fraction of the liable amount taken
periodicity *P*	periods elapsed between assessments
threshold *θ*	the exempt amount, in multiples of the mean of the base

Everything collected in a period is redistributed per capita in the same period. The levy is therefore a **pure transfer**: aggregate wealth is untouched and only dispersion changes. This is verified to machine precision rather than assumed: `test_the_levy_is_a_pure_transfer` in `tests/test_redistribution.py` holds the implementation's reported `transfer_error` below 1e-12, so that no result below can be an artefact of the levy quietly changing the growth rate.

2.3 · Realisation

ρ is the share of a period's capital gain that is recognised as flow, and therefore enters a flow levy's base. ρ = 1 is a mark-to-market levy on accruals; ρ = 0 is the pure rentier whose gains accrue and are never realised.

ρ is not a free parameter fitted to the result. It is a **stated structural property of every real tax system** — realisation-based taxation is not a modelling convenience but the near-universal practice — and it is **swept rather than chosen**: §3.1's flow rows are stated at ρ = 1 and labelled as such, §3.2 sweeps the axis, and the paper's central result is a statement about the whole ρ axis rather than about a value of ρ chosen to make it come out right.

2.4 · What is measured

Gini, of the wealth vector, in the exact sorted-rank form rather than a Lorenz approximation. **Top decile share**, because the Gini saturates (§3.4). And **κ**, the share of aggregate wealth actually moved per assessment — the levy's *compressive budget.* It is a budget and not a mechanism: §3.1 matches two levies at κ and finds them compressing unequally, and §3.3 removes a quarter of κ at no measurable cost.

A fourth quantity is reported wherever the question is whether a levy reached the *generator* of the process rather than its outcome: **Var[log *a*]**, the variance of the log per-period multiplier normalised by aggregate growth — written $a(\eta)$, a different object from §2.1's wage *a*, with which it unhappily shares a letter. It is a property of the process that produces next period's distribution, where the three statistics above are properties of this period's.

Bounded is the verdict the tables below carry, and it is a criterion rather than a statistic: a run is bounded when its Gini has settled over the final quarter of the path **and** its top decile sits below 0.90. The second condition is the one that does the separating, and §3.4 is the result that establishes why a settled Gini alone will not.

3 · Results

3.1 · The base sets a ceiling; the rate moves you within it

levy	Gini	κ	top 10 %	bounded
none	0.994	—	1.000	**no**
stock, $r = 0.025$	0.443	0.0250	0.336	yes
stock, $r = 0.100$	0.222	0.1000	0.193	yes
flow, $r = 0.025$	0.812	0.0025	0.734	yes
flow, $r = 0.100$	0.596	0.0102	0.481	yes
flow, $r = 1.000$	0.125	0.1026	0.138	yes

The flow rows are assessed at full realisation, $\rho = 1$ — §2.3's mark-to-market case and the implementation's default; §3.2 is the sweep that lowers it. These six rows are a selection: the rate sweep behind them is wider on both bases, and §3.4 quantifies over all of it.

At a matched rate the two bases sit roughly an order of magnitude apart in κ, at every rate tested. The budget is the table's κ column and is not a fitted relationship:

- for a **stock** base at zero exemption, $\kappa = r$ exactly — §3.3 raises the threshold and κ falls;

- for a **flow** base, $\kappa = r \cdot \mathrm{E}[\eta^{+}]$, where $\mathrm{E}[\eta^{+}]$ is the *gross positive* growth rate — because a levy cannot rebate a loss. In closed form, $\mathbf{E[\eta^{+}] = \mu\Phi(\mu/\sigma) + \sigma\varphi(\mu/\sigma) = 0.1073}$ for the parameters above. The simulated κ runs **4–7 % below** that form across the full rate sweep behind the table — −4.3 %, −4.6 %, −5.7 % at the three flow rates tabulated here (r = 1.000, 0.100, 0.025), reaching −6.8 % at the sweep's lowest rate, r = 0.010. The residual is flat between r = 1.000 and r = 0.500 and widens monotonically below it, which makes it a denominator convention rather than noise: the implementation measures κ against post-growth wealth. *These residuals are computed from the unrounded κ rather than from the four-decimal values the table displays; at* r* = 0.025 that display quantum is ±2 % of κ itself, which is wider than the spread being reported.* The test suite asserts agreement within 10 %.

So a **confiscatory** levy on flow has approximately the compressive budget of a **10 % levy on stock**. The base is not a detail of implementation. It sets a ceiling, and the rate — the coordinate that receives essentially all public attention — moves an economy only within the ceiling its base has already fixed.

The two bases do not merely differ in budget. They act on different objects. Matched at $\kappa \approx 0.10$, the two levies compress the cross-section unequally — Gini 0.222 against 0.125 — but the more telling comparison is what each does to §2.4's fourth quantity, Var[log a] — the generator of the process rather than its outcome. Unlevied, Var[log a] = 0.076542. Under the **stock** levy at that budget it is **0.076536** — a change of 6×10^{-6}, which is to say none at all. Under the **flow** levy it is **0.051189**, a third lower. A levy on stock rescales what a holder has and leaves the process that got them there exactly as it found it; a levy on flow reaches into the multiplicative term itself. The stock base truncates the outcome, the flow base damps the generator — and both register as a smaller Gini, which is why the distinction is invisible in the statistic normally reported. An outcome measure records that the distribution was compressed. It does not record whether the mechanism producing next period's distribution was touched.

This contrast is not new. Bouchaud and Mézard (2000) carry a flow levy, a stock levy and the per-capita redistribution of each in a single wealth balance, and give the stationary Pareto exponent in closed form in all four of their own tax parameters — a rate and a redistributed fraction for each base. They write that exponent μ — a different object from §2.1's growth drift μ, with which it unhappily shares a letter. Their ranking is the one measured here, and they state it more strongly: income taxes *"tend to reduce the inequalities of wealth (i.e., lead to an increase of μ), even more so if part of this tax is redistributed"*, while *"quite surprisingly, capital tax, if used simultaneously to income tax and not redistributed, leads to a decrease of μ"*. Their stock levy can *reverse* the sign of the effect; the one measured here merely buys less compression per unit of budget. What this section adds is not the contrast but the pair of witnesses for it — κ, which says how much budget a base has, and Var[log a], which says whether the levy spent it on the outcome or on the generator — in a discrete process where the two can be matched and separated. §6 states what that leaves.

A prediction that half-failed. The claim this section was built to test was stronger: that a levy on flow does not oppose the multiplicative term *regardless of rate.* That is **false as stated**, and the sweep is what falsified it. At full mark-to-market realisation a flow levy does bound the Gini; it is merely weak. Rate 1.00 on flow reaches Gini 0.125, which a stock levy reaches at rate 0.25. The reachable frontiers are **stock 0.000 < flow 0.125** — the bases occupy *nested* regions, not disjoint ones. The surviving claim is narrower and is in the next section.

3.2 · Realisation is the crux

The multiplicative term operates on the **stock**. A flow base reaches it only through whatever share of the period's gain is recognised as income. That share is ρ, and it is the quantity that was doing the work all along:

realisation ρ	reachable Gini (flow base)
1.00	0.125
0.25	0.395
0.00	**0.994**

At ρ = 0 — the holder whose gains accrue but are never realised — a **100 % levy on flow leaves the wealth vector exactly unchanged**: Gini 0.994 against 0.994, top decile 1.000 in both, and the two paths agree agent by agent rather than merely on the summary statistics. The identity is structural. The levy is still assessed — at ρ = 0 the flow base is not empty but is the accrued **wage**, and the assessments do fire — but the wage is identical for every agent, so the levy takes the same amount from each and returns it per capita. A uniform assessment with a uniform rebate is the identity on the wealth vector. What ρ = 0 removes is not the levy but the **dispersion in its base**.

That is the true "regardless of rate" result, and note what kind of statement it is. It is a claim about what a base is able to **observe**, not about how hard it squeezes. A rate is an intensity; realisation is an *observability*, and the observability binds first.

Unrealised appreciation is wealth whose growth the assessing layer has not been asked to recognise, and what the ρ axis measures is how much of a period's gain that layer can see at all.

A stronger reading was available, was tested, and is withdrawn here. One might hold that a levy which cannot see an accrual and a financial statement which does not record a degradation *are the same structure*, seen from two sides. Put to a cross-scale check against the companion work on the reporting layer, that identification does not hold. The share of a change a reporting filter fails to recognise is **deferred** — it accumulates and is released later at a stated rate — while the share of a gain this model's base fails to recognise is **never assessed at all**. Deferred arrival and non-arrival are different operators, and a shared adjective between them is not an equation. The check, its thresholds, and its preregistration — the withdrawal was registered before the check was run — are recorded in `docs/RESULT-END-TO-END-001-E1.md`.

3.3 · Periodicity and threshold are trim, not structure

Both remaining coordinates modulate the *effective rate* rather than opening or closing a region, which is what leaves base and rate as the two structural axes.

Periodicity. On a stock base, holding the average rate constant at 0.02 per period, assessing every P periods at rate $0.02{\cdot}P$ moves the stationary Gini from 0.486 (P = 1) to 0.456 (P = 20). A lumpier assessment is very slightly **stronger** over that range, because it catches dispersion that has had time to accumulate. The effect is not monotone in P, and the sweep behind those two endpoints says so: the minimum is **interior**, 0.451 at P = 30, and by P = 50 — where holding the average rate at 0.02 requires the maximum rate, 1.00 — the Gini has returned to 0.469, above its P = 20 value. The whole sweep spans 0.035, which is the operative fact: an annual assessment is not a watered-down continuous one.

Threshold. On the same base at r = 0.025, monotone and smooth, with no cliff: Gini 0.443 at zero exemption rising to 0.770 at 20× the mean of the base. The interesting part is the near end. A threshold at **0.25× the mean costs nothing measurable** in compression (0.444 against 0.443) while reducing κ by a quarter. Exempting small holders removes a quarter of the assessed volume and none of the effect, because the compression is performed by transfers at the top of the distribution.

A threshold that exempts the poor is therefore not a concession that weakens the mechanism. It is close to free.

κ is necessary and it is not sufficient, and there is a witness on each side. §3.1 matches the two bases at $\kappa \approx 0.10$ and finds them compressing unequally, 0.222 against 0.125; the threshold sweep above supplies the converse from the other side, removing a quarter of κ at no measurable cost in compression, 0.444 against 0.443. κ can hold while the outcome moves and move while the outcome holds, so **no function of κ alone reproduces §3.1's table.** κ is what a base makes available to spend — which is why the bases sort, and why the closed form is worth having — but what the spending buys is fixed by the object the levy acts on, which is the distinction §3.1's two-witness comparison draws.

3.4 · A methodological result: saturating statistics cannot detect convergence

The boundedness criterion was first written as a drift test — the Gini has settled if its mean over the last quarter of the path exceeds the previous quarter's by less than a tolerance. It scored the **unopposed** process, the one the entire exercise exists to contrast against, as *bounded*.

The Gini of N agents is capped at $(N-1)/N$. A fully condensed economy therefore also stops rising — not because it reached a stationary distribution but because it ran out of headroom. At $N = 800$ and $T = 1200$ the unopposed process reads Gini 0.994 and flat — short of the 0.99875 ceiling it is pinned against — while its top decile holds 1.000 of everything. The drift test was measuring the ceiling.

§2.4's criterion pairs the settled Gini with a top decile below 0.90 for exactly this reason, and it is the second condition that does all of the separating. Across §3.1's full rate sweep — wider than the six rows tabulated there — the bounded runs' Gini spans 0.000–0.891 against the condensed run's 0.994 — a gap of 0.103 whose upper edge is a *saturated reading* and not the 0.99875 ceiling it falls short of, so any Gini threshold would have to be drawn inside it and redrawn for every N; their top decile spans 0.100–0.861 against 1.000, clearing the 0.90 threshold with 0.039 to spare.

The general rule, which is not confined to this model: a summary statistic with a hard ceiling cannot distinguish "converged" from "saturated." Before using one as a convergence criterion, ask what its maximum is and whether the failure mode you are trying to detect drives it there.

4 · Abandoned approaches

A result reported without the routes that failed is a result the reader cannot calibrate — they are shown the one path that worked and left to assume it was the only one considered. Everything here was actually attempted.

Classification by institutional origin. The first framing compared named systems. It was abandoned because it is unanswerable as posed: any ranking of institutions is read as an argument about traditions, and correctly so. The parameter-space framing replaces "which system do you favour" with "which regions bound the Gini below unity", which is a question the models answer and no one can call advocacy. The cost is that the paper can no longer say anything about any actual institution, and it does not.

"Regardless of rate." §3.1. The original claim was sharper and false. The shape of the failure is informative: the direction was right, the mechanism was misidentified, and the surviving claim is the narrower one about realisation.

Defining "flow" so that the claim came out right. Once "regardless of rate" failed, an obvious repair was available: redefine the flow base so that it excludes the components that rescued it. This was refused. A definition adjusted until the prediction survives is a free parameter wearing different clothing. The claim was narrowed instead.

The drift-only boundedness test. §3.4. It looks like a convergence check. It was caught by asking what the statistic's maximum was — and it is now pinned by a test named `test_a_flat_gini_does_not_mean_a_bounded_one`, so that any future simplification of the criterion fails loudly instead of quietly re-scoring condensation as success.

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5 · Limitations

1. **ρ is exogenous here, and in the world it is not.** Realisation is chosen, and it responds to the rate: raising a flow levy gives holders a reason to realise less. Endogenising ρ would make the flow base *weaker* than reported, but it is unmodelled, and a reader should treat the ρ axis as a comparative static rather than a policy response function.
2. **This is a model-class result.** No causal claim about any historical or contemporary institution is made, and none is supported. No field evidence is used, required, or available.

3. **No production, no labour supply, no portfolio choice, no behavioural response to the levy.** The Lucas critique applies in full force and is not answered here.
4. **One good, one asset, no prices.** The process is a wealth process, not an economy.
5. **Finite N, one seed per reported figure, and a fixed parameter neighbourhood.** $N = 800$, and every *simulated* number above is a mean over a tail window of a **single** path at `seed = 0` rather than an ensemble average — the exceptions are the five closed-form quantities §7 names: §3.1's $E[\eta^+]$ and its three $Var[\log a]$ values, which are quadrature, and §3.4's Gini ceiling, which is arithmetic in N. Seed-robustness is asserted separately rather than averaged in: `test_the_result_is_not_a_lucky_seed` holds two configurations inside their stated bands across five seeds, at the reported $T = 1200$ as well as at the suite's $T = 600$. The qualitative separations are large relative to those bands, but the third decimal is not defended.
6. **Growth shocks are Gaussian and i.i.d. across agents and time.** Heavy tails and autocorrelation both plausibly matter and neither is explored.
7. **Positive, not normative — and that boundary is a constraint on the reader too.** Nothing here implies that bounding inequality is desirable. It implies only that certain regions of the parameter space do it and others do not.

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6 · Relation to existing work

The condensation result is standard in kinetic exchange (Chakrabarti, Chatterjee, Chakravarty and the surrounding literature), where the effect of saving propensity, taxation and redistribution on stationary wealth distributions has been examined from several directions. **Two works in that literature are prior to this paper's central contrast.**

Bouchaud and Mézard (2000), whose wealth balance and closed-form Pareto exponent §3.1 sets out, are prior to the stock-versus-flow ranking used there. The contrast between the two bases — in terms of what each does to the shape of the stationary distribution — is theirs, and the per-capita rebate fraction is a parameter in their solution rather than an extension awaiting one. Their solution is continuous-time and carries neither a periodicity nor a threshold, so §3.3's two trim coordinates are outside it.

Benhabib, Bisin and Zhu (2011) supply three further results that bound what is left. Their Proposition 3 has the tail index rising in both the estate tax and the capital income tax, so the *nested* frontiers this paper reaches in §3.1 were already visible in a different metric and a different model. Their Proposition 4 has tail inequality rising with a mean-preserving spread of the return process, which is the general form of §3.1's finding that the flow levy reaches the dispersion of the multiplier and the stock levy does not. And their §4.1 notes that an economy whose multiplier is bounded below one has a stationary distribution bounded above, with no power-law tail at all — a claim this paper does not make and does not need, but the first one any extension of §3.1 toward tail indices, the object Gabaix (2009) surveys, would meet.

What remains is narrower than the contrast. It is not that redistribution opposes condensation, and it is not that the two bases act differently on the shape of the distribution. It is that the mechanisms sort by **observability of the base** (§3.2) rather than by rate or institutional form; that the budget through which they operate has a closed form (κ) rather than being a simulation regularity, though the sorting is not a function of that budget alone (§3.1); and that a single sweep separates the two by measuring the generator and the outcome side by side. Three further differences are of construction rather than of claim, and none is offered as a result: the levy here is on the **realised gain only**, with no loss offset, so the multiplier is asymmetrically truncated rather than symmetrically contracted toward one; the two bases are compared at matched compressive **budget** rather than at matched rate; and the process is a discrete Kesten-type recursion with an explicit per-period budget identity rather than a continuous-time mean-field one.

The realisation result touches the public-finance literature on the choice of base (Kaldor, 1955) and on realisation-based versus mark-to-market taxation (Auerbach, 1991; Toder and Viard, 2016; Saez and Zucman, 2019) from an unfamiliar angle: not from the incentive or valuation side, but as a statement about the information available to the assessing layer. On the stock-versus-flow axis, the paper is silent about optimal taxation and deliberately so; it characterises reachable regions, not desirable ones.

7 · Data and code availability

All results in this paper are produced by open code and no proprietary or restricted data is used, because no empirical data is used at all — every measured number is generated by simulation, save the four closed-form quantities the next bullet names and §3.4's Gini ceiling $(N-1)/N = 0.99875$, which is arithmetic in N and is printed by no command here.

- **Repository:** `https://github.com/jasoncbraatz/wealth-tensor` (public)
- **Module:** `src/wealth_tensor/redistribution.py`
- **Regenerate every number in §3:** `python3 scripts/wt030_report.py` — except §3.1's four closed-form quantities: $\mathrm{E}[\eta^+] = 0.1073$ and the three Var[log a] values, which are quadrature over the multiplier's distribution rather than simulation output and come from `python3 scripts/wt077_tail_index.py`, and except six quantities neither command prints in any precision: §3.4's Gini ceiling $(N-1)/N = 0.99875$, which is arithmetic in N; §3.4's 0.90 top-decile criterion, which is a chosen threshold and not an output; §3.3's 0.035 periodicity span and §3.4's 0.103 Gini gap, each a difference of two values `wt030_report.py` prints; §3.4's 0.039 top-decile margin, the distance from that command's printed 0.861 to the 0.90 threshold above; and §3.1's 6×10^{-6} change in Var[log a], the difference of two values `wt077_tail_index.py` prints.
- **Test suite:** `python3 -m pytest tests/ -q` runs the whole repository; the **18** tests in `tests/test_redistribution.py` are the ones that hold this paper's claims in place, and that count is the one quoted in the abstract and in §1.
- **The two tests that make overclaiming fail loudly here — the suite holds others:** `test_a_flat_gini_does_not_mean_a_bounded_one`, which pins §3.4's boundedness criterion so that any future simplification of it fails instead of quietly re-scoring condensation as success, and `test_excess_demand_is_monotone_here_so_this_is_not_an_SMD_result`, which constrains this programme's price-formation manuscript — that manuscript is superseded and withdrawn, and the guard outlives it — in a companion module of the same suite, `tests/test_excess_demand.py`; it is a different companion from §3.2's work on the reporting layer.
- **Commit for the results reported here: 3b11f23** — the last commit touching `src/wealth_tensor/redistribution.py`, and therefore the state of the module that produced §3's simulation output. The pin is **per file** deliberately: a pin on the last commit touching `src/` as a whole is a sentence whose truth changes whenever any unrelated module moves, and nothing in the repository watches it. It does

not cover the two `scripts/` commands named above, which produce §3 numbers from outside `src/`. *A head-of-repository SHA will additionally be pinned when this paper is posted, and it is what covers them.*

Known limitations of this review. Every number above is regenerated by the two commands named here, and a test suite pins the claims; the cross-references, the citation pointers and the reference entries were checked by a sequence of independent reads rather than mechanically, and the simulation figures rest on a single seed per configuration with seed-robustness asserted separately rather than averaged in. Two things were not checked at all and are named rather than left to be found: §1's characterisation of zakat is not supported here by a primary source, and §3.1's *"4–7 %"* band is an under-claim — the sweep's largest residual is 6.831 %, so the band is true as written and looser than the evidence requires.

Pinning the last commit that touched the module rather than a bare placeholder is deliberate: it is non-circular (a paper cannot cite the commit that adds the paper), it is verifiable today, and it names the object a replicator actually needs — the state of the code, not the state of the prose.

The repository's `docs/` directory is deliberately public and contains the project's working notebook, including the entries in which the claim of §3.1 half-failed and was narrowed. It is part of the record rather than an appendix to it.

References

Bibliographic details for the entries marked ✓ were verified against live sources on 2026-08-10. The two marked ✓⧗ were re-verified against their Crossref records on 2026-08-17 and name, in the entry, the pre-publication version actually read, per `REFERENCE-POLICY` *§4. The remainder are standard works whose details are to be re-checked at submission per* `docs/papers/PREPRINT-CHECKLIST.md`.

Kinetic exchange and wealth condensation

Bouchaud, J.-P., & Mézard, M. (2000). Wealth condensation in a simple model of economy. *Physica A: Statistical Mechanics and its Applications*, 282(3–4), 536–545. `doi:10.1016/S0378-4371(00)00205-3` ✓⧗ *(issue and pagination checked against the Crossref record, 2026-08-17. Text consulted: arXiv* `cond-mat/0002374`*, read in full. The quotations in §3.1 are attributed to that preprint and may not appear verbatim in the article of record. Consulted 2026-08-17 / published 2000.)*

Chakrabarti, B. K., Chakraborti, A., Chakravarty, S. R., & Chatterjee, A. (2013). *Econophysics of Income and Wealth Distributions.* Cambridge University Press.

Chakraborti, A., & Chakrabarti, B. K. (2000). Statistical mechanics of money: how saving propensity affects its distribution. *The European Physical Journal B*, 17(1), 167–170. ✓

Chatterjee, A., & Chakrabarti, B. K. (2007). Kinetic exchange models for income and wealth distributions. *The European Physical Journal B*, 60(2), 135–149.

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Yakovenko, V. M., & Rosser, J. B. (2009). Colloquium: Statistical mechanics of money, wealth, and income. *Reviews of Modern Physics*, 81(4), 1703–1725. ✓

Wealth dynamics and inequality

Benhabib, J., Bisin, A., & Zhu, S. (2011). The distribution of wealth and fiscal policy in economies with finitely lived agents. *Econometrica*, 79(1), 123–157. `doi:10.3982/ECTA8416` ✓⧗ *(page range checked against the Crossref record, 2026-08-17. Text consulted: NBER Working Paper 14730 full text, read in full; §6's characterisation of Propositions 3 and 4 and of §4.1 is taken from that version and the numbering may differ in the article of record. Consulted 2026-08-17 / published 2011.)*

Gabaix, X. (2009). Power laws in economics and finance. *Annual Review of Economics*, 1, 255–294.

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Public finance: base, realisation, and mark-to-market

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Saez, E., & Zucman, G. (2019). Progressive wealth taxation. *Brookings Papers on Economic Activity*, 2019(2), 437–533. ✓

Toder, E., & Viard, A. D. (2016). Replacing corporate tax revenues with a mark-to-market tax on shareholder income. *National Tax Journal*, 69(3), 701–731.

Methodological

Lucas, R. E. (1976). Econometric policy evaluation: a critique. *Carnegie-Rochester Conference Series on Public Policy*, 1, 19–46.

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§1's characterisation of zakat — a levy assessed on stock held above a threshold across a full year — is not supported here by a primary source. The paper's argument does not depend on it: the institution is a measurement, not a premise, and §1 says so.

Timeliness and durability are not separately identified from a reported series

Jason C. Braatz
Independent researcher
jason@braatzresearch.com

Draft — not yet submitted. Version 0.6, 2026-08-24.

Declaration of interest. The author is employed by a company building accounting software for very small businesses. This work was conducted independently, on personal time, and without company funding, data or direction.

Use of AI assistance. Anthropic Claude Opus 5, at high reasoning effort, was used throughout as a research and drafting assistant: literature retrieval, adversarial review, code review and prose drafting. All claims, results and final text are the author's, and every computational result is produced by committed code in the repository named in §11.

Revision note: ***v0.6****, this draft — independent review passes; results, numbers, claims and*
citations were corrected where those passes found them wrong.

Abstract

A balance sheet is an instrument, and instruments have transfer functions. Model the reporting layer as a low-pass filter on a physical layer that degrades whether or not anyone records it: a share $\boldsymbol{\varphi}$ of each true change passes through at once, the rest released at rate $\boldsymbol{\alpha}$ from an unrecognised gap.

The triples $(\alpha, \delta, \varphi)$ and $(\delta, \alpha, \varphi\delta/\alpha)$ generate the identical reported series, where δ is the physical decay rate: the roots exchange, preserving $\varphi\delta$ exactly. Timeliness and durability are therefore not separately identified, and where the asset's physical scale is unobserved (every firm-level series) the identified set is a continuum, a factor of **1.67** in that scale spanning the whole unit interval.

The corollary is cross-sectional: **classes are ordered by $(1 - \varphi) \odot \delta$, divided elementwise by $(\alpha - \delta)$, not by φ**. Across the four GAAP classes, δ leverage is **4.2** times the level at which recovery fails, and the composite **inverts** the intended ranking, Kendall $\tau = -1$ at the calibrated rate; drawing δ independently, the ordering survives in **11.5%** of 4,000 ladders. This constrains cross-sectional conditional-conservatism measures.

The framework's sharpest prediction, recognition lag ordered by GAAP asset class, was pre-registered, tested on 688 EDGAR-derived events in two pre-declared sectors, and **failed** (Jonckheere–Terpstra $z = -0.290$, -0.095; power 0.95–1.00). The repair follows from the theorem, not the failure: **disclosed useful lives supply δ from outside the series**, restoring φ for the two amortised classes; at $\delta = 0$ goodwill's parameter is absent.

Keywords: identification · conditional conservatism · reporting lag · impairment · pre-registration · asset life · deferred information

JEL classification: M41, D80, C18, G14, E01

1 · Introduction

A balance sheet is not a window. It is an instrument, and instruments have transfer functions.

That sentence is the whole paper, and the rest of it is an attempt to make the sentence cost something — to state the transfer function precisely enough that it forbids outcomes, to derive a consequence that can be checked, and then to check the sharpest one against public data and report what happened.

The starting observation is old and belongs to Soddy: a physical asset and a financial claim on that asset are different kinds of object, and confusing them is how societies come to believe they are wealthier than they are. What this paper adds is that the confusion has a *dynamics.* If the physical component degrades whether or not anyone records the degradation, and the claim component changes only when someone records something, then the gap between them is an accumulating quantity of a specific kind: **information the reporting layer owes and has not delivered.**

Read that way, several things stop being metaphors. Technical debt becomes off-balance-sheet entropy, and deferred maintenance becomes an unrecognised liability with a measurable integral — exactly proportional to $(1 - \varphi)$, where φ is the share of degradation the reporting layer can see.

The paper's substance, however, is not that model. It is what the model implies about the *readers* of reported numbers, including this paper's own attempt to be one.

The filter has two roots: the rate at which the reporting layer releases what it has withheld, and the rate at which the physical layer decays. **Those two roots are exchangeable, and the exchange preserves the product of timeliness and decay.** A reported series therefore contains $\varphi\delta$ and nothing further about φ — a prompt reporter of a durable asset and a slow reporter of a perishable one emit the *same series*, to fourteen decimal places. Timeliness and durability are not separately identified from reported numbers.

That constraint is not a property of this framework. It is a property of the object every conditional-conservatism measure takes as input, and its cross-sectional form is sharp enough to be uncomfortable: indexing asset classes, the ranking a reader can compute is the ranking of $\boldsymbol{(1 - \varphi) \odot \delta \oslash (\alpha - \delta)}$ — with $\odot$ the Hadamard product and $\oslash$ elementwise division, both across classes — so a comparison across classes with different asset lives is not a comparison of timeliness. Across the four GAAP classes, with the decay rates the standards themselves imply, the composite does not blur the intended ranking — it **inverts** it — and the ordering the design imposed survives in **11.5%** of ladders even when no durability ordering is imposed at all.

The framework's sharpest empirical prediction was preregistered, tested twice on EDGAR-derived impairment data with a stopping rule declared in advance, and it failed. §5 reports the failure at full length. §4 reports what the identification result says about that failure, and the answer is that it **does not explain it**. The one observable the composite spares is precisely the one the registration ordered — and it is not computable from public filings, so the instrument measured a substitute whose relation to the model was never written down. A theorem about identification, a registered null, and the discovery that the two are less connected than they look, are three different results, and this paper reports them as three.

Contributions.

1. **An exact observational equivalence, with its proof and its reach** (§4.2). The filter's two roots exchange, preserving $\varphi\delta$, so timeliness is not recoverable from a reported series by any estimator. The structure is the Bateman function's, and the accounting instance is placed against its known analogues in pharmacokinetics and compartmental identifiability. Where the physical scale is unobserved the degeneracy widens from two points to a continuum spanning all of φ.
2. **The cross-class corollary, in Hadamard form** (§4.3–4.4). With classes indexed the reporting layer is diagonal, the observable ranking is the ranking of $(1 - \varphi) \odot \delta \oslash (\alpha - \delta)$, and on the ladder GAAP supplies that ranking is the

reverse of the ranking of φ. The validity condition for a timeliness-ordered design is itself a statement about decay rates.

3. **A constraint on the conditional-conservatism measures** (§4.6), stated with its three qualifications, together with the specific circumstance under which those measures remain sound. The returns those measures condition on do break the equivalence (§4.7) — and the strength with which they break it is a property of the asset rather than of the design: its level is set by the gap between the book's amortisation rate and the asset's true decline, its response to news is set by the decay rate and reverses sign as that rate nears zero, and no horizon attains the root-T rate. The corner where the repair is weakest is the one the standards decline to schedule.
4. **A repair that needs no new data** (§4.7). Disclosed useful lives supply δ from outside the series, which is what the theorem requires; comparing timeliness within a life band reads φ.
5. **A pre-registered severe test and its failure** (§5) — registered before the data were touched, replicated in a second sector declared in advance, controlled against a label-permutation null, powered at 0.95–1.00, and lost, with the stopping rule honoured.
6. **A bridge discipline** (§6.2) arising from that loss, and now with an argument behind it: any identification of a model parameter with a measurable must be written as a proposition a competent critic could deny.
7. **A survivals ledger** (§7), including the row that overturned this paper's own preferred reading of its null.

The failed prediction is in the body and in the abstract, not in *Abandoned Approaches*: a pre-registered prediction that was tested and lost is a **result**, and §5 reports it as one.

The framework the filter was built inside — three propositions about the composition of wealth (**P1** composition, **P2** decay, **P3** atomism) and the coupling they oblige — is set out in **Appendix A**. No result in §§2–5 depends on it. §6 accounts for the appendix alongside the body when it states what the demotion leaves standing, §7's survival ledger records the appendix's own checks, and §9 names P1's domain where it bounds what may be claimed.

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2 · The filter

Two layers and one parameter that matters.

The physical layer decays at an entropy rate net of maintenance:

$$\mathbf{E(t+1) = E(t) \cdot (1 - d{\cdot}(1 - m))}$$

Of each true change, a share **φ** is *observable* — announced capital expenditure, a disclosed impairment, a write-down someone had to sign — and reaches the claim layer immediately. The remaining $(1 - \varphi)$ is deferred maintenance and technical debt: real, accruing, and absent from the statements. It accumulates in the gap and is recognised only at rate α per period:

$$\mathbf{C(t+1) = C(t) + \varphi{\cdot}\Delta E + \alpha{\cdot}gap(t),\ gap(t) = E(t) - C(t)}$$

When the unrecognised gap exceeds a threshold share θ of physical wealth, the deferral becomes unsustainable and the claim layer snaps to the physical one. **That discontinuity is the crisis, and its magnitude is exactly the information that had been withheld.**

Terminology. The discrete event is called a **recognition event** throughout, and where the referent is literally ASC 350 it is called an **impairment loss**. It is deliberately **not** called a *correction*: in finance a correction is a price decline of a specified magnitude from a peak, and in accounting ASC 250 — *Accounting Changes and Error Corrections* — reserves the word for the repair of an **error**, which a change in estimate driven by later information is not. Using *correction* for the event would assert, in the technical register of the standard §5 is built on, that the prior statements required retrospective restatement. The word **crisis** is used for the phenomenon §2 models and for nothing wider. It is **not** in the title, which names the identification result; §8.2 gives the framing it belongs to and the paper that will carry it. The systemic, country-level sense used in the banking-crisis literature is not intended anywhere here.

Throughout: $E_0 = 100$, $d = 0.05$, $m = 0.6$ (effective decay 0.02 per period), $\alpha = 0.05$, $\theta = 0.25$, 400 periods. Where the filter is examined in isolation the recognition mechanism is disabled ($\theta = \infty$), because otherwise the snap timing truncates the measurement window and the lag statistic reports the recognition schedule rather than the filter.

Two conditions bound what this filter models. The first makes the wedge one-signed: reported value may fall and may not rise. Under US GAAP — the regime this paper's sample files in —

there is no upward revaluation of property, plant and equipment and no impairment reversal for goodwill or indefinite-lived intangibles. IAS 36 requires reversal for non-goodwill assets and IAS 16 permits revaluation, so the condition holds for these filers and is not general, and a physical layer that only degrades is not by itself enough to produce it. §10 gives the reason. The second restricts the domain: the claim above is restricted to degradation on which conservatism has nothing further to bite — carrying no impairment trigger, no estimable expected loss and no observable event to key recognition to. **Where a loss is estimable, recognition is faster than the market and this model predicts nothing.** §10 gives the accounting literature these two conditions belong to.

φ is not a fudge factor, and the distinction is load-bearing. φ is the *observability of the degradation*, and it is what makes this model survive the objection that would otherwise kill it outright — see §8.1.

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3 · What the filter does

3.1 · Lag and deferred information scale with unobservability

Filter isolated, θ = ∞. **Recognition lag** is the offset, in periods, that maximises the cross-correlation between ΔE and ΔC — how long the claim layer takes to track a change in physical value. **Smoothing** is the ratio of the standard deviation of ΔC to that of ΔE, taken over periods carrying no recognition event: 1.000 where the claim layer moves exactly as the asset does, falling as more of the movement is withheld. Smoothing is measured over different periods in the two regimes: with the mechanism disabled there are no events and it runs over every period, the **inter-period** column below; §3.2 turns the mechanism on, leaving the periods between events, and heads the same quantity **inter-event**.

φ	recognition lag (periods)	inter-period smoothing	deferred information
1.0	0	1.000	0.0
0.8	3	0.928	399.8
0.5	14	0.845	999.5
0.2	22	0.798	1599.2
0.0	26	0.791	1999.0

Deferred information is exactly proportional to (1 − φ), and this is a closed form rather than a simulation regularity.

With the recognition mechanism disabled, substituting $E(t+1) - C(t) = \mathrm{gap}(t) + \Delta E$ into the two recursions gives

$$\mathbf{gap(t+1) = (1 - \alpha)\cdot gap(t) + (1 - \varphi)\cdot \Delta E(t)}$$

so with $\mathrm{gap}(0) = 0$ the gap at every t is $(1 - \varphi)$ times its value on the $\varphi = 0$ path. Since $\Delta E < 0$ throughout, every term shares a sign and the absolute integral inherits the factor exactly:

> $\mathbf{D(\varphi) = (1 - \varphi) \cdot D(0)}$, $D(0) = 1998.99$ for the parameters above.

The simulation reproduces this to a relative error of 10^{-15} across φ, and the test suite asserts it. **A doubling of unobservability doubles the integral of what the statements owe, exactly.**

The lag does not share that simplicity: it is sigmoidal in $(1 - \varphi)$, rising slowly at first ($\varphi = 0.9 \longrightarrow$ lag 1), then steeply through the middle of the range ($\varphi = 0.8 \longrightarrow 3$, $\varphi = 0.5 \longrightarrow 14$), then saturating ($\varphi = 0.1 \longrightarrow 24$, $\varphi = 0.0 \longrightarrow 26$). So the *quantity* of undelivered information is linear in unobservability while the *delay* in delivering it is not, and the two should not be expected to move together.

With the recognition mechanism live, recognition-event frequency at fixed observability ($\varphi = 0.3$) sorts by entropy rate:

entropy rate d	sketch	recognition events in 400 periods
0.01	warehouse retail: re-provisions on a decade	**0**
0.05	industrial	**16**
0.20	software: faces zero-days continuously	**100**

The contrast between a warehouse retailer and a firm exposed to continuous obsolescence is therefore a **position in a parameter space**, not a rhetorical flourish. Two firms with identical observability and different entropy rates live in different worlds, and the model says by how much.

3.2 · Volatility is not suppressed, it is relocated

This programme previously predicted that the claim layer would be *smoother* than the physical one. Measured across a whole path, **that prediction is false wherever the mechanism is actually active**: at $\varphi = 0.5$, 0.2 and 0.0 the full-path ratio of reported to physical volatility is 1.56, 2.71 and 3.27 — the claim layer is *more* volatile, because the recognition events dominate the variance. The prediction survives only at $\varphi \geq 0.8$, where the gap never reaches the recognition threshold and there are no recognition events to dominate anything; there the ratio is 1.00 and 0.93. **So the original claim was not merely wrong, it was right only in the regime where the model has nothing to say.**

The true behaviour is more interesting than the prediction it replaced. Measured with the recognition mechanism live, the claim layer is far smoother than the underlying **between** recognition events, while the share of all reported movement occurring **inside** recognition events rises toward one:

φ	inter-event smoothing	share of reported movement inside recognition events	events
1.0	1.00	0.00	0
0.8	0.93	0.00	0
0.5	0.80	0.69	8
0.2	0.52	0.96	19
0.0	0.35	**0.99**	25

At zero observability, **essentially every reported movement is a recognition event.** The signature is therefore not a quiet system but one that is quiet for long intervals and then abruptly is not — which is a more recognisable description of the historical record than uniform smoothing.

The qualitative result is not new and is not claimed here. Bleck and Liu stated it in 2007: historic cost accounting "stabilizes asset prices in the short term. Under the veil of this apparent stability, volatility actually accumulates only to hit the market at a later date," transferring volatility across time and raising it overall. **What the table above adds is the parameterisation** — the same claim indexed by a continuous observability parameter, with the smoothing and concentration measured separately rather than argued. The provenance and the difference in mechanism are set out in §10; the table's headline result is nineteen years old.

It is not, however, a usable empirical target. Against filed accounting data the concentration statistic is unfalsifiable by construction: the asset class with no amortisation schedule has essentially all of its recognised change arrive discretely *as a matter of accounting definition*, not of firm behaviour. A test of that is a test that cannot fail, and a test that cannot fail is not a test. Neither pre-registration included it, on exactly those grounds, before either was run. The relocation result is therefore a property of the model and a candidate description of the world, and it is

not offered as the replacement severe test. The framework does not currently have one.

Two statistics, not one. Smoothing is measured on the periods between recognition events and concentration on the recognition events, because a statistic asked to carry two claims answers neither.

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4 · Timeliness and durability are not separately identified

§3 established what the filter does to one asset. This section establishes what a reader of the filter's output can learn from it. The answer is a constraint on the instrument rather than a caution about estimation: it holds for every estimator, it has a four-line proof, and it applies to measures built without reference to this model or any like it.

The weak form is a conditioning statement — φ is recoverable, but the estimator's variance grows like $1/\delta^2$. That form is true and too kind. In the filter regime the degeneracy is exact.

4.1 · The class index, and why the product is elementwise

A firm holds several classes of asset, and the accounting standards treat them differently on purpose. Index the classes *i*. Each carries its own physical decay rate δ_i, its own observable share φ_i, its own recognition rate α_i and its own threshold θ_i, and §2's two recursions become one line in vectors:

$$\mathbf{C}(t+1) = \mathbf{C}(t) + \boldsymbol{\varphi} \odot \Delta\mathbf{E} + \boldsymbol{\alpha} \odot \mathbf{gap}(t), \quad \mathbf{gap}(t) = \mathbf{E}(t) - \mathbf{C}(t)$$

with $\odot$ the Hadamard product and, below, $\oslash$ elementwise division. The elementwise form is a substantive claim rather than a notational economy: **the reporting layer is diagonal in class space.** A dollar of unrecognised deterioration in a distribution centre does not force recognition against a trademark. Each class's filter reads its own gap and nothing else.

That claim is an assumption and it is false in detail; §9 says so and proposes the test. Without the class index the identification result below is a remark about a single parameter. With it, the remark acquires a corollary about *rankings*, and rankings are what the empirical literature on this subject estimates.

4.2 · The theorem

Take one class and disable the recognition mechanism, so the filter is examined in isolation. Write δ for the effective decay rate d(1 − m) of §2. Substituting ΔE = −δE(t) collapses the pair of recursions to a single line:

$$C(t+1) = C(t)(1 - \alpha) + E(t)(\alpha - \varphi\delta),\ E(t) = E_0(1 - \delta)^t$$

φ appears once, in the product φδ, and nowhere else. With the books opening square — $C(0) = E(0) = E_0$, an asset carried at cost on the day it is acquired — this solves in closed form. Writing A = 1 − α and D = 1 − δ for the two roots,

$$\mathbf{C(t) = E_0 \cdot [\ \delta(1 - \varphi)\ A^t - (\alpha - \varphi\delta)\ D^t\ ] / (\delta - \alpha)}$$

and the reported series is a linear combination of two geometrics whose exponents are the reporting rate and the physical decay rate.

Now exchange them. Send (α, δ, φ) ⟶ (δ, α, φ′), which swaps A and D, and ask what φ′ must be for the series to come back unchanged. The A^t coefficient requires δ(1 − φ) = δ − φ′α. The D^t coefficient requires α − φδ = α(1 − φ′). Both reduce to

$$\boldsymbol{\varphi'\alpha = \varphi\delta}$$

Two equations, one unknown, and they agree. That coincidence is the theorem: the system is overdetermined and consistent, so the exchange is not approximate, not local, and not a matter of conditioning.

> **Observational equivalence.** The parameter triples **(α, δ, φ)** and **(δ, α, φδ/α)** generate the *identical* reported series. The filter's two roots — the reporting rate and the physical decay rate — are exchangeable, and the quantity preserved by the exchange is exactly **φδ**.

And this is not a property of the particular filter. Any model in which reporting lag attenuates a physical signal will multiply a timeliness parameter by an asset-life parameter somewhere, because the observable is a rate times a duration. The model's contribution is to make the product explicit enough to be checked.

The numerical confirmation runs alongside the proof rather than in place of it: the largest deviation between a series and its mirror is 8×10^{-16}, the arithmetic and not the model, against 3

$\times 10^{-1}$ when the mirror's φ' is replaced by the value that would preserve the *unrecognised gap* instead of the reported series — a discrepancy of fourteen orders of magnitude between the right conserved quantity and a plausible wrong one. Admissibility requires $\varphi\delta \le \alpha$, which holds at every parameter setting used anywhere in this paper.

So a reported series does not merely make φ hard to recover. **It does not contain φ.** It contains φδ, and it cannot distinguish a timely reporter of a durable asset from a laggard reporter of a perishable one.

What the two worlds disagree about is the firm, not the filing. The mirror is not a mathematical curiosity with no economic content. It is a slow reporter of a fast-decaying asset: at $t = 400$ its physical stock stands at 4×10^{-6} of the original world's, a book value sitting above an asset that has all but evaporated. That is a recognisable kind of company. It files the same statements, to fourteen decimal places, as the prompt reporter of the durable asset.

The mathematics is old. Subtract E(t) from the closed form and the unrecognised gap is

$$G(t) = E_0 \cdot (1 - \varphi)\,\delta \cdot S(t),\ S(t) = (A^t - D^t)/(\delta - \alpha)$$

a scalar amplitude carrying every trace of φ, multiplied by a shape function that is invariant under exchanging the roots. S is the Bateman function, written down in 1910 for the daughter activity of a radioactive decay chain (Bateman, 1910), and its exchange symmetry is the reason pharmacokinetics has a name for this: **flip-flop**, the interchange of an absorption rate constant with an elimination rate constant in a one-compartment model, which leaves the concentration–time profile unmoved (Garrett, 1994). Kuan, Wright and Duffull (2023) place it as an issue of *local* identifiability, "in that there exists a finite set of parameter values (rather than a single set) that solves the problem" — which is precisely the two-point structure above. Their own caution should be carried across with the result: they hold that the competing solutions are "not simply a function of swapping the rate constants" but a partial permutation of the parameter set, with $n + 1$ of them for an n-compartment model. The accounting case here, where the exchange **is** a clean swap of two roots, is therefore the simplest member of that family rather than the general one. The general framework is Bellman and Åström's (1970), which defines structural identifiability by what the input–output map determines and tests it through the transfer function — and a transfer function fixes its poles as a set, which is the exchange above stated once and for all.

Economics has its own instance, and it sits closer to this paper than either. Nerlove's supply-response model (1958) stacks adaptive expectations at rate β on partial adjustment at rate γ; eliminating the two unobservables leaves a reduced form in which **every systematic coefficient is a symmetric function of β and γ** — $[(1-\beta)+(1-\gamma)]$ on the lagged dependent variable, $-(1-\beta)(1-\gamma)$ on its second lag, $\beta\gamma a_1$ on the price. The conditional mean is exactly invariant under exchanging the two behavioural rates. What distinguishes them is the disturbance, $\gamma[u(t)-(1-\beta)u(t-1)]$, which is not symmetric. That the tie is broken by the error process rather than by the systematic part is a property of the Nerlovian model which the filter here does not inherit, and §4.7 says what breaks it instead.

The ambiguity bites here for the same reason it bites in pharmacokinetics and not in radiochemistry: it needs a **free scale parameter** to hide in. A decay chain's parent activity is measured independently, so the exchange changes the observed curve by a detectable factor and nothing is lost. A plasma profile has an unknown volume of distribution, and a balance sheet has an unknown $(1-\varphi)$. Accounting is on the pharmacokinetic side of that line, and the next result is what it costs.

The result is stronger than a two-point ambiguity. The closed form has two roots and two amplitudes: four numbers, and that is everything a reported series contains. The model has five parameters — α, δ, φ, the physical scale E_0, and any gap g_0 already open when the observation starts. Five into four does not go, and the shortfall lands on φ.

Two consequences, both exact. First, opening the books with a gap already in place does not rescue identification: the mirror survives with the same g_0 under the shifted map $\varphi' = [\varphi\delta + g_0(\alpha-\delta)]/\alpha$, and the conserved quantity generalises to $(\varphi - g_0)\delta = (\varphi' - g_0)\alpha$. The obvious escape — *real firms are not observed from acquisition* — makes matters worse rather than better.

Second, and this is the sharp form: **when the physical scale is not observed, φ is not two-valued. It is free.** Fix a reported series generated at $\varphi = 0.60$ in a world whose books already open 15% above the physical asset — the shifted map above, not §4.2's square opening — and ask what other parameter vectors reproduce it exactly. Assuming a physical scale of 0.76 implies $\varphi = 0$; assuming 1.27 implies $\varphi = 1$; every assumption in between implies an intermediate φ, and every one of them regenerates the observed series to 2×10^{-16}. **A factor of 1.67 in the unobserved physical scale spans the entire unit interval of timeliness.** Each member of that family also carries its own opening gap, and the $\varphi = 0$ end requires books opening **51% above** the physical asset

— the state impairment accounting exists to prevent. Bounding that gap at ten per cent still leaves an identified set covering **31.7%** of the unit interval, which is fatal to a cross-sectional ranking and not an artefact of an unbounded freedom. The reported series is consistent with a firm that recognises everything at once and with a firm that recognises nothing.

The condition under which this bites is worth naming precisely, because it is the empirical norm rather than an edge case. E_0 is observed when an asset is followed from acquisition, where cost fixes the physical scale and the gap opens at zero. It is not observed for a firm-level series, which aggregates assets of many vintages, so the scale that would pin φ is exactly what a cross-sectional study does not have.

The earlier conditioning result stands underneath all of this and is worth keeping for its size: fitting the model to a synthetic series recovers φ with a median absolute error of 0.211 when δ is estimated jointly and 0.00073 when δ is pinned at its true value, a **291-fold** difference, with a noise-free series giving 0.211 as well. That is what an exact degeneracy looks like from inside a numerical optimiser: not a cliff, a canyon.

One channel does break the equivalence, and it is closed. The recognition events' trigger reads gap/E, and across five mirror pairs the event counts differ sharply enough to separate them (16 against 66, 25 against 80, 36 against 133, and two pairs where one side is silent entirely). **The trigger reads E, which no filing reports.** The information that breaks the degeneracy arrives through a channel the reported series does not have.

4.3 · What a cross-class ranking reads

The class index now earns its keep. What a series identifies is $\varphi_i\delta_i$, so a cross-class ranking computed from reported numbers is a ranking of the product — and a ranking of $\varphi \odot \delta$ is not a ranking of φ unless δ is constant across the classes being ranked.

The model's own measure of how much a class defers sharpens this. The steady-state ratio of unrecognised gap to physical value has a closed form:

$$\mathbf{R}_i = (1 - \varphi_i)\,\delta_i \,/\, (\alpha_i - \delta_i)$$

which simulation reproduces to the transient bound, 2×10^{-4} after 400 periods, against a witness of 1.0 when φ is misstated by 0.1. Across classes with a common α this is a Hadamard product again, $(\mathbf{1} - \boldsymbol{\varphi}) \odot \boldsymbol{\delta}$, divided elementwise by $(\alpha - \delta)$. Decay reaches the ranking through two channels, neither of them the parameter of interest.

R is the model's deferral measure and not an observable — it is a ratio to E, and E is the series nobody reports. Under the mirror it does not shrink or stretch: it **changes sign**, from +0.267 to −1.267, because the mirror world is one in which the books outrun the asset. A quantity that reverses sign under an exchange the data cannot detect is not a quantity a reader recovers. What §4.4 uses R for is what it is good for: computing, inside the model, which way a ranking runs.

Timeliness sets the ranking only when durability is constant across the classes being ranked — and the classes accounting standards distinguish are, almost by construction, the classes whose durability differs. That is why the standards distinguish them.

4.4 · The design has a validity region, and the disclosed numbers fall outside it

The natural expectation is that a confound of this kind adds noise to a ranking. It does something more specific. There is an exact region of δ-space inside which a φ-ordered cross-section recovers the ordering it imposed; the region has a closed-form boundary in quantities the design already declares; and it is small. The case this paper's own registration used sits outside it, and so does every firm whose asset lives are read off its filings.

Order the four GAAP classes as the registration did — property, plant and equipment; finite-lived intangibles; indefinite-lived intangibles; goodwill — and assign the observability shares the registration assumed, falling up the ladder as the standards' willingness to put a class on an amortisation schedule falls. Then assign the decay rates the same standards imply, which fall up the same ladder, because a class is placed on a schedule precisely when its decline is predictable enough to schedule. Goodwill sits at the end of both: least observable, and with no degradation schedule at all.

Two of the table's columns are evaluated at rates §5.4 measures rather than at the calibration. The **measured** rate is the censored geometric maximum-likelihood estimate of the recognition rate PRE-002's instrument identifies, on the registered sample, each event carrying the interval from the onset of deterioration to the charge and right-censored at twenty quarters. The **unregistered adverse cut** is that same estimate refit with the 175 events charged one quarter after the peak dropped — aimed at the doubt REG-003 §3.3 registered in advance, which is what makes it the adverse one.

tier	φ	$1-\varphi$	δ	$(1-\varphi)\delta$	**R** at α = 0.05, the calibration	**R** at $\hat{\alpha}$ = 0.408, measured (§5.4)	**R** at $\hat{\alpha}$ = 0.327, the unregistered adverse cut	R at a common δ
0 · property, plant and equipment	0.80	0.20	0.030	0.00600	**0.3000**	**0.0159**	**0.0202**	0.1333
1 · finite-lived intangibles	0.60	0.40	0.020	0.00800	**0.2667**	**0.0206**	**0.0261**	0.2667
2 · indefinite-lived intangibles	0.40	0.60	0.010	0.00600	**0.1500**	**0.0151**	**0.0189**	0.4000
3 · goodwill	0.20	0.80	0.002	0.00160	**0.0333**	**0.0039**	**0.0049**	0.5333

The right-hand column is the world the design assumed: classes differing in observability and in nothing else. There the deferral measure rises monotonically up the ladder exactly as predicted, Kendall $\tau = +1$. The two **R** columns are the world the standards describe, at the recognition rate calibrated here and at the one §5.4 goes on to measure. There the deferral measure is monotone too — **running the other way.** Kendall τ = **−1** at the calibrated rate, and **−0.67** at both the measured rate and the unregistered adverse cut, where the first rung alone turns over; the rung that separates them is identified below.

The design and its own observable are anti-aligned across the ladder. A confounded design returns noise; this one returns the reverse of what it ordered. On the model's own arithmetic, the class predicted to defer most defers least, because it barely deteriorates, and there is little to defer.

The condition deciding the direction is worth writing out, because of what it contains. Taking logs of R,

$$\log R = \log(1-\varphi) + \log\delta - \log(\alpha-\delta)$$

so the deferral measure rises from one tier to the next exactly when

$$\Delta\log(1-\varphi) + \Delta\log\delta - \Delta\log(\alpha-\delta) > 0$$

The first term is the design. **The other two are facts about δ**, and on a falling ladder they carry the same sign, so they add. At every rung of the ladder above, the combined δ contribution (−0.81, −0.98, −1.79) outweighs the design term (+0.69, +0.41, +0.29), and the decomposition predicts the direction of every step. Checking only the log δ term would have got the first rung right for the wrong reason: there the design term is the larger of the two, and the step still falls, because δ enters twice.

So the validity condition for a φ-ordered cross-sectional design is a statement about δ — the quantity §4.2 says the reported series does not contain. A researcher cannot establish that the design is sound without already possessing what the design was built to avoid needing.

Dispersion and ordering do different damage, and the two are worth separating. Draw 4,000 four-class ladders under the design's own constraint alone — observability falls up the ladder — with δ drawn independently across classes. The deferral measure recovers the registered ordering in **11.5%** of them and exactly reverses it in **1.1%**, mean Kendall τ **+0.32**. Hold δ common across the four classes instead and redraw: recovery is **100.0%**. Now impose the standards' falling ladder on the same draw and recovery falls to **1.9%** while exact reversal rises to **23.8%**, mean τ **−0.41**. So δ *dispersion* is what destroys the ranking, and the *ordering* is what turns the wreckage into a reversal. The table above is what the confound does at one corner of the region; losing the ranking is what it does across the region.

The boundary is exact, and it is drawn in quantities the design already declares. Write the design's *budget* as the mean per-rung Δlog(1 − φ) and the ladder's *δ leverage* as the mean per-rung |Δlog δ − Δlog(α − δ)|. Over the same 4,000 draws the probability that the design fails to recover its ordering, fitted as a logistic in log(leverage / budget), has slope **+1.58** (se 0.081, z = +19.5; the same fit on a permuted outcome returns z = 0.23) and crosses one half at a leverage-to-budget ratio of **0.61**. A φ-ordered cross-section is more likely than not to read what it ordered only while per-rung δ leverage stays under about three fifths of the design budget. The ladder tabulated above sits at **2.58**.

The tabulated ladder is a knife edge in its own top rung. Holding the first three tiers fixed, the deferral measures of goodwill and indefinite-lived intangibles cross at

$$\delta_3^* = K\alpha/(1 + K),\ K = R_2/(1 - \varphi_3)$$

which is **0.0079**, a half-life of eighty-seven periods. Five per cent above it the ladder is no longer monotone and Kendall τ moves from −1 to −0.67. The table assigns goodwill 0.002. The exact reversal therefore needs goodwill to lose half its value no faster than in eighty-seven years, and the standards do not say that: a class is left off an amortisation schedule when its decline cannot be *scheduled*, which is a statement about predictability and not about speed.

Unpredictable is not slow, and the difference is measurable in the same direction. Drive a class at $\delta = 0.20$ with probability 0.05 and at zero otherwise — an identical mean decay rate of 0.010, delivered in rare jumps — and its realised deferral is **1.30 times** the closed form evaluated at that mean rate (se 0.002 over 2,000 paths). An unscheduled class defers *more* than a scheduled one of the same average durability, and the lumpy path defers as a smooth class at $\delta =$ **0.0123**, above the crossing rate. Both halves of the inference from an absent schedule therefore push the same way.

Two rungs need no inference at all, and they are the two that break the table. ASC 360 and ASC 350-30-50 require disclosure of useful lives for property and for finite-lived intangibles, and a disclosed life L fixes a write-down rate $1/L$. The first rung falls only when

$$\delta_1 < \alpha\delta_0/(2\alpha - \delta_0)$$

which tends to $\delta_0/2$ as α grows. At the tabulated $\delta_0 = 0.030$ that boundary sits at $\delta_1 =$ **0.0214**, a life of 46.7 years, and the table assigns 0.020 — inside by a fourteenth **at $\alpha = 0.05$, and outside it at the measured rate**, where the same boundary is 0.0156. That is the rung the table's τ turns on, and it turns on the recognition rate's *level* rather than its shape: the top three rungs are unchanged at either rate, and §4.9 puts the shape's contribution to the crossing below it at 0.13%. Disclosure, however, amortises finite-lived intangibles over materially *shorter* lives than property, so $\delta_1 > \delta_0$ is what a filing presents. Measured on the filings themselves — 665 admissible firm-year pairs across 577 firms, both lives read off one page — **the first rung rises in 65.9% of them**, in both cycles separately (63.3% and 68.2%). Swept uniformly and independently over the rectangle those lives were *assumed* to span — ten to forty years for property, three to twenty for finite-lived intangibles — the same test returns 99.8% at the same rate, and 86.1% of the disclosed pairs fall outside that rectangle.

And the binding constraint is the model's domain, not the ordering. R is defined only for $\delta < \alpha$. Past that the deferred gap grows without bound relative to the asset — the ratio reaches 10^{69} by period 400 at $\delta = 0.20$ — and there is no steady-state deferral measure to rank. At the $\alpha = 0.05$ calibrated here **the entire asserted rectangle lies outside the domain**: every useful life short enough to appear in a filing implies a decay rate at or above the recognition rate. Half of the rectangle is admissible only at $\alpha \approx 0.19$, and all of it above $\alpha = 0.33$. **That made the recognition rate, not the ordering, the quantity to establish first — and §5.4 establishes it.** On the registered sample the recognition rate PRE-002's instrument identifies is **$\hat{\alpha}$ = 0.408 per year**, 95% interval [0.383, 0.432], on both known biases' inflating side: the calibration used here is low by an order of magnitude, and the asserted rectangle lies inside the domain at the measured rate and across its 95% interval, where **0.974** of the 683 disclosed pairs are admissible and **0.959** at the interval's lower bound. At the unregistered adverse cut above, **0.327**, the rectangle's own fastest disclosed rate of 0.3333 is no longer cleared, and **0.814** of the pairs remain admissible. The domain restriction is a property of the calibration and not of the disclosure, and what its remaining margin turns on is the onset bridge rather than the sample.

The shape of this argument is not new, and its best-known instance is one field over. Fisher and McGowan (1983) argued that an accounting rate of return cannot be used to infer economic profitability, because the reported ratio depends on the depreciation schedule and the firm's growth rate as well as on the economic return it is supposed to measure — a reporting-rule parameter and an asset-life parameter confounded inside a published number, forty years before this paper, and it detonated an industrial-organisation literature. Two things about its reception instruct the present one. Their demonstration was numerical rather than a theorem, and the analytical core of it belongs to earlier work (Kay, 1976). And the sweeping inference they drew — that accounting returns carry almost no information about economic ones — did not survive: it was rebutted on the arithmetic and on the representativeness of the chosen examples (Long and Ravenscraft, 1984), and superseded by a literature that recovers conditional usefulness once growth and capitalisation policy are corrected for. **The claim here is deliberately the narrower kind** — an exact equivalence with a stated domain and a repair in §4.7, rather than a verdict of futility. The ancestor is cited for the shape of its confound, and its fate is cited as the reason not to overreach with one.

4.5 · One statistic survives, and it is the one nobody can compute

Everything above concerns a *magnitude* — how much a class defers. The registration did not order magnitudes. It ordered **lag**:

how long a class takes. And the lag statistic does not invert.

tier	lag, standards' ladder	lag, common δ
0 · property, plant and equipment	2	3
1 · finite-lived intangibles	10	10
2 · indefinite-lived intangibles	18	17
3 · goodwill	**36**	22

Monotone under both, and the falling-δ ladder makes the ordering *steeper* rather than flattening it. The reason is that the two monotonicities compound: lag falls in φ at every δ, and rises as δ falls at every φ, so on a ladder where both move the way the standards say they do, **the two effects add.** Across 400 randomly drawn admissible ladders the lag ordering holds in **100%** of them, against 1.9% for the magnitude measure. **Part of that margin belongs to the ladder rather than to the statistic.** Drop the durability ordering and draw δ independently across classes, as §4.4 does, and the lag ordering holds in **66.2%** (2,000 ladders, se 0.011) against **11.5%** for the magnitude measure. Lag is the more robust of the two by a factor of six, and it is robust in that ratio rather than in the way a figure of 100% suggests.

The identification result does not, by itself, wreck a design ordered on lag. Any claim that the registered prediction was doomed by the φδ confound is claiming more than the arithmetic gives.

What wrecks it is the next line. **§3.1's lag statistic is a cross-correlation between ΔE and ΔC, and ΔE is the change in physical value, which no filing reports.** The one statistic the confound spares is the one that cannot be computed from public data. Any empirical instrument must substitute something else — §5's substituted the interval from the onset of a decline in a firm-level signal to the recognition of a charge — and the relation between that substitute and the model's lag has never been written down.

That is a second identification gap, upstream of the first, and it is the one that bit. The framework's response to it is §6.2's bridge discipline, which the registered null motivated and this section supports.

4.6 · The field's instruments read the same product

The constraint is not local to this model.

Conditional-conservatism measurement estimates how promptly accounting recognises economic losses. The standard instruments — Basu's asymmetric-timeliness coefficient (1997), Khan and Watts's C_Score (2009), Ball and Shivakumar's accrual–cash-flow piecewise measure (2006), Givoly and Hayn's accumulated negative accruals (2000), Bushman and Williams's DELR (2015) — differ in construction and share an input: a reported series, and the requirement to infer a recognition property from it.

If a timeliness parameter reaches a reported series only in product with an asset-life parameter, then a cross-sectional comparison of any of these measures is a comparison of the product. **Two industries with identical recognition practice and different asset lives will score differently, and two industries with identical scores may differ arbitrarily in practice.** The sign of the induced difference is not fixed by the measure; §4.4 shows it can invert a ranking rather than attenuate it.

The literature has already met the confound and read it the other way up. Khan and Watts (2009) report that firms with longer investment cycles score as more conservative, and treat the association as an economic determinant — a demand for verification that rises with the horizon. Ball, Kothari and Nikolaev (2013) state plainly that firms with shorter asset maturity are expected to exhibit lower timely loss recognition, and read that dependence as the measure behaving correctly. Under §4.4 those are the readings a δ channel would produce whether or not any recognition practice differed at all. The data cannot separate the two accounts, which is the whole of the present claim; it may well be both.

One paper has the mechanism itself, in signed form, and it is the nearest accounting-native ancestor this result has. Beaver and Ryan (2005) model unconditional conservatism *preempting* conditional conservatism, and name the channel exactly: "the unconditionally conservative nature of accelerated depreciation creates unrecorded goodwill for tangible assets that preempts conditional conservatism as long as shocks to the market value of those assets are not negative enough to use up that goodwill." A depreciation schedule suppressing measured timeliness — in accounting, in print, in 2005. What it is not is an identification claim. Preemption is a signed comparative static with a stated mechanism, and it leaves the two parameters separately meaningful; the claim here is that the reported series does not contain the difference between them. The lineage is closer than the pharmacokinetics of §4.2 and is owed the same acknowledgement.

The distinction from the standing critiques is what makes this worth stating. The econometric objections to the asymmetric-timeliness coefficient — truncation from conditioning on the sign of returns, scale effects, return-variance dependence — are **esti-**

mator problems, and their remedies are the estimator's: controls, fixed effects, interactive corrections, a debiased functional form. A degeneracy is not a bias. When two parameter vectors generate the identical series, no control recovers the difference between them, because there is nothing in the series to recover it from. The existing repairs are aimed one level up from where the problem is.

The sharpest of those critiques deserves separating from this one precisely, because it shares a title-word with it and almost nothing else. Dutta and Patatoukas (2017) decompose the asymmetric-timeliness coefficient into a component that survives when recognition is symmetric and a component attributable to conservatism, and show that the first is positive whenever the return distribution is skewed, while the second moves with three properties of the news process — expected returns, cash-flow persistence, and the skewness itself — at a fixed degree of conservatism. Two things make that a different claim from this one. Their confounders are properties of the **news process**; the confounder here is a property of the **asset**, its decay rate, set against a reporting rule — and their firm is a cash-flow stream with no capitalised asset in it to carry one. And their recognition parameter stays recoverable in their own setting, from the spread between bad-news and good-news accrual variances, which is the repair they propose. **A claim that a better statistic can repair is a claim about a statistic.** The claim here is that the reported series is itself invariant, so no statistic computed from it separates the two worlds. Ryan (2006), whose title is nearly the same as theirs, uses *identifying* in the empirical sense of detecting conservatism in practice and makes no claim of the econometric kind.

The notation overlaps: in Dutta and Patatoukas, δ is the fraction of bad news recognised — this paper's φ. Here δ is the physical decay rate, and has no counterpart in their model.

Three qualifications. First, the mapping from this filter to each of those estimators is not established here: they are not fitting this model, and the composite they read need not be $\varphi\delta$ exactly. Second, the magnitude-versus-timing distinction of §4.5 matters, and these measures sit on both sides of it — Basu's coefficient is a slope on returns rather than a delay, and is closer to this paper's magnitude case than its timing one. Third, **the theorem is proved for the reported series alone**, and the returns-based measures condition on a second series. That second series does break the equivalence. §4.7 gives the result and its price, and the claim of this section is correspondingly narrow: what is said here is said about a comparison of reported series, and a design holding returns is repairing the problem rather than inheriting it.

What the paper claims is that the burden has moved. **A cross-sectional conservatism ranking now requires an argument that asset life is constant across the compared groups, a correction for it, or an auxiliary series that identifies it** — and the ranking most often compared, across GAAP asset classes, is the one where the first of those is least defensible.

4.7 · The repair

The way out is visible in the theorem's own statement: the series determines $\varphi\delta$, so **anything that supplies δ from outside the series restores φ.** Two things do, and they are priced very differently.

The first is the one the field already holds: returns. A market series drawn from the same asset breaks the two-point equivalence immediately and by a wide margin. Under the exchange the mirror firm's asset decays at α rather than δ, so two worlds whose books agree to fourteen decimal places differ in return by $\alpha - \delta$ in every period — three percentage points a year, indefinitely. The ambiguity of §4.2's first half does not survive contact with a second series drawn from the asset.

It survives the continuum, and the reason is one line. **A return is a ratio, and the residual degeneracy is a degeneracy in the unobserved physical scale.** Grant an analyst both roots exactly — strictly more than returns supply — and the one-parameter family is untouched: φ still sweeps $[0, 1]$ with the reported series reproduced to 2×10^{-16}, and every member of that family emits the *identical* return series, bit for bit. A scale divides out of a ratio, so no quantity of returns data bears on the parameter the scale is concealing.

What breaks the continuum is not the returns but the **news** they carry. The degeneracy is a property of a noiseless economic path: when the asset's value decays geometrically and does nothing else, the reported series has a single geometric driving term and a scale factor absorbs any rescaling of φ. Let the value receive innovations and the realised rate of decline varies from period to period; matching the driving term then requires $c\alpha = \alpha$ and $c\varphi' = \varphi$ *simultaneously*, which forces $c = 1$. Regressing the reported series on its own lag, the return-implied path and that path's first difference recovers α, E_0 and φ to 10^{-16} at a return volatility of 0.15, against a design matrix that is exactly singular at zero volatility.

So the answer is yes — something outside the reported series does restore φ — and the price is a rate rather than a proof. Identification does not switch on at the first innovation; it fades in. Over a twelvefold range of return volatility both the design's collinearity and the standard error on $\hat{\varphi}$ degrade as power laws in σ, with weak-identification bias visible in the mean by $\sigma = 0.025$. **Neither exponent is a constant of the model, and neither should be read as one.** Across nine (α, δ) settings spanning the four decay rates §4.4 attributes to the standards, the collinearity exponent runs from −1.07 to −0.38 and the standard error's from −0.78 to −0.09. What holds in all nine is the sign: identification always degrades as the asset quietens.

What the exponents track is more useful than their values, and the two rates do different jobs. The decay rate governs how strongly identification *responds* to volatility; the gap between the two rates governs its *level*. Holding the gap fixed and sweeping δ, the volatility exponent runs from −0.39 at a property-like δ of 0.030 to **+0.16** at a goodwill-like δ of 0.002 — a change of sign rather than a flattening, so below roughly $\delta = 0.01$ further news stops helping and begins to hurt. Holding δ fixed and sweeping the gap moves the *level* instead, by a factor of 6.8, as $(\alpha - \delta)$^−0.70. The two must be kept apart or they read as a contradiction: a sweep at one volatility says the decay rate hardly matters, a sweep across volatilities says it decides everything, and they are statements about different quantities. §4.8 gives the arithmetic.

The sample cannot compensate either. The standard error **never attains the root-T rate at any horizon**: quadrupling the panel from 50 to 200 periods buys a factor of 1.22 where root-T would buy 2.00, and from 400 to 1,600 periods it buys nothing measurable at all. Every term in the estimating equation — signal, regressors and accrual noise alike — is proportional to the asset's remaining value, so once the asset has decayed the later periods are not noisy observations but absent ones. **The information about recognition speed is a property of the asset: how much its value moves, and how long it goes on existing. The analyst chooses neither.**

Two things follow, pointing opposite ways. The first is a defence of the field's specification arrived at from outside it. The variation that identifies φ in this filter is return variation, which is the same variation Basu's regression requires in order to run at all; an instrument conditioning on returns is drawing on exactly the right information, and the return-variance corrections the literature reached for empirically are operating on the identification-strength parameter rather than on a nuisance. The second is where that leaves the assets anyone argues about. The corner in which every term above is worst is a quiet asset whose book amortisation rate sits close to its true rate of decline — small σ **together with a small gap between the two rates**. Slow decay on its own is not the hazard it looks like: at a fixed rate gap it is mildly *helpful*, because a slow asset stays alive to be observed.

§4.8 separates the two and gives the arithmetic. What holds of both terms is that a design cannot buy its way out of either — the panel saturates within a few half-lives whatever the volatility, and the response to news flattens and then reverses as decay slows, so more years and more news fail together rather than in sequence.

And the design this licenses already exists. Beaver and Ryan (2000) decompose the book-to-market ratio into a persistent **bias** component and a **lag** component by regressing it "on the current and six lagged security returns with fixed firm and time effects," taking the firm effect as bias and the returns-associated portion as lag. The regression is Ryan's (1995) — the book-to-market ratio on current and lagged market-value changes with firm and time effects — and the bias reading is not. Ryan's model assumes conservatism away by construction: his assumption (A8) "eliminates the possibility of conservative accounting," and his firm effects enter as a control for what that assumption leaves unmodelled. Beaver and Ryan supply the reading that turns a lag regression into a two-component decomposition. That is this section's repair, run empirically twenty-six years ago: a second series used to separate a persistent understatement from a delay, which is exactly the separation §4.2 shows a reported series alone cannot make. The theorem supplies a warrant the design did not have. The measurements above supply its boundary — the strength of the separation belongs to the asset, and is weakest where the amortisation rate sits near the asset's true rate of decline, a condition a firm effect cannot report.

The second repair does not require the asset to be noisy, and for most classes it is already published. This is the accounting form of a move the pharmacokinetic literature has long made: the tie is broken by information the profile itself does not contain. Kuan, Wright and Duffull (2023) observe that it is precisely "in the absence of intravenous data" that covariates describing elimination can load onto absorption parameters, and their own proposals — a mechanistic model of the two processes, or an estimated cutoff at which the rate constants exchange — share that shape. An outside determination of one root releases the other, and the scale with it.

For two of the four classes, the standards already supply that outside determination. Finite-lived intangibles and depreciable property carry **disclosed useful lives and amortisation schedules** — an estimate of the physical decay rate, made by the firm, audited, published, and *not* derived from the series whose timeliness is in question. Pinning δ rather than estimating it jointly is precisely the 291-fold improvement quoted in §4.2. A design that uses disclosed useful lives as an independent δ, and compares timeliness only within a life band, is reading φ rather than φδ, and it runs on the sample §5 already collected — **151 property events across 98 firms** on the tier-0 tag list §5.4 repaired, against 55 across 38 on the list as first collected, with 110 of the 151 joining to a disclosed life. Across the one-year life bands that design requires, those 110 events occupy sixteen bands and **exactly one clears the registered floor of 30 — the minimum number of events a life band must carry before a within-band comparison is run** — thirty-six events from twenty firms at a five-year life — with none clearing on firms rather than events. Filling the coverage §5's two cycles leave between them — the seven intervening cycles, run — raises the join to **133 of the 151** and leaves the same single band clearing: the second band reaches twenty-seven against the floor of thirty. The registered reading is the only one that gives one — the two other cycle choices give two, as does the nearest-cycle rule under the opposite tie-break.

Three properties recommend that design over the one this paper registered. It is diagonal-safe: no comparison crosses a class boundary, so the diagonality assumption of §4.1 is not load-bearing. It holds δ approximately constant by construction, which is the condition §4.4 identifies. And it has a built-in negative control — the same comparison across life bands, where the theorem says the ranking should degrade — which is the kind of prediction that can embarrass the framework rather than decorate it.

The analogy also marks its own weak joint, and the joint is real. An intravenous dose is exogenous in the strong sense: a different physical administration of the same compound, whose elimination rate is set by physiology and not by the analyst's question. **A disclosed useful life is chosen by the same management whose timeliness is being measured.** Audited and published is not the same as exogenous, and a firm that reports slowly may also amortise slowly. Three things bound the consequence — useful lives are anchored by industry convention and by tax and regulatory schedules, they are sticky within a firm across the horizon over which timeliness is measured, and the design can be run on industry-median lives rather than firm-specific ones, at the cost of resolution. The sign of any residual endogeneity is toward finding *less* timeliness variation than exists, not more.

The repair does not reach the two unamortised classes, and it fails them for different reasons. Indefinite-lived intangibles are tested for impairment rather than amortised (ASC 350-30-35-15), so no life is disclosed and there is nothing to pin δ to — a gap in the evidence, and one a standard could close. **Goodwill's is not a difficulty of measurement.**

4.8 · The goodwill limit, and what it is a limit on

At $\delta = 0$ the physical layer does not move; $\Delta E = 0$; the term $\varphi \odot \Delta E$ vanishes identically; the gap is **exactly** zero at every φ — not small, zero, at all eleven values swept — and no recognition event occurs at any φ in 400 periods. **At zero decay, φ is not ill-conditioned. It is absent from the dynamics.**

That limit is narrower than it looks. The run requires two conditions, not one: $\delta = 0$ *and* an asset whose value does not otherwise move. Set $\delta = 0$ and let the value receive news, and the gap reopens and φ is recovered exactly — to 3×10^{-15} — from the reported series and returns together. **The limit belongs to a motionless asset, not to a slowly-decaying one.** An asset whose value never changes for any reason is not goodwill. Impairment testing exists because goodwill's value does change; the standards decline to *schedule* that change, which is not the same as denying it.

What decides whether a class is readable is the gap between the two rates, not either rate alone. Hold $\alpha - \delta$ fixed and sweep δ over a fifteenfold range: the standard error on $\hat{\varphi}$ moves by a factor of 1.24, and in the direction that favours *slow* decay, since a slow asset stays alive to be observed. Hold δ fixed and sweep $\alpha - \delta$ over a sixteenfold range: it moves by 6.8, as $(\alpha - \delta)$^−0.70. At a realistic amortisation rate the goodwill decay rate is no harder to read than property's — 0.021 against 0.023. **The unreadable case is the firm whose book amortisation rate sits close to its asset's true rate of decline**, which is hard for the plainest reason in econometrics: the two numbers being told apart are nearly the same number. At a gap of 0.002 the standard error is 0.13, so φ is readable to ±0.26 — the whole interval.

The two rates do different jobs and both are needed. The gap sets the level; the decay rate sets how strongly that level responds to volatility, and the response **changes sign**. At the fixed gap above, the volatility exponent runs from −0.39 at a property-like δ of 0.030 to +0.16 at a goodwill-like δ of 0.002. Below roughly δ = 0.01, a noisier asset is read *less* accurately, not more — which is the one place in this paper where the repair of §4.7 runs backwards, and it is the corner the standards decline to schedule.

That claim has two properties the goodwill version lacks. It is checkable by a reader against a disclosed useful life, which §4.7 argues the standards already publish. And it does not require inferring a physical decay rate from a reporting rule.

What survives about goodwill specifically is a fact about this model rather than about goodwill. Within the deterministic filter, the class the standards decline to amortise produces no recognition events, so the model has nothing to say about it — and the registered test drew its largest single share from that class. That much stands and §5 pays for it. But the cause is the model's determinism (Limitation 3), not goodwill's nature: a filter admitting stochastic degradation would speak about goodwill as readily as about anything else, and would find it neither the hardest class nor the easiest. **Which classes this model can speak about is decided by the δ ladder before any hypothesis about φ is entertained** — and §4.4 states what that ladder rests on.

None of this was known when the registration was written, and all of it was derivable. §5 reports what was registered and what happened; §6 states what may now be claimed.

4.9 · The closed form does not need the constant hazard, and what it does need is a tail

§5.4 measures the recognition rate PRE-002's instrument identifies and, in the same fit — a discrete lag distribution fitted to that sample's recognition intervals — rejects the shape the model assumes: discrete Weibull $\hat{k}$ = 1.210, 95% interval [1.135, 1.285], stable under truncation at eight, twelve and sixteen quarters. A constant hazard is k = 1, so the estimate rejects it upward. **R = (1 − φ)δ/(α − δ) is derived by summing a geometric**, and a geometric is the one lag distribution whose hazard does not depend on how long the gap has been open. Everything §4.3 and §4.4 read off that expression — the cross-class ranking, the top-rung crossing, the domain — inherits whatever the assumption was doing. REG-004 asks what it was doing.

It survives, and it survives as a transform. Let a gap cohort created at time s be recognised after a lag $T \geq 1$ periods, so it sits in the gap over s+1 … $s+T$. The gap is then the flow convolved with the lag's survival function, and with $E(t) = E_0(1 - \delta)^t$ and $z = 1/(1 - \delta)$,

R = (1 − φ) δ Σ_{a ≥ 1} z^a P(T ≥ a) = (1 − φ) · (Π(z) − 1), Π(z) = **E**[z^T]

because Σ_{a≥1} z^aP(T ≥ a) = z(Π(z) − 1)/(z − 1) and z − 1 = δ/(1 − δ). The generating function is evaluated **outside the unit disc**, so it is a moment generating function and not a Laplace transform, and that is precisely why it can fail to exist. With T geometric at rate α, Π(z) = αz/(1 − (1 − α)z), which at z = 1/(1 − δ) is α/(α − δ) — and the published form returns exactly. **The constant hazard was never doing structural work. It was supplying a closed form for one transform.**

Three checks, each of which could have ended the section. The general form reproduces the published one at a geometric lag to **2**

$\times$ **10^{-13}**. An age-structured simulation — the gap carried as cohorts, each aged one period and multiplied by its own $P(T \geq a+1)/P(T \geq a)$, no closed form anywhere in the loop — reproduces it at the fitted lag distribution to **2×10^{-13}** against the 2×10^{-4} transient bound §4.3 already publishes, while **rejecting** the substitution $\alpha \leftarrow 1/\mathbf{E}[T]$ at ten times that bound. And $R(\varphi)/R(0) = (1 - \varphi)$ to **zero**, at every φ on a tenth-grid: φ is a pure scale under age-dependence exactly as it is under memorylessness, so §4.2's proportionality result and every ranking statement resting on the $(1 - \varphi)$ channel alone are untouched.

What the assumption *was* doing is holding up the domain. R is finite exactly when $\Pi(1/(1 - \delta))$ is, which is a condition on the **radius of convergence** of the lag's generating function and therefore on its **tail** — not on its mean, and not on any single rate. For a geometric lag that radius is $1/(1 - \alpha)$ and the condition reads $\alpha > \delta$, which is §4.4's domain verbatim. For a lag whose hazard *rises*, the survival function outruns every geometric and the generating function is entire: at $\hat{k} = 1.21$ the transform is finite at every decay rate swept, up to $\delta = 0.80$ per year where it is 7.5×10^{25} and still a number. **The existence condition has no analogue at the measured shape.** The general statement is the classical one for distributions with monotone hazard: the transform converges below the limit inferior of the hazard rate and diverges above its limit superior (Barlow, Marshall and Proschan, 1963, Theorem 6.3). A constant hazard puts both limits at α. An increasing one puts the first at infinity — and a *decreasing* one puts it at zero, so a lag distribution with a fattening tail would admit no steady-state deferral measure at any positive decay rate at all. **The interval [1.135, 1.285] is therefore doing more than rejecting a null.** It is what makes the model well-posed across the disclosed range, and had the same fit returned $\hat{k} < 1$ the closed form would have had no domain to be restricted to.

This does not rescue the asserted rectangle, and the statistic that would look as though it did is withdrawn rather than reported. §4.4 reports the share of the rectangle inside the domain. If the domain is everything, that share is one by construction and its complement is empty; a share of an empty set is arithmetic, not evidence. What replaces it is the level of R, which is defined everywhere and moves in both directions — and does.

The correction is negligible where this paper's ladder sits and material where the filings sit. Against a geometric with the *same mean* — which is the comparison the constant-hazard form implicitly makes — the measured distribution defers **less**, at every decay rate, on the direction registered in advance from the standard reliability bound for a distribution that is new-better-than-used in expectation (Marshall and Proschan, 1972). The size is a different question from the sign:

disclosed life	δ per year	R at the measured shape	R under a constant hazard	overstatement
40 years	0.025	0.0248	0.0249	**0.6%**
20 years	0.050	0.0523	0.0530	**1.2%**
10 years	0.100	0.1180	0.1215	**2.9%**
5 years	0.200	0.3152	0.3445	**9.3%**
3 years	0.333	0.9145	1.3156	**43.9%**

(φ held at 0.5 throughout, since it is a common scale.) Across §4.4's tabulated four-tier ladder, whose fastest rate is 0.030, the worst overstatement is **0.67%**. Across the rates ASC 360 and ASC 350-30-50 disclosure actually spans it reaches **43.9%**, at the three-year life the second of those routinely carries. The mechanism is visible in the transform: z^a with $z > 1$ weights the tail, and the tail is what a constant hazard gets wrong. This is not an artefact of approaching a singularity — the constant-hazard form's pole sits at 0.435 per year, a 2.3-year life, outside the rectangle, whose own fastest disclosed rate is 0.333 — and above §5.4's measured peak-to-charge recognition rate of 0.408 per year as well.

An effective rate exists and it is not a constant. Writing $\alpha\_\mathrm{eff}(\delta) = \delta\,\Pi(z)/(\Pi(z) - 1)$ returns the published form verbatim, $R = (1 - \varphi)\delta/(\alpha\_\mathrm{eff} - \delta)$. But α_eff runs from **0.437** per year at a forty-year life to **0.476** at a three-year one — **nine per cent** of itself across the asserted rectangle, in the direction that a faster-decaying class behaves as though recognition were faster. Across the four-tier ladder it moves by six parts in a thousand, which is why the magnitudes there barely move. **A recalibration is therefore available and is not a repair:** any comparative static that holds α_eff fixed while moving δ is using the wrong derivative, and a single recognition rate quoted for a cross-section of asset lives misstates one end of it.

The crossing is insensitive to the shape and sensitive to the level, and §4.4 says which. §4.4's $\delta_3^* = K\alpha/(1 + K)$ generalises exactly, to $\Pi(1/(1 - \delta_3^*)) = 1 + K$. REG-004 named the two channels in advance and they oppose: a lower R_2 lowers K and pushes the crossing down, a flatter transform pushes it up. They very nearly cancel — the shape moves δ_3^* by **0.13%**, from 0.00755 to 0.00754, and goodwill's tabulated rate stays a factor of 3.8 inside it. The *level* moves it by 4.3%, from 0.00789 at the calibration — §4.4's closed form evaluated there — to 0.00755 at the measured rate, and moves something else besides, which §4.4's tier table states.

One thing the shape does not do is break the exchange. An age-dependent world sits 5×10^{-4} from its own constant-hazard match in the reported series — four orders of magnitude below the 3×10^{-1} that separated the right conserved quantity from a plausible wrong one in §4.2 — and exactly as far from that match's mirror, which is forced rather than discovered, since the two constant-hazard worlds are an exact mirror pair to begin with. **§4.2's degeneracy is not repaired by age-dependence.** Whether the shape parameter itself is recoverable from a reported series is a different question, and §4.10 measures it.

The mathematics is old here too, and the accounting instance is what is new. That a stationary stock equals arrival rate times *mean* delay, whatever the delay's shape, is Little's law (Little, 1961), and its insensitivity to shape is a property of a *constant* flow. This model never has one: the flow is $(1 - \varphi)\delta E(t)$ against a base decaying at δ, and the moment a base grows or shrinks geometrically the stock-to-flow ratio becomes a transform of the delay evaluated at the growth rate rather than its mean — the same substitution that turns a stationary population's mean age into the Laplace transform that stable population theory carries. **The accounting literature has the delay and not the shape.** Goodwill write-offs are known to lag economic impairment by three to four years on average, with the delay extending to ten for a third of firms (Hayn and Hughes, 2006); impairment timing has been modelled as a first-event hazard with covariates (Potepa and Thomas, 2023); and conditional conservatism is measured throughout as a timeliness coefficient. In none of it is the recognition lag's **shape** estimated rather than assumed. §5.4 estimates it, and this section is what the estimate costs the model that motivated it: less than one per cent where the paper's own ladder sits, forty-four per cent at a disclosed three-year life, and one domain restriction that was never a fact about disclosure.

4.10 · The shape is identified, and the price of admission is four significant figures

§4.2 proves an impossibility by counting. A reported series is a sum of two geometrics, so it holds four numbers — two roots and two amplitudes — against five parameters, and the shortfall lands on φ. That count was taken under a constant recognition hazard. §4.9 replaced the constant hazard with an arbitrary lag distribution, which is not one more parameter but an **infinite-dimensional** object, so the count has to be taken again: does the extra structure leave a trace in the series, or does a constant-hazard world reproduce an age-dependent one exactly?

It leaves a trace, and the trace has a size. REG-005 registered the question, four falsifiers and five ladders before `wt091` existed. The measurement is made in §4.2's *favourable* setting — the books open square and the physical scale is granted, which is precisely what §4.2 says a firm-level series does not supply — so a null result would have held a fortiori and a positive one carries the condition that the asset is followed from acquisition.

The best constant-hazard world reproduces the measured world's reported series to 3.9 × 10^{-4} per quarter over ten years at a ten-year life, rising to **4.1 × 10^{-3}** at a three-year one, with the best mimic an admissible firm at every rate and every φ swept rather than a fit escaping into inadmissible parameters. That number is the whole answer in one figure: **it is the precision a reported series must carry to reject the constant hazard at all.** Coarser than that and the two worlds are the same series.

The same statement made from the other side is the set of shapes a series of given precision cannot separate from the measured one, fitting the remaining three parameters freely at each shape:

precision of the reported series	shapes it cannot separate from $\hat{k}$ = 1.21	width	against the 0.150 width of §5.4's [1.135, 1.285]
10^{-6} per quarter	1.21 alone	0.00	—
10^{-4}	**[1.16, 1.26]**	**0.100**	**0.67 ×**
10^{-3}	[0.60, 1.87]	1.27	8.5 ×
10^{-2}	the whole range swept	≥ 1.40	≥ 9.3 ×

(The lower two rows run into the boundary of the pre-registered sweep, so their widths are lower bounds; on a sweep extended to [0.2, 3.0] the 10^{-3} interval is [0.50, 1.86] and the reading is unchanged. The search's own floor at the true shape is 2.7 × 10^{-8}, thirty-seven times below the finest tolerance reported, so the top row measures the model and not the optimiser.)

At one part in ten thousand the reported series is a better instrument for the shape than the event dates are — an interval of 0.100 against the 0.150 width of §5.4's [1.135, 1.285], from hand-collected impairment lags. At one part in a thousand it is an order of magnitude worse. The identification is real and it is expensive.

And what lies inside the interval matters more than how wide it is. At one part in a thousand the registered sweep's set reaches **k = 0.60**, below one, a *decreasing* hazard — and §4.9's tail condition says a decreasing-hazard lag admits no steady-state deferral measure at any positive decay rate, because its generating function diverges inside the disc the transform is evaluated on. A series matched to a tenth of a per cent per quarter **cannot separate the world in which this model is well-posed from one in which it has no steady state at all.** That is a sharper limit than any width, and it is the sense in which [1.135, 1.285] was doing structural work rather than rejecting a null.

The identified set and the estimator answer different questions, and both were registered. The widths above are deterministic and worst-case: every shape that *could* have produced the series. Fitting the shape to a series carrying independent noise at the same level — 200 draws, seed recorded — recovers $\hat{k}$ with a median of **1.211** and an interquartile range of **0.125**, narrower than the event-date interval. The two are not in tension. Forty observations average independent noise down by a factor of six, and the estimator sees the identified set at the resulting effective tolerance rather than at the raw one. At one part in a hundred it breaks: the interquartile range is 1.19 and a third of the draws pile on the boundary of the swept range, which is reported as a fraction rather than as the quantiles of a censored distribution.

A longer series does not help, and the reason is where the information sits. The interval is not monotone in the observation window: 1.40 over five years, 1.26 over ten, **0.98 over twenty**, and 1.32 over a hundred. The narrowest window is twenty years and a longer one is *worse*. Once the gap reaches its steady state each further quarter repeats a single number — the deferral measure itself — so extending the window adds redundancy to a mean and dilutes the transient that carries the shape. This is an identification property rather than a sample-size one, which is the general result for any finite-dimensional approximation to a lag space: the approximating families are meagre in it and their approximation error "cannot, in other words, be made asymptotically negligible" (Sims, 1971, §5), a statement he makes of exactly the rational lag distributions of which a constant hazard is the lowest-order member (Jorgenson, 1966).

Three recognition rates now live in this paper and they are three different quantities.

disclosed life	δ per year	$\hat{\alpha}$, the event dates	α_ser, the series	α_eff, the deferral measure
40 years	0.025	0.408	0.4383	0.4368
20 years	0.050	0.408	0.4385	0.4388

disclosed life	δ per year	$\hat{\alpha}$, the event dates	α_ser, the series	α_eff, the deferral measure
10 years	0.100	0.408	0.4388	0.4431
5 years	0.200	0.408	0.4370	0.4538
3 years	0.333	0.408	0.4037	0.4758

The series-matching constant is nearly flat across the rectangle at 0.438 per year, which is the **reciprocal mean lag** $1/\mathbf{E}[T] = 0.435$ to within a per cent; α_eff rises with the decay rate because the transform weights the tail; and $\hat{\alpha}$ is the geometric maximum-likelihood summary of the same sample. They agree to **five parts in ten thousand** at a twenty-year life and part company at a three-year one, where they differ by **15%** and move in opposite directions from $\hat{\alpha}$. Least squares on the series matches the mean, the transform matches the tail, and the likelihood matches the event dates: three functionals of one distribution, with no obligation to coincide. §4.9 says a single effective rate misstates one end of the asserted rectangle; the series adds that a single *recognition rate* does not name one quantity.

§4.2's exchange survives into all of this, and it is forced rather than discovered. The mimic search returns the mirror pair $(\alpha, \delta, \varphi)$ and $(\delta, \alpha, \varphi\delta/\alpha)$ at an identical objective, which it must, since the theorem makes the two worlds one series and any third series is equidistant from both by construction. What the mirror costs is the mimic's own parameters: at a forty-year life the set of worlds fitting within one part in a million of the best spans **0.128 in each root and 0.577 in φ**. **The best-fitting constant hazard is not a world. It is a pair, and φ inside it is as free as §4.2 says it is.**

So §4.9's open question closes with a number rather than a verdict. The shape correction of §4.9 is recoverable from a reported series, at a precision of one part in ten thousand per quarter that audited financial statements do not carry for the relevant quantity — which is why §5.4 dated impairments rather than fitting a series, and why the correction travels with the lag distribution rather than with the filings. §4.2's count is not made worse by age-dependence. The infinite-dimensional lag does not disappear into the four numbers; it leaves a residue, and the residue is measurable by anyone who can read a balance sheet to four significant figures.

Two predictions registered in advance were wrong, in the same direction. REG-005 predicted the shape would be invisible below one part in a thousand and that the shape interval would be a hundred times the event-date interval, reasoning from the density of rational lag distributions in lag space and from the classical ill-conditioning of exponential sums, where a three-term signal is reproduced by two terms to a few parts in ten thousand with rates bearing no relation to the truth (Lanczos, 1956, as quantified by Varah, 1982). The measured residue is four times larger than that reading allows and the interval is nine times, not a hundred. Approximation theory describes what a family can do in the limit; this is one distribution over one horizon, and it leaves more behind than the general argument suggests. The literature's positive identification results do not close the gap either, since recovering hazard shape there is bought with a covariate that varies, a proportionality restriction and a moment condition (Elbers and Ridder, 1982) — and a single reported series supplies none of the three, which is why the answer here had to be measured rather than cited.

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5 · The severe test: registered, run twice, and lost

What §4 bears on here. The test reported here ordered asset classes by expected timeliness. §4.4 shows that a *magnitude* reading of such a ladder is inverted by the composite; §4.5 shows that the *timing* reading — the one this registration used — is not. So the identification result is not an excuse for the null, and it is not offered as one. It bears on this section at one point only, and that point is §4.5's second half: the model's lag is defined against a physical series that no filing reports, so the instrument below measures a substitute. §6 states what may be concluded from §5.

5.1 · What was predicted, and when

§3 establishes a property of a model. Whether the property holds of the world is a separate question. The framework's sharpest available prediction is the one that follows directly from §3.1:

> **Recognition lag scales with the unobservability of degradation.**

To test it, unobservability must be identified with something measurable. The identification chosen was **GAAP asset class**, on the reasoning that the categories accounting standards decline to place on an amortisation schedule are precisely the categories whose degradation is hardest to observe. The classes and their schedules are those of the FASB *Accounting Standards Codification*: Topic 360 for property, plant and equipment, Topic 350 for intangibles and goodwill — where the indefinite-lived classes are tested for impairment rather than amortised, which is what makes the recognition moment discretionary — and Topic 280 for the segment disclosures §9 identifies as the unit of observation this test did not have. That yields a four-tier ordering, predicted in advance to be monotone in lag:

tier	asset class	predicted
0	property, plant and equipment	shortest lag
1	finite-lived intangibles	↓
2	indefinite-lived intangibles	↓
3	goodwill	longest lag

Every event in this test is a recognised impairment, which places the sample on the boundary of §2's domain restriction rather than inside its complement: a charge is the moment degradation became estimable, so §2 governs the accumulation that precedes it and the event marks where that accumulation ends — which is why an interval is measurable on these events and on no others.

The registration preceded the data. PRE-001 was committed **alone**, at commit 9722342, and pushed, before any lag was computed; the analysis code did not yet exist. The git history is the timestamp, and it is the entire evidence that the prediction preceded the outcome — which is why the registration and the code were deliberately not batched into one commit.

A disclosure, because the discipline just described does not fully cover the test this paper reports. PRE-001 timestamps the *prediction*. The result reported in §5.3 comes from PRE-002, which specifies a different *instrument* — and PRE-002's registration shipped in the same commit (d655501) as the implementation of that instrument. The result itself came later, at a subsequent commit, so the git history still establishes that PRE-002 was registered before its outcome existed. What it does **not** establish is that the instrument's details — the onset rule's window, the tie-break direction, the materiality floor — were fixed before anyone had seen what they produced. That is precisely where the remaining researcher degrees of freedom live.

No claim is made here that they were exploited, and the author's account is that they were not. The point is that this is an *account* rather than a demonstration, whereas for PRE-001 it is a demonstration. A reader is entitled to weight the two differently. A registration that precedes the **instrument's code**, and not merely the result, closes this gap; PRE-002's did not.

5.2 · The first instrument failed, and so did its diagnosis

The first test (PRE-001) returned a null in both universes, and the two nulls did not even agree with each other. The Jonckheere–Terpstra statistic sums the pairwise Mann–Whitney counts across the ordered tiers, so a tier ordering carrying no information returns z near zero from either direction — which is what happened, twice, in opposite directions. The pilot retained 120 events across 72 firms and gave Jonckheere–Terpstra **z = −0.177** (one-sided p = 0.570), with goodwill's median lag sitting *below* PP&E's — 4.0 quarters against 5.0, the reverse of the predicted ordering. The replication universe, declared in `PRE-001` §4.2 before the pilot was run, retained 202 events across 106 firms and gave **z = +0.634** (p = 0.263), with goodwill and PP&E tied at 3.0. A weak negative and a weak positive, neither significant: the signature of a measurement carrying no information about the ordering rather than of an effect in either direction.

The instrument was then examined, and it had a defect. Across the 322 events retained by both universes combined there was **zero right-censoring**, 69% of pilot lags fell at six quarters or fewer, and **1,047 charges were discarded for having no measurable onset**. An onset rule requiring an unbroken decline in a firm-level signal measures the volatility of that signal, not the phenomenon: the rule can only find an onset when the signal happens to fall monotonically, which is common over short windows and vanishingly rare over long ones. The lag distribution was therefore pinned against a ceiling the instrument itself imposed.

This diagnosis was not permitted to rescue the result. A second, **separately numbered** registration (PRE-002) was written with a different onset instrument (peak-to-charge), a label-permutation negative control, a power curve to be reported whatever happened, the significance level tightened to 0.025 for the second look, and — decisively — an explicit **stopping rule** stating in advance that there would be no third instrument.

5.3 · The second instrument worked, and the prediction failed anyway

Pilot — retail trade (SIC 5200–5999). 244 events across 121 firms.

tier	n	median lag (quarters)	IQR	mean
0 · PP&E	21	5.0	3.0–9.0	7.05
1 · finite-lived intangible	34	4.0	1.0–8.0	5.71
2 · indefinite-lived intangible	34	5.5	1.2–9.0	6.12

tier	n	median lag (quarters)	IQR	mean
3 · goodwill	155	5.0	1.0–9.0	5.93

Jonckheere–Terpstra **z = −0.290**, permutation p = **0.590**. median(t_3) − median(t_0) = **0.0 quarters**, CI [−4.0, +2.0].

Replication — computer and data processing services (SIC 7370–7379). 444 events across 190 firms.

tier	n	median lag	IQR	mean
0 · PP&E	34	5.0	1.2–9.8	6.62
1 · finite-lived intangible	102	4.5	1.2–10.0	6.12
2 · indefinite-lived intangible	46	6.0	2.2–11.0	6.85
3 · goodwill	262	5.0	1.0–10.0	6.46

Jonckheere–Terpstra **z = −0.095**, permutation p = **0.520**. median(t_3) − median(t_0) = **0.0 quarters**, CI [−4.0, +2.5]. The registered sensitivity analyses per universe are in the run logs (`docs/preregistration/RESULT-002-*-run.log`); none reverses the verdict.

The instrument was demonstrably better this time, and that is what makes the null bite:

	PRE-001 (streak onset)	PRE-002 (peak onset)
events retained, both universes	322	**688**
charges discarded for no onset	**1,047**	**0**
right-censored	0%	7.8% pilot, 14.2% replication
right-censored by tier, 0 · 1 · 2 · 3	0% throughout	pilot 4.8 · 17.6 · 11.8 · **5.1** %; replication 17.6 · 15.7 · 17.0 · **12.5** %
pilot IQR width, tier 3	3.0–7.0	1.0–9.0

The lag distribution now spans the registered range instead of piling against a ceiling imposed by signal volatility, and censoring is non-zero — which is what an instrument capable of observing long lags looks like. **And the ceiling is not where the gradient went:** goodwill, the tier predicted to lag longest, carries the least censoring of the four in the replication and the second-least in the pilot, so the twenty-quarter cap is not hiding its tail.

Negative control. Tier labels permuted 1,000 times with the lag distribution held fixed:

universe	null z mean	null z sd	observed z	empirical p
pilot	+0.007	1.025	−0.290	0.590
replication	−0.002	1.000	−0.095	0.520

The permutation distribution is centred on zero with unit spread in both universes. The pipeline does not manufacture a gradient, and the empirical p-values do not lean on a normal approximation, so they are untroubled by tier sizes of 21/34/34/155.

Power, reported because a null without its detectability attached is not a result:

true effect	power, pilot	power, replication
0.5 quarters per tier	0.65	0.87
1.0 quarter per tier	**0.95**	**1.00**
2.0 quarters per tier	1.00	1.00

This design would have detected a one-quarter-per-tier gradient with 95% probability in retail and with certainty in computer services, and it found nothing.

Three qualifications bound that sentence:

1. **The power simulation assumes the onset instrument measures lag without error.** It resamples the observed lag distribution and adds a noiseless per-tier shift. The peak rule always finds an onset — which is why zero charges were discarded, and which is a property of the rule rather than proof of its quality. Any measurement error in the onset attenuates a true gradient, and the reported power does not model that attenuation. The true power is therefore lower than 0.95–1.00, by an unquantified amount.
2. **The 688 events come from 311 firms**, and the test statistic, the permutation control and the power simulation all treat events as independent. The effective sample is smaller than 688 and the reported power is an upper bound.
3. **One quarter per tier was never derived from the model.** It is a plausible round number, not a minimum interesting effect computed from the framework — and it could not have been, because deriving one requires the very φ-to-tier bridge that §6.2 concludes was unsound. A design cannot be well-powered against an effect size the theory never specified.

What the result supports, stated at the strength the evidence carries: a well-powered, pre-registered, replicated test found no gradient, and if a gradient of the size tested exists it is unlikely to have been missed twice. **What it does not support** is the stronger reading that the framework's own predicted effect has been ruled out, because the framework never named that effect in the units the test measured. Calling this *evidence of absence* would contradict §6.2 in the same paper: the bridge cannot be too broken to license a prediction and sound enough to license its refutation.

The shape of the failure, stated as sharply as the data allow. Both z-statistics are negative, meaning the point estimates ran *opposite* to the predicted ordering in both universes, as they had in the PRE-001 pilot. And the ladder does not merely fail to be monotone — it is wrong in a specific and instructive place: **tier 2, indefinite-lived intangibles, carries the longest median lag in both universes, with goodwill sitting below it** (5.5 against 5.0 in the pilot, 6.0 against 5.0 in the replication). Those are point estimates, and the ordering is the claim: **the predicted ordering appears in 5.7% of firm-clustered resamples in the pilot and 5.8% in the replication**, resampling firms rather than events because §5.4's own finding says events are not the unit.

The stopping rule fired. There is no third instrument. A hypothesis that requires one on the same data is a hypothesis being fitted.

5.4 · The same sample answers two questions it was not collected for

The stopping rule bars a third instrument for the *lag gradient*. It does not bar asking the sample questions it was never asked. Two were registered in `REG-003`, committed and pushed before the instrument existed, and both returned. Neither is a re-test of §5.1's prediction and neither may be read as one.

The sample rebuilt at 99.0% agreement. `companyfacts` serves each firm's latest view of its own history, so a re-pull is not the original pull. Rebuilt: **695 events across 313 firms** against 688 across 311, with three of four tier counts identical in the pilot and censoring at 7.7% against 7.8%. Those firm counts are per-universe sums — 122 + 191 against 121 + 190. Read as a **union** out of the committed `data/pre-002-events.json` the count is **307**, six registrants having changed SIC between 2013 and 2024 and so entering both universes, and that union is the n of the one-event-per-firm sensitivity below. The registered reconciliation rule, fixed before the count was known, is 95% agreement in total n with no tier moving by more than 20%; at 99.0% and 1.4% it admits this as the registered sample.

The peak-to-charge recognition rate is 0.41 per year on

both known biases' inflating side, and the calibration was low by an order of magnitude. Each event carries the interval from the onset of deterioration to the charge, right-censored at twenty quarters — which is α's definition, measured once per event, by an instrument built to look at something else. The censored geometric maximum likelihood estimate is **$\hat{\alpha}$ = 0.1227 per quarter (se 0.0046), 0.408 per year, 95% interval [0.383, 0.432]**; the median observed gap is five quarters. Retail gives 0.433 and computer services 0.394. The three sensitivities registered with PRE-002 give 0.397, 0.499 and 0.413. As unregistered robustness, administratively censoring the sample at eight, twelve and sixteen quarters instead of twenty gives 0.396, 0.398 and 0.404. **Every cut lands in the same regime,** and the calibrated 0.05 is outside the interval of all of them.

The shape was fitted rather than assumed, and it is not the model's shape. A discrete Weibull in Nakagawa and Osaki's (1975) parameterisation, whose hazard is increasing exactly when its shape parameter exceeds one, so that the nesting is a boundary case and not a coincidence of fit, gives **$\hat{k}$ = 1.210, 95% profile interval [1.135, 1.285]**, excluding the constant hazard the model assumes. The non-parametric hazard shows why: a quarter of the sample (175 of 695) is recognised one quarter after the peak, and the rest faces a hazard rising from 0.09 to about 0.25 over the following five years. **The longer a gap has been open, the likelier it is to close** — which is the opposite of the memorylessness a single α encodes, and it means $\hat{\alpha}$ is an average over a window and not a constant of the technology.

Two biases push this estimate up, and one pushes it down; the direction of each was registered before the number. A gap that opened and was never recognised leaves no filing, so conditioning on a charge over-represents short intervals. If revenue peaks after economic value has turned — the ordinary case for a business whose customers have not yet left — the measured interval is short of the true one. Against those, the sample contains no lag of zero, so fitting on a support that includes it understates $\hat{\alpha}$; the unregistered shifted estimate is 0.460. **The one unregistered cut that removes the mass where the onset bridge is least credible — the 175 events charged one quarter after the peak — gives 0.327, still an order of magnitude above the calibration.** The result does not rest on its most suspect quarter.

And the reporting layer is not diagonal. §9's ninth limitation states the assumption and names its test: a diagonal layer predicts that recognition in one class does not force recognition in another, so events should be independent across classes within a firm-quarter. Taking each firm's per-class impairment frequency as given and redrawing which quarters they land in — 10,000 draws within each firm's own eligible-quarter set — firm-quarters carrying two or more classes are

universe	observed	null mean	central 95%	observed/expected	two-sided p
retail	**30**	7.3	[3, 12]	**4.12**×	0.0002
computer services	**44**	21.8	[15, 29]	**2.02**×	0.0002

Both universes, the same direction, at the resolution 10,000 draws can report; the design detects an injected excess of five per cent of events with probability 1.00. The pairwise cells put the strongest coupling on goodwill with indefinite-lived intangibles in retail (5.83×) and on goodwill with finite-lived intangibles in computer services (2.22×), and it is these two intangible-with-goodwill cells that replicate across both sectors — 5.83× and 2.41×, 3.33× and 2.22×, all four surviving Holm correction. Property with goodwill was published at 4.35× and 4.03× on a tier whose tag list omitted the element most filers use for it; `REG-006` repairs the omission and re-derives that cell at **3.99**× and **2.17**×, against **3.63**× and **4.14**× from the same crawl unrepaired — so the repair *raises* the retail cell and takes the computer-services one below significance, and its cross-sector agreement does not survive the repair. The headline does: **4.01× and 2.10**× repaired, against 4.01× and 2.01× from the same crawl unrepaired. So does the intangible cells' agreement: **5.86× and 2.34**×, **3.35× and 2.22**× repaired, against 5.83× and 2.34×, 3.34× and 2.21× from that crawl unrepaired.

The mechanical reading has to be excluded before the economic one is available, §9 already named it, and it is two readings rather than one. The ordering is imposed by ASC 350-20-35-31, which requires that any other asset or asset group of a reporting unit be tested before goodwill, and by ASC 350-20-35-32, which extends that requirement to every asset tested rather than only to those within ASC 360-10 — so it governs the intangible cells above as well as the property one. On one channel the rule creates joint *testing*: ASC 350-20-35-3C(f) names the testing for recoverability of a significant asset group as an event requiring an interim goodwill test, so a single trigger fires two tests. On the other it suppresses joint *recognition*: the other charge is recognised first and reduces the reporting unit's carrying amount, and under ASC 350-20-35-2 and 35-8 the goodwill charge is the excess of that carrying amount over fair value, so the prior charge is subtracted from it one for one until zero or the goodwill cap binds. **Under the ordering alone the two charges are substitutes at the margin, and this sample shows them as complements.** Signing the net requires the two charges at the reporting-unit level, which US filings do not disclose; `REG-006` registered an entity-level test of the suppressing channel. **It failed as registered: no consistent sign in either sector.** The natural fallback is the triggering *disclosure* rather than the charge, and `REG-007` measures why that route does not identify either. ASC 350-20-50-2(a) compels a description of the facts and circumstances leading to the impairment only *for each goodwill impairment loss recognized*; where a test is run and nothing is charged the Codification compels nothing, and disclosure falls to MD&A. A triggering-event population assembled without that restriction is therefore selected on the outcome under study. Inside the window where the mandate does fall — and falls on both arms alike, which is what makes the comparison available — filers naming the standard's own internal trigger run at **0.436**, against **0.403** among firm-years that took a non-goodwill charge and no goodwill charge, on whom the mandate does not fall at all. Three points across 1,833 classified firm-years measures the vocabulary of the disclosure, not the mechanism behind it. `REG-008` sharpens the instrument to the sentence and to the *named* reporting unit, and the separation from the placebo more than doubles — **0.103 against 0.030** — while the joint-versus-goodwill-only difference stays at **+0.014** (p 0.60) in a design that could have detected 0.068. The reason is countable: **not one of the 281 firm-years taking both charges writes a sentence naming a reporting unit, a trigger, and any of the standard's own (f)-family language, and exactly one of the 644 in the mandated window does**, and the two phrases the Codification uses for that family appear in none of the 1,925 firm-years. The disclosure does not carry the quantity the decomposition needs.

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6 · What may now be claimed, and what may not

6.1 · The demotion, stated exactly

Not supported: that recognition lag scales with the unobservability of degradation, where unobservability is identified with GAAP asset class, in US-listed retail trade (SIC 5200–5999) or computer and data processing services (SIC 7370–7379), among registrants filing in 2013–2024, on charges recognised 2012 Q2 – 2026 Q2, at the firm level, at effect sizes of one quarter per tier or larger.

Unaffected: every result in §A.2 and §§2–3. Those are properties of a stated model, established by simulation and held in place by a test suite. A model result is not made false by the failure of an empirical identification, and it is not made true by one either.

That is a problem, not a reassurance. If nothing in §A.2 or §§2–3 was at risk, then nothing in §A.2 or §§2–3 was on test — and a framework that retains every claim after losing its only public bet is in exactly the position §6.3 accuses Odum's of occupying. The accounting is therefore:

- **What was at risk and lost:** the conjunction of the model, the bridge, and the firm-level unit of observation. That conjunction was the framework's *entire empirical content* as of this writing. It is gone.
- **What was never at risk:** everything in §A.2 and §§2–3, because those are theorems about a simulation. They were never capable of losing and no result of the severe test could have retracted one of them. Calling them "unaffected" states a fact about their logical type, not a survival.
- **What follows: the framework currently has no confirmed empirical claim.** It has a model with derived consequences, one registered prediction that failed, and a stated method for building the next one. A reader who wants to know what this paper has established about the world should read that list literally.

The framework would have been *confirmed* had the gradient appeared; it was not, and the correct posture is not that the theory survived but that the theory has not yet been given a test it can pass. Designing one is §6.2's business and it is unfinished work, not a conclusion.

Demoted: the lag-scaling claim moves from *a prediction the framework makes* to *a prediction the framework made, at one level of aggregation, with one bridge, and lost.* Any surviving version must state its measurable and its bridge before it is tested again.

Three post-hoc conjectures about where the conjunction broke are recorded in `docs/preregistration/RESULT-002-wt026.md` §4. They are excluded from this paper's argument deliberately: **each arrived after the number, none is evidence for anything, and any of them that is ever tested must be registered from scratch.** One is worth naming here only because it generalises into a discipline rather than a defence, and that is §6.2.

6.2 · The bridge discipline

The registration contained a tier table and no **bridge proposition**. It never wrote down, as a deniable claim, the sentence connecting the model to the world:

> *φ, the observability of degradation, is identified with the presence or absence of a GAAP amortisation schedule, because…*

Had that sentence been written, its weakness would have been visible before the data were touched. The observability of *degradation* and the observability of the *accounting treatment* are different quantities, and they may even be anti-correlated: goodwill carries no schedule, but its impairment is triggered by conspicuously public signals — a share-price fall, a missed segment, a lost contract. The physical condition of a distribution centre carries a schedule and is visible to essentially nobody outside the firm.

The general form of the error is a type error in a second costume. §A.1.1 recorded one: a structure cannot be promoted to a proposition by rewording. This is its empirical twin — **a quantity in the model was matched to a quantity in the world that shares its name and not its meaning.** The lesson is not about accounting. It is that a bridge from a parameter to a measurable must itself be stated as a proposition and checked, and this programme now requires it of every registration.

6.3 · The comparison this paper is not entitled to make

This paper is not entitled to the argument that the framework has escaped the trap that closed on Odum's emergy programme — that emergy's fatal defect was making no risky predictions, and that losing a registered bet therefore counts as a kind of methodological success.

That argument is withdrawn, on three counts a sceptical reader would have reached first. It selects its own reference class: introduce a comparator that made no predictions and any loss becomes a comparative victory. It is an assessment of the author's conduct rather than of the world, and it arrives at the end of the section reporting the failure, so it would be the last thing a reader carried away from it. A paper does not get to grade its own integrity; a reader grades it, or does not.

What remains after the withdrawal is a fact and not an evaluation. **A prediction was registered before the data were seen, it was tested, and it failed.** Whether that is worth anything is a judgement this paper leaves to whoever is reading it.

7 · What was tested and survived

A paper that reports only its failures gives a reader no way to weigh them. This programme's public record has been unbalanced in that direction: every test run is reported, and until this section existed, none of the ones that held were collected anywhere a reader could find them. Here they are, with what would have killed each.

claim	test	what would have killed it	outcome
$D(\varphi) = (1 - \varphi)\cdot D(0)$	closed form against simulation, φ swept	any φ at which the ratio departs from (1 − φ)	held to $\mathbf{10}^{-15}$
$(\alpha, \delta, \varphi) \sim (\delta, \alpha, \varphi\delta/\alpha)$	mirrored simulation, five parameter settings	any visible separation between a series and its mirror	$\mathbf{7 \times 10}^{-14}$, against 4×10^{-2} when the mirror's φ is perturbed by 0.05
φδ is the conserved quantity, not (1 − φ)δ	both candidate maps run against the reported series	the two maps agreeing, which would make the check vacuous	mirror $\mathbf{8 \times 10}^{-16}$, rival map $\mathbf{3 \times 10}^{-1}$ — the gap is preserved by one, the filing by the other
An open initial gap does not restore identification	mirror rebuilt at $g_0 = 0.15$ with the shifted map	the shifted map failing, or the $g_0 = 0$ map still working	shifted map $\mathbf{7 \times 10}^{-16}$, naive map $\mathbf{5 \times 10}^{-2}$; invariant $(\varphi - g_0)\delta$ held exactly
Unobserved physical scale ⇒ φ free over [0, 1]	one-parameter family constructed and regenerated	any member of the family failing to reproduce the series, or the family collapsing to one φ	nine members spanning φ ∈ [0, 1], **all exact to** $\mathbf{2 \times 10}^{-16}$
Returns kill the two-point exchange	mirror rebuilt with its own asset, return series compared	the two worlds' returns agreeing, which would leave §4.7's question open the other way	books agree to $\mathbf{7 \times 10}^{-16}$, returns differ by $\boldsymbol{\alpha - \delta}$ = **0.0300 every period**

claim	test	what would have killed it	outcome
Returns cannot touch the scale continuum	the nine-member family regenerated, return series compared across it	any member emitting a different return series	**2×10^{-16} — bit for bit identical** across a family spanning $\varphi \in [0, 1]$
News, not returns, restores identification	regression on lag, return-implied path and its first difference, $\sigma = 0.15$	recovery failing, or the $\sigma = 0$ design being well conditioned	α, E_0, φ recovered to **10^{-16}**; cond(X) **11.8** at $\sigma = 0.15$ against **4×10^{16}** at $\sigma = 0$
The repair's strength is the asset's, not the analyst's	σ swept 12×, T swept 32×, at nine (α, δ) settings	the panel buying the root-T rate, or the σ and T channels agreeing	T: 50⟶200 buys **1.22×** where root-T buys 2.00×, 400⟶1600 buys **1.00×** — regime-independent; the σ exponents are not, and §4.7 gives both ranges
Neither degradation exponent is a model constant	both re-fitted over nine (α, δ) settings on the GAAP ladder	the nine agreeing to within fitting error, which would license quoting a number	collinearity spans **−1.07 to −0.38**, se($\hat{\varphi}$) **−0.78 to −0.09** *— the check that removed two numbers from §4.7*
The response to news flattens as decay slows		exponent	ranked against $\delta(\alpha - \delta)$
The goodwill limit needs a motionless asset, not a slow one	$\delta = 0$ rerun with the asset's value allowed to move	the gap staying zero once news is on, which would make the limit about δ	gap **0.204** against an exact 0.0, and φ recovered to **3×10^{-15}** *— the check that rewrote §4.8*

claim	test	what would have killed it	outcome
The rate gap governs readability; the decay rate does not	each held fixed while the other is swept, 15× and 16×	the decay rate dominating, which is what §4.7 had asserted before this check	δ at fixed gap: **1.24×**, favouring slow decay. Gap at fixed δ: **6.8×**, as (α − δ)^−0.70
The two rates do different jobs	volatility exponent re-fitted across δ at a fixed rate gap	the exponent being flat across δ, which would collapse the two findings into one	exponent spans **−0.39 to +0.16** — *a change of sign*; level spread 1.24× at σ = 0.15 against 2.16× at σ = 0.025
R = (1 − φ)δ/(α − δ)	closed form against simulation	departure beyond the transient bound	held to **2 × 10^{-4}**, the bound the geometric transient predicts; **1.0** when φ is misstated by 0.1
The ranking inverts, not just blurs	4,000 ladders drawn on the two qualitative facts alone	the intended ordering surviving often enough to be a design	recovered in **1.9%**; **100.0%** when δ is held common — the witness that the construction is not vacuous
The inversion spares the lag statistic	400 admissible ladders, lag ordering checked	lag inverting like the magnitude measure, which would have made the story tidier	lag ordering held in **100%**
The inversion belongs to the ordering; the destruction belongs to the dispersion	4,000 ladders with δ drawn independently, no durability ordering imposed	mean τ staying negative, which would have made the inversion the general case	mean τ **+0.32**, recovery **11.5%**, exact reversal **1.1%** — against −0.41 / 1.9% / 23.8% ordered
τ = −1 is a knife edge in its top rung	closed form for the crossing rate, verified by bisection to 1 × 10^{-9}	a crossing rate far above any defensible goodwill decay	**δ_3^* = 0.0079**, an eighty-seven-year half-life; the table assigns 0.002

claim	test	what would have killed it	outcome
Lumpy defers more than slow at an identical mean rate	compound-Poisson decline, 2,000 paths, mean rate matched exactly	the ratio at or below 1, which would license reading "unscheduled" as "slow"	**1.30**× (se 0.002), a δ-equivalent of 0.0123 — above the crossing rate
The design's validity region has a fitted boundary	logistic of failure on log(leverage / budget), 4,000 ladders	a slope indistinguishable from zero	slope **+1.58**, z = **+19.5**; the same fit on a permuted outcome gives z = 0.23
The *asserted* rectangle lies outside the model's domain *at the calibrated rate*	useful lives spanning disclosure practice against α = 0.05, and the 683 disclosed pairs against the measured rate	any part of it admitting a steady-state deferral measure	**0%** admissible at α = 0.05; **all** of the asserted rectangle and **0.974** of the disclosed pairs at the measured $\hat{\alpha}$ = 0.408
The rectangle's 99.8% is a property of the assumed support, not of the disclosure	§4.4's first rung evaluated on 683 disclosed firm-year pairs across 577 firms, against the same test on the asserted rectangle at the same rate	the two agreeing within the clustered interval, which would have made the rectangle an adequate stand-in for the disclosure	rises in **0.659** of admissible pairs [**0.621**, **0.696**] against **0.998** on the rectangle; only **0.139** of the pairs fall inside it
The peak-to-charge recognition rate is an order of magnitude above the calibration	censored geometric MLE on 695 registered events, two universes, three sensitivities, four truncations	any cut returning a rate near the swept 0.05	**$\hat{\alpha}$ = 0.408/yr** [0.383, 0.432], on both known biases' inflating side; range **0.327–0.499** across every cut including the unregistered robustness, none containing 0.05
The constant hazard the model assumes is rejected	discrete Weibull fitted, not assumed, with a profile interval	$\hat{k}$ = 1, which would have left α a constant	**$\hat{k}$ = 1.210** [1.135, 1.285]; the hazard rises with the age of the gap

claim	test	what would have killed it	outcome
The closed form survives an age-dependent hazard	general form against an age-structured simulation carrying the gap as cohorts, no closed form in the loop	departure beyond §4.3's published transient bound	held to **2×10^{-13}** against a 2×10^{-4} bound; the same simulation **rejects** $\alpha \leftarrow 1/E[T]$ at 2×10^{-3}
φ is a pure scale under age-dependence too	$R(\varphi)/R(0)$ against $(1 - \varphi)$, φ swept on a tenth-grid	any φ at which the ratio departs	held to **exactly 0.0**
The domain restriction is the constant hazard's, not the disclosure's	the transform evaluated to a proven remainder bound across the disclosed rates	divergence anywhere the bound admits	finite at every rate to $\delta = 0.80$/yr, where it is 7.5×10^{25}; the geometric form is **infinite** at $\delta = 0.60$
The shape correction is small on the ranked ladder and large at disclosed lives	measured lag distribution against a geometric of the same mean	the two agreeing, which would make the fitted shape decorative	**0.67%** across the tabulated ladder, **43.9%** at a disclosed three-year life — and the constant-hazard pole sits outside the rectangle, so this is not a pole artefact
The reporting layer is not diagonal	10,000 within-firm permutations of which quarters each class's impairments land in	co-occurrence at the independence rate, which the Hadamard form requires	**4.12**× and **2.02**×, both $p = 0.0002$, power 1.00 at a 5% injected excess
The sample rebuilds from a live endpoint	full re-pull of both universes a week after the original	drift large enough to make the original unrecoverable	**695 events against 688**; three of four tier counts identical

claim	test	what would have killed it	outcome
Lag's 100% is partly the ladder	2,000 ladders, durability ordering dropped	lag holding at 100% regardless, which would have made §4.5 unconditional	**66.2%** (se 0.011) against 11.5% for the magnitude measure
Results are dimensionless	η — Appendix A's numeraire conversion, currency per joule — swept over **twelve orders of magnitude**	any dimensionless output moving with η	spread **exactly 0.0**
...and not because η is unused	mutation testing	a mutant that leaves results unchanged	**every substituted vacuous witness killed its run**
Recognition frequency is driven by δ	sweep at fixed φ	δ having no effect on event counts	0 ⟶ 16 ⟶ 100 events
The tier instrument has no baked-in ordering	label permutation	a non-null under randomised labels	z-mean **+0.007**, sd 1.025
The registered design had power	power analysis, to be reported whatever the outcome	power too low to interpret a null	**0.95–1.00**, with three stated qualifications making it an upper bound
The lag's shape leaves a trace in the reported series	best admissible constant-hazard mimic, five disclosed lives x four φ	a mimic reproducing the measured shape to machine precision — the shape would not be identified at any precision	residue **3.9×10^{-4}** per quarter at a ten-year life, **4.1×10^{-3}** at a three-year one
The T = 0 mass is invisible in the reported series	conditioning on T ≥ 1 against a compensating φ, five lives x three φ	any series moving after the substitution	held to **5×10^{-16}** — and the same conditioning moves α_eff by **6%**

claim	test	what would have killed it	outcome
A decreasing-hazard lag is NOT mimicked by a constant one	k = 0.5 witness at matched δ and φ	the metric fitting a world with no steady state as easily as the measured one	**5.4×10^{-3}**, a **14**× separation
Three recognition rates are three quantities	series match vs. deferral match vs. event-date MLE across the asserted rectangle	the three agreeing everywhere, making the distinction empty	agree to **5×10^{-4}** at twenty years, **15%** apart at three
The framework's guards can fail	audit of the guards themselves	a guard that could not fail passing silently	**six found and retired**, before publication, recorded in `METHOD-001`
The departure from diagonality is not an artefact of tier 0's tag list	§5.4's permutation re-derived with the omitted element restored, both arms on one crawl	the lift moving with the tag list, which would make it a property of the instrument	**4.01**× ⟶ **4.01**× and **2.01**× ⟶ **2.10**×; every cell not involving tier 0 identical to two decimals
Testing another asset first REDUCES the goodwill charge	the single-step measurement run against a published worked example	the sequenced and goodwill-first branches agreeing, which would make the ordering inert	a $850 prior charge converts a $700 goodwill impairment to **$0**; the offset is one-for-one inside the region
The suppressing channel is not visible in entity-level filings	censored slope of the goodwill charge on the other charge, by sector and by ASU 2017-04 regime, with a placebo date	a consistent negative slope, or a regime contrast the placebo could not reproduce	**failed as registered** — no consistent sign, and the placebo moved further than the true date

claim	test	what would have killed it	outcome
The disclosed trigger does not separate the two channels	rate at which filers name the standard's own internal trigger, joint-charge against goodwill-only, inside the window where ASC 350-20-50-2(a) compels the description — against a placebo arm on whom it does not	a window-versus-placebo gap wide enough to show the measure reads events rather than accounting policy	**0.436 against 0.403** on 1,833 classified firm-years; the difference is **+0.041** (*p* 0.38) or −**0.032** (*p* 0.24) according to a coding choice fixed in advance so that the sign could not be chosen afterwards
A sharper disclosure instrument finds the quantity absent, not merely unresolved	sentence-level co-occurrence of a registered trigger phrase with a *named* reporting unit, joint-charge against goodwill-only, inside the same mandated window and against the same placebo	a window-versus-placebo gap no wider than the keyword families', which would leave the null attributable to the instrument rather than to the filings	the separation more than doubles — **0.103 against 0.030**, where the families gave 0.436 against 0.403 — and the difference stays at **+0.014** (*p* 0.60) in a design that could have detected **0.068**; **0 of 281** joint firm-years name a unit and an (f)-family trigger in one sentence

Two rows carry more than their cells.

The row on the inversion sparing the lag statistic records that the identification result does not explain the registered null. The check in that row was built to show that it did and refused, in every one of 400 draws, and the claim it would have supported is not made.

The row on the guards' own audit states a fact of a different kind. The claim is not that the work was careful. It is that the guards were audited against the possibility of being unfalsifiable, that the audit found six that could not have failed, and that they are named.

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8 · Abandoned approaches

Every route below was actually taken and then abandoned. The section is placed in the body, not an appendix, and for the reason Papers I and II state at the head of theirs: a result reported without the routes that failed is a result the reader cannot calibrate — they are shown the one path that worked and left to assume it was the only one considered.

"First principles" as undeniable truths. The original formulation asserted the framework's foundations were undeniable. Abandoned on the argument of §A.1.1: an undeniable axiom is a definition, and definitions forbid nothing. Several attempts to repair it by rewording failed before the type error was identified, and that failure is instructive — a wording problem responds to wording, and this one did not.

The pure-delay reading of the reporting layer. The claim layer as a simple lag on the physical one. Abandoned because it is falsified by any case where financial signals *lead* physical change, and those cases plainly exist. Replaced by the asymmetry of §8.1, which is strictly sharper: it predicts *where* the lead-versus-lag boundary falls rather than denying that leads occur.

The over-smoothing prediction. §3.2. The framework predicted the claim layer would be less volatile than the physical layer. Measured over a whole path this is **false**, and the probe caught it. The shape of the failure is informative: the direction was wrong, the mechanism was right, and the corrected claim (relocation, with 99% of reported movement occurring inside recognition events at $\varphi = 0$) is stronger than the one it replaced.

The streak-onset instrument (PRE-001). §5.2. A genuine methodological dead end and the one part of the severe-test story that belongs in this section: an onset rule requiring an unbroken decline in a firm-level signal measures that signal's volatility rather than the phenomenon, which is why 1,047 charges were discarded and zero events were censored.

Re-specifying the onset rule until it worked. After PRE-002 also failed, a third instrument was available and was not built, because PRE-002's stopping rule had been registered in advance.

Answering the efficient-markets objection with φ. §8.1. This was the paper's reply to its most serious standing objection until the severe test removed its support, and it is abandoned in §8.1 rather than defended. It is listed here as well because the failure mode is the general one this programme keeps meeting: an unmeasured parameter partitioning a space so that the objection lands only where nothing can be checked.

Estimating φ from the reported series alone. The obvious route to closing §8.1's gap — fit the model's parameters to an observed reporting layer and read φ off the fit. Abandoned because the recovery is ill-conditioned by construction rather than by bad luck: φ reaches the observable only through the product φδ, so estimating it means dividing by an effective decay rate that is itself being estimated, and the variance grows like $1/\delta^2$. §4 gives the algebra and the measured degradation. The route is recorded because its failure identifies its own successor, and the successor has two handles rather than one: **φ becomes tractable as δ grows, and — at any δ — as δ is known more precisely from outside the reported series.** Fast-decaying assets are therefore partly self-rescuing, since at the top of the range tested φ recovers usably even with δ estimated jointly; slow-decaying assets are not, and for those an independent δ is the only handle available. What that independent determination should be is deliberately *not* specified here — an instrument named in a paper before it is registered is an instrument that has escaped its registration.

Adding a free parameter to absorb an objection. Faced four times across this programme and worth recording as a class, since each instance looked locally reasonable: introducing a scaling constant to rescue the dimensional argument; defining a levy's base so a companion paper's claim came out right; letting λ vary freely rather than in a shaped way; and, as §8.1 now concedes, leaning on an unmeasured φ — three refused, the fourth not. A quantity that can accommodate any observation forbids nothing.

8.1 · The efficient-markets reply, withdrawn — in full

The strongest standing objection to any lag thesis is efficient-markets in its plainest form: prices already discount deferred maintenance, so a systematic lag should not survive. Note that a **pure-delay** model has no answer to this and is straightforwardly falsified by cases where financial signals *lead* physical change — markets price an announced technology transition before a single unit of capital is built, and such cases obviously exist.

This model is not a pure-delay model, and the asymmetry is the whole mechanism. **The claim layer leads on what is disclosed and lags on what is deferred.** At $\varphi = 1$ the filter is a perfect window: zero lag, zero deferred information, coupling identically 1, no recognition events. So the objection holds *exactly* where the model predicts no lag, and has nothing to act on where degradation is undisclosed. The two are not in competition; they partition the space by φ.

That reply is not available to this paper.

The partition is drawn along φ, and φ is not measured anywhere in this work — §4 concedes it is swept, not estimated. So “the objection holds where φ is high and the model holds where φ is low” concedes every case an efficient-markets reader could check and claims every case nobody can, along a coordinate no one has observed. That is a free parameter absorbing an objection, which is the move this programme has refused three times in other costumes (§8) and should have refused here.

The requirement was recognised before the severe test was run and was written down: a framework conceding everything the efficient-markets reading claims and retaining only the unobserved residue **must be able to identify that residue in the world**, with evidence rather than assertion. §5 is the attempt to supply that evidence. It failed.

So §8.1 is an open problem, not a reply. The asymmetry remains a coherent structure and a genuine improvement on pure delay — a pure-delay model is refuted outright by observed leads, and this one is not. But until φ is independently measurable the asymmetry cannot be used to answer the efficient-markets objection, and this paper does not use it that way. §2’s assertion that φ is not a fudge factor rests on the argument in this section; with that argument withdrawn, the assertion is a statement of intent about how φ is to be treated, and it should be read as one.

8.2 · The crisis framing, and the paper it belongs to

The model in §2 was built to say something about crises: a deferral has one available ending — arrival — so within the model a crisis is the reporting layer delivering its accumulated error at once, and the magnitude of the discontinuity is exactly the information that had been withheld. That reading survives everything in this paper. What it does not yet have is a price, an agent, an equilibrium, or an asymmetry that a crash-risk reader would recognise as one, and that gap is real.

So the framing is not defended here and not deleted either. The material that supported it — the volatility-relocation result and its 2007 antecedent, the three-cell taxonomy locating the wedge in an incentive, a belief or the measurement rule, and the priority concessions to Jin and Myers and to Bleck and Liu — is retained in §§3.2 and 10, where it does honest work as description and attribution rather than as a claim this paper can support.

The crash paper is a later paper in this corpus, written with a price line and after the reading list in `docs/papers/paper-III-dual-tensor/POSITIONING-002-second-pass.md` §6 — which that file still marks undischarged — is discharged. **A framing that has to be argued for in the paper that introduces a theorem is a framing that will be argued about instead of the theorem.**

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9 · Limitations

1. **The severe test failed and this paper does not know why.** Three post-hoc explanations exist — the theory is wrong; the bridge was wrong; the unit of observation was wrong — and **the data do not distinguish them.**

2. **The unit mismatch is real, unfixed, and was unfixed by both registrations.** The impairment charge is asset-level; the deterioration signal used was firm-level. A firm can impair a failing reporting unit while consolidated revenue rises. Fixing this requires segment-level disclosures and is a different project with a different registration — which **may not cite the present failure as support for anything.**

3. **The filter model is deterministic and single-firm.** No stochastic degradation, no heterogeneity, no interaction between firms, no market. Every empirical signature it suggests is therefore a qualitative target, not a fitted one. **The determinism is not innocuous, and §4.8 is where it bites**: the goodwill limit reported there is a consequence of the physical layer being noiseless, and it dissolves once that layer is allowed to move for reasons other than a schedule. Admitting stochastic degradation is the single change to this model most likely to alter what it says, which is why it is named here rather than in a list of extensions.

4. **φ and θ are not measured; they are swept — and for φ the reason is §4.** α is measured, but for the quantity PRE-002's instrument dates rather than for the model's α: §5.4 estimates that rate at 0.408 per year on the registered sample, against the 0.05 swept through the body and on both known biases' inflating side, and finds the constant hazard the model assumes to be rejected. The bridge from that rate to the model's α is the one §6.2 requires of every registration, and this paper has not written it. §4.9 settles

what that rejection costs: the closed form is the recognition lag's moment generating function evaluated at the decay rate, so it survives with an effective rate that is a function of δ rather than a constant, and what the constant hazard was supplying was the domain. The paper reports how outcomes vary across the sweep and does not claim any firm's φ is known. That is no longer a concession about this construction: §4.2 establishes that **no** estimator recovers φ from a reported series, because the series does not contain it. The consequence is stated there rather than softened here, together with the one repair available — an independent determination of δ, for which disclosed useful lives are a candidate this programme has not yet used. (Method, scripts and full figures for the conditioning result that preceded the theorem: `docs/notes/NOTE-001-phi-identifiability.md`. Synthetic data only. It is **not** evidence about §5's null, which used an entirely different, non-parametric estimator.)

5. **Λ^{-1} and SDG 7.3.1 are the same quantity dimensionally, not empirically.** Λ is Appendix A's coupling — the claim measure per unit of physical measure, in currency per joule — so its inverse has the dimensions of energy per unit of recorded output. The SDG series is a national aggregate over primary energy and PPP output; the model's coupling is a firm-level ratio between a physical capacity measure and a claim measure. The correspondence licenses "this dimension is one institutions already report," not "this series measures the model."

6. **P1's domain excludes purely contractual objects** whose referent is another claim, and a large share of modern financial wealth is precisely that. The framework applies to the base of that stack, and the composition of layers above it is a question this paper does not address.

7. **A Duhem–Quine problem is present and is narrower than the usual invocation.** A failed test here cannot distinguish a false theory from a bad observability proxy — that is genuine, and §5 is an instance of it. It should not be confused with heterogeneity in entropy rates across industries, which is an ordinary identification problem with ordinary remedies and not a philosophical one.

8. **The framework claims necessary conditions, not uniqueness.** Any adequate account must distinguish physical from claim components, must let the second lag the

first asymmetrically, and must make the residue accumulate. This construction satisfies those conditions; it is not argued to be the only one that does.

9. **The diagonality of the reporting layer is an assumption, it was testable, and it is false.** §4.1 writes the reporting layer as a Hadamard product, which asserts that recognition in one asset class does not force recognition in another. Real practice couples them: a goodwill test under ASC 350-20 runs at the reporting-unit level, and the triggering event that forces an ASC 360 recoverability screen on property is frequently the same event. §5.4 puts the resulting prediction — independence across classes within a firm-quarter — to the registered sample and rejects it in both universes in the same direction, at **4.12**× and **2.02**× the independence expectation. The consequence is bounded rather than open: the Hadamard form is an approximation whose error is now measured, and §5's treatment of the events as independent draws overstates their information content by a factor this paper can state. What the design cannot do is separate an economic coupling from the sequencing the standards impose — though that sequencing, imposed by ASC 350-20-35-31 and extended to every asset class by 35-32, is itself two channels of opposite sign, one creating joint testing and one suppressing joint recognition, and §5.4 says so where the number is. The disclosure route that would separate them is closed twice over — by a selection argument, and by a count: sharpened to the sentence and to the named reporting unit, the disclosure ties the standard's own internal trigger to the unit it fired in **once in the 644 firm-years of the mandated window and not once in the 281 that took both charges** (`RESULT-REG-008` §2.2); §7 carries the second figure and §5.4 both. Any test of that route must be registered before its instrument is coded.

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10 · Relation to existing work

Soddy is the origin of the composition claim and P1 is his observation made axiomatic. The present contribution is not the distinction between physical wealth and claims on it but its *dynamics*: that the gap between them is an accumulating quantity with a measurable integral and a characteristic release.

Georgescu-Roegen is a hostile witness inside this framework's own bibliography. He is the most-cited authority in the biophysical tradition this work draws on, and he explicitly refused the physical-to-monetary reduction that a naive reading of Λ proposes: for him the source of economic value is the subjective enjoyment of life. That refusal is not answered here — it is *adopted*. Measuring the wedge between physical throughput and financial claims does not assert that energy determines value. It measures the gap Georgescu-Roegen said would exist, and treats its drift as the object of study. On this reading he is not an obstacle to the construction but the reason the construction measures a wedge rather than proposing a conversion.

Piketty is not contradicted; he is relocated. The datasets and the rigour are not in question. The disagreement is about *layer*: those measurements are taken on the claim component — the financial abstraction — and are, on this framework's reading, silent about the physical component beneath. $r > g$ may hold exactly as described at the abstraction layer while saying nothing about the atomic one. This converts the relationship from refutation into a **scope statement**, which makes the two accounts complementary rather than rival.

Austrian business cycle theory shares this framework's architecture while disagreeing about its cause, and that is worth more than agreement would be. Hayek's knowledge problem holds that prices transmit dispersed information and that the characteristic failure is *informational* rather than moral; §2 is a formalisation of exactly that concern. Mises's malinvestment — misallocated capital accumulating unrecognised through a boom, revealed and liquidated in the bust — is structurally identical to §3.2: unrecognised accumulation followed by discontinuous recognition event. The causes assigned differ (credit expansion there, undisclosed physical degradation here) and the claim made here is a structural analogy, not an identity. What it offers is a single mechanism reproducing a phenomenon that mutually hostile traditions each describe in their own vocabulary.

Stock-flow consistent macroeconomics (Godley and Lavoie) shares the insistence that accounting identities constrain dynamics. The difference is what the accounts are taken to *be*: consistent and complete there, consistent and **systematically incomplete** here, with the incompleteness being the object of study.

The efficient-markets literature, in Fama's canonical statement of it, supplies the objection §8.1 answers — and, as §8.1 now concedes, does not yet receive an answer. What the model offers is a partition rather than a refutation: the model concedes disclosed information entirely and retains only the undisclosed residue.

The stock price crash risk literature is this paper's near-

est neighbour, it arrived twenty years earlier, and the concession owed to it is larger than the contribution claimed against it. Jin and Myers model a firm whose insiders absorb firm-specific bad news up to a limit and then, when a long enough run of it arrives, give up and release the accumulated stock at once; the frequency of large negative firm-specific return outliers rises with how opaque the firm is to outsiders. Hutton, Marcus and Tehranian supplied the firm-level opacity measure and the panel, and an active literature has extended both continuously since. **That is §3.2 in a different vocabulary, and on evidence this paper is much the weaker of the two accounts**: crash risk is measured on prices, tested on large panels and supported, while §2 is a property of a simulation whose one registered prediction failed. The asymmetry is theirs before it is this paper's, too — their COUNT measure nets downside outlier frequencies against upside ones and their COLLAR trade shorts a call against a put, so the crash-not-jump direction is already inside the quantity they measure.

And the concession runs one step further than the paragraph above, into a case Jin and Myers themselves set out and this paper claims no priority over. Before their model begins, they consider an opaque firm run by "a saintly manager who always acts in shareholders' interest, never taking a dollar more or less than deserved," and ask what such a firm's returns look like. Their third possibility is this paper's mechanism:

> "If a stable lag is implausible, think of good or bad news accumulating within the firm until the difference between intrinsic value and share price reaches a critical value. The news would then be released all at once, like a pressure vessel letting off steam."

Accumulation to a threshold and release all at once, with the agency conflict switched off, was written down in 2004. No claim of priority over it is made here.

What that case retains, and §2 removes, is an informed party. "Saintly" qualifies capture and not information: the manager still observes the hidden component, and it is *investors* who "cannot see the news as it happens." Their friction is verifiability toward outsiders; §2's manager knows no more than the market does. **Deliberately withheld known news, honestly held unverifiable news, and unrecognised unknown degradation are three objects, and only the third is §2's.** Two further differences follow from the same page rather than from a defence of it. Their case is **two-sided** — "good or bad news accumulating," against a threshold on a signed difference — and they assign it long tails rather than crashes, entering kurtosis as a control variable against which their agency-driven crash results are then identified. And it has no accounting layer of any kind; the working paper contains no occurrence of *goodwill*, *intangible*, *impair*, *GAAP*, *book value* or *historic cost*. The operating asset neither depreciates nor is reinvested in — "For simplicity, we ignore depreciation and reinvestment" — by declared assumption, and the footnote attached to that assumption concedes only depreciation "according to a pre-defined schedule" — which, being common knowledge, enters value and price identically and opens no wedge at all.

That epistemic difference is observationally fragile. From the price's point of view a wedge that widens because someone will not speak and a wedge that widens because nobody yet knows resolve identically, and the obvious discriminating tests cut against §2 rather than for it: deliberate withholding predicts correlation with insider incentives, insider selling, litigation exposure and regulatory regime, and the post-SOX dissipation Hutton, Marcus and Tehranian report is exactly that pattern. Zhu (2016) runs those discriminating tests on the accruals of long-lived operating assets — the classes §5 measures — and the agency account survives them: the crash relation concentrates in the least reliable accrual components, strengthens where CFO option incentives are higher and where monitoring is weaker, and is absent in non-current operating *liability* accruals. That is the accounting layer §10 notes Jin and Myers lack, supplied for the competing explanation and not for this one.

The trend the agency account is losing its grip on is nonetheless real, and it is where a mechanism without a lying manager would matter if one were established. Andreou, Lambertides and Magidou document that idiosyncratic crash occurrences among US-listed firms rose from 5.5% of firm-years in 1950 to 27% in 2019 — 23% across the CRSP universe and 27% once the sample is narrowed to CRSP–Compustat–Execucomp — while the opacity– and overinvestment–crash relations they test come back non-significant, particularly in the period after Sarbanes–Oxley. Hutton, Marcus and Tehranian report the same dissipation in their own abstract. Those authors read their own nulls as the *conduct* declining rather than the explanation failing, and this paper does not recruit them against that reading. **What the trend establishes is that the space is open, not that §2 occupies it.**

Two things must be conceded here, or §2's claim is not narrow but wrong.

The first is that §2's asymmetry is assumed and not derived, and that the thing which would derive it belongs to

someone else. Jin and Myers obtain one-sidedness from symmetric primitives: the quantity of good news insiders can absorb is unbounded because they can capture it, and the quantity of bad news is not, so the bound is one-sided for a reason internal to the model. §2 assumes a physical layer that only degrades. That assumption is not by itself sufficient — degradation at a stochastic rate around a booked rate produces a two-signed reporting error, which is Jin and Myers' case again, long tails and no skew. What makes the wedge one-signed is §2's first condition, that reported value may fall and may not rise. **That condition is conditional conservatism, it is Basu's object, and §2 uses it as machinery rather than contributing it.** Basu is not an obstacle to be scoped around; he is the part of the mechanism this construction stands on. §2's domain restriction — to degradation on which conservatism has nothing further to bite — is the same point read from the model's side.

The second is that the reported layer accumulating hidden deterioration and releasing it as a price crash is a published result, and §3.2 quantifies it rather than discovering it. Bleck and Liu model the accounting regime itself: historical cost gives management a "veil," poor performance "accumulate[s] and only eventually materialize[s]," and greater opacity produces more frequent and more severe crashes. Their statement of the volatility result is §3.2's, nineteen years earlier and in prose — historic cost "stabilizes asset prices in the short term. Under the veil of this apparent stability, volatility actually accumulates only to hit the market at a later date," transferring volatility across time and raising it overall. **§3.2's contribution is the parameterisation, not the finding.** Their manager, however, is strategic and fully informed, keeping a project alive for a private benefit while knowing it will not recover, and their regimes are two discrete alternatives rather than a continuum. The separation from §2 is the same one that survives Jin and Myers, which is either reassuring or the last plank.

What is left is a claim about form. Beaver and Ryan decomposed the divergence between book value and economic value into a **bias** component and a **lag** component twenty-six years ago, and Bushman and Williams connect delayed expected-loss recognition to the risk profile of banks. That literature models conditional conservatism as a contemporaneous asymmetric response; §2 models it as threshold-crossing accumulation under a continuous observability parameter, which is a form that can carry a recognition lag, a jump magnitude and a location for the variance where a response coefficient cannot. **The ordering remains the whole of the claim — the recognition event is the**

cause and the price crash is the effect — and it remains a research programme and not a finding.

On pre-registration and severe testing, this paper is a straightforward application of the standard argument — that a prediction's evidential weight depends on the test having had a real chance to fail — to a domain where the practice remains uncommon. The lineage is Popper's demarcation — that a claim earns its standing from the risk it ran — Mayo's severity requirement, which makes that standing quantitative by asking how probable it was that the test would have detected the error had it been present, and the preregistration case made by Nosek and colleagues. Mayo's account is cited here at its origin rather than at its restatement, and the volume in which she and her critics argue it out directly is listed beside it, for the same reason this paper ships with an adversarial referee report on itself (§11): an argument is easier to judge when its objections are in the room. The registration, the negative control, the power analysis and the stopping rule are all conventional, and are claimed as nothing more than that.

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11 · Data and code availability

Every simulation result in §A.2, §§2–3 and §4 is produced by open code, and the bullets below name the command that prints it. Where a figure is printed by no command here, the bullet says so. The severe test in §5 uses only public data.

- **Repository:** `https://github.com/jasoncbraatz/wealth-tensor` (public)
- **Modules:** `src/wealth_tensor/lag.py` · `src/wealth_tensor/lambda_sensitivity.py` · `src/wealth_tensor/edgar.py`
- **Regenerate §3 (and §A.2.4):** `python3 scripts/wt027_report.py` — **five blocks**: §3.1's table, §3.1's three prose-only figures (the sigmoidal lag at $\varphi = 0.9$ and $\varphi = 0.1$, and D(0) to four decimals), §3.2's table together with the full-path volatility ratios §3.2 quotes in prose, the entropy-rate sweep, and §A.2.4's sawtooth. Every simulation figure §3 states, tabulated or in prose, is printed by that one command. §2 states the model and reports no simulation result.
- **Regenerate §A.2.3:** `python3 scripts/wt002_lambda_report.py`
- **Regenerate §4.2:** `python3 scripts/wt084_identification_closed_form.py` — the two coeffi-

cient equations, the mirror agreeing to $\mathbf{2 \times 10^{-16}}$ where the rival map fails by fourteen orders of magnitude, and §4.2's one-parameter family: the factor of **1.67** in the unobserved physical scale, the implied φ at each assumed scale, and the **51%** opening gap the φ = 0 end requires. The **31.7%** share is printed by no command here: it is that same family restricted to opening gaps within ten per cent, and the restriction is applied in §4.2's prose rather than in the script.

- **Regenerate §4.4 and §4.5:** `python3 scripts/wt083_tier_ladder_antialignment.py` prints §4.4's four-tier ladder at the calibration — φ, δ, (1 − φ)δ and **R** both simulated and in closed form — with both Kendall τ and the log-decomposition §4.4 reads the step direction from. `python3 scripts/wt088_disclosed_ladder.py` prints §4.4's crossing rate $\boldsymbol{\delta_3}^*$ = **0.007895**, the asserted rectangle's admissible shares, and §4.5's ladder statistics under both draws — the lag ordering and the magnitude measure, with the durability ordering imposed and dropped, at each ladder count §4.5 quotes. The tier table's two measured-rate columns are printed by neither: they are §4.4's own $R = (1 - \varphi)\delta/(\alpha - \delta)$ evaluated at §5.4's $\hat{\alpha}$ and at the adverse cut, over the φ and δ the first columns carry.
- **Regenerate §4.6, §4.7 and §4.8:** `python3 scripts/wt085_returns_conditioning.py` is §4.6's run: returns break the root swap, and the scale continuum survives them because a return is a ratio. `python3 scripts/wt086_exponent_robustness.py` re-fits §4.7's two exponents across nine regimes, which is why §4.7 quotes neither as a constant of the problem. `python3 scripts/wt087_goodwill_gradient.py` prints §4.7's δ-sweep at a fixed root gap — the volatility exponent from **−0.386** at a property-like rate to **+0.163** at a goodwill-like one — and §4.8's arithmetic, the level moving by **6.81**× as (α − δ)^−0.700.
- **Regenerate §4.9:** `python3 scripts/wt090_age_dependent_alpha.py` prints the constant-hazard pole at **0.435** per year, the overstatement column across the disclosed lives — **0.55%** at a forty-year life to **43.87%** at a three-year one — α_eff across that same rectangle, and δ_3^* at both shapes, **0.00755** published and **0.00754** measured, beside the **0.00789** §4.4's closed form returns at the calibration.
- **Regenerate §4.10:** `python3 scripts/wt091_lag_shape_identifiability.py` — ladders I, P, W, S and N exactly as registered in `docs/preregistration/REG-005-p3-lag-`

`shape-identifiability.md`, committed at **6f0e7be** before that script existed. It prints §4.10's precision table, the identified set at every reported precision, the search's own floor at the fitted shape, and the profile at the constant hazard. Minutes on a commodity CPU.

- **Regenerate §5:** `python3 scripts/wt026_severe_test.py --universe pilot --onset peak` and `--universe replication --onset peak`. **That command reproduces the instrument and not the sample.** It re-pulls `companyfacts`, which serves each firm's *latest* view of its own history. §5.3's 244 and 444 events come from the original pull, whose full output is committed at `docs/preregistration/RESULT-002-pilot-run.log` and `RESULT-002-replication-run.log` — those logs, not a command, are the record of §5.3. The 2026-08-12 rebuild is committed as data (below) and returns Jonckheere–Terpstra z = **−0.223** and **−0.083** against §5.3's −0.290 and −0.095; a pull later still returns different counts again, and once the replication z crosses zero, §5.3's observation that both statistics are negative is a fact about the analysed sample rather than about any re-pull. Nothing in this repository re-derives §5.3's figures from committed data.
- **Regenerate §5.4:** `python3 scripts/wt089_recognition_and_offdiagonal.py`. §5.4 asks two questions of §5.3's sample that it was not collected for, and neither is answered by the command above: `wt026_severe_test.py` is §5.3's instrument and prints nothing about the recognition rate or the off-diagonal. This command re-pulls `companyfacts` as §5.3's does, so it reproduces the instrument and not the sample; its own §2 reconciles the rebuild against `RESULT-002` before it computes a statistic.
- **Test suite:** `python3 -m pytest tests/ -q` runs the whole repository; at the pinned commit **d655501** — the state that produced every result in §A.2 and §§2–3 — that suite held **100** tests, of which the **62** in `tests/test_edgar.py`, `tests/test_lag.py` and `tests/test_lambda_sensitivity.py` are the ones that hold this paper's claims in place. The remaining 38 hold the companion papers' claims and are named here only because a suite total is a property of the repository and not of any one paper in it. Both counts are derived from the repository rather than asserted, and `tests/test_paper_test_counts_are_derived.py` fails if either drifts. The suite at the head of the repository is much larger — it grows with every registration in `docs/preregistration/`, which is why the paper-scoped count is the one quoted; **six** of its addi-

tions, all in tests/test_lag.py, guard claims this paper makes and change no model code: two for §3.1's closed form $D(\varphi) = (1 - \varphi)\cdot D(0)$ and its accompanying negative claim that the lag is *not* linear; one asserting the algebraic collapse §4 publishes — which had no test until an audit found the published form using the entropy rate where it meant the effective decay; and three added at commit **cc1d198** for §4.4, holding the crossing rate δ_3^*, the first-rung boundary, and the absence of a steady-state deferral measure once decay outruns recognition.

- **Hardware:** none required. Every figure in §A.2 and §§2–3 regenerates on a commodity CPU in seconds. The fits reported in §4 use two thousand synthetic firms at four hundred gradient steps in double precision; a larger reference fit of ten thousand firms at three hundred steps completed in **76 seconds on two 2.8 GHz cores**, a machine chosen deliberately so the figure is an upper bound. No accelerator is used and none is needed.
- **Empirical data source:** SEC EDGAR companyfacts (XBRL), and the SEC Financial Statement Data Sets for the CIK⟶SIC mapping including dead registrants. No proprietary or restricted data is used.
- **Pre-registrations:** docs/preregistration/PRE-001-wt026-observability-lag.md, registered at commit **9722342** — a single-file commit containing the registration and nothing else · docs/preregistration/PRE-002-wt026-peak-to-charge.md, registered at commit **d655501**, which **also contains the implementation of PRE-002's instrument**; see the disclosure in §5.1. Results and full run logs are in the same directory.
- **Code state for the results reported here:** the per-file pins — src/wealth_tensor/edgar.py at commit **d655501**, src/wealth_tensor/lag.py at **ad779eb** (its only commit) and src/wealth_tensor/lambda_sensitivity.py at **b9089c7** — each verifiable with git log -1 --format=%h <sha> -- <path>. src/ as a whole has moved since, on companion-paper modules and, at **93a159b**, on one addition to edgar.py that appends the REG-006-corrected tier-0 tag list beside the registered one without editing it: the TIER_TAGS block that selected §5's published sample is byte-identical at **d655501** and at the head of the repository. No head-of-repository SHA is pinned; the per-file pins above are what a replicator needs, and each is verifiable now.
- **Data retrieval, pinned.** companyfacts serves each firm's *latest* view of its own history, so a re-pull is not the original

pull and the sample grows with every filing — §5.4 reports 688 ⟶ 695 over one week. The events analysed here were retrieved on **2026-08-12** and are committed rather than re-fetched, at commit **0569ab6**: `data/pre-002-events.json`, SHA-256 **`974d156b53dfb915f48bdc3df99d7f2f2dd146427fa537fd7bdff4a503006bf0`**, and `data/pre-002-riskset.json`, SHA-256 **`60627429d937bc42b9cf96a1be4bcfb78d198948a3212ac8f8c5e4412a436cce`**. Charges in that file run **2012 Q2 – 2026 Q2** and onsets **2010 Q1 – 2026 Q1**; the 2013–2024 window selects registrants, not events.
- **Drop accounting for §5**, as required by the registrations: the attrition from candidate charges to the 688 analysed events is in the run logs at `docs/preregistration/RESULT-002-*-run.log`, one log per universe, counted per drop bucket. It is reported there whether or not it flatters the result. **The count is per bucket and not per tier.** `edgar.py`'s `drops` counter is keyed by bucket alone, so *does attrition differ systematically by tier?* — the one selection channel capable of manufacturing the reported null — is not answerable from these logs. What the registrations put in its place is the label-permutation control printed in those same run logs (`docs/preregistration/RESULT-002-*-run.log`), which permutes the tier labels while holding the lag distribution fixed; in the pilot its null runs at mean **+0.007**, standard deviation **1.025** over 1 000 draws, against an observed z of **−0.290**.

Two tests are worth naming because of what they forbid rather than what they check. `test_pre001_constants_are_what_was_registered` fails if any registered constant is edited, so a registration cannot be amended by stealth. Companion modules in this programme carry `test_excess_demand_is_monotone_here_so_this_is_not_an_SMD_result` and `test_a_flat_gini_does_not_mean_a_bounded_one`, both of which exist solely to make overclaiming fail loudly.

Known limitations of this review. The model results are checked mechanically — the closed forms against simulation, the exchange against its mirror, the invariances against a numeraire sweep, and all of it against a test suite that fails if a registered constant is edited. The empirical results rest on one pull of a live endpoint, committed rather than re-fetched, and §11 says plainly that nothing here re-derives §5.3's figures from committed data. The reference list, the cross-references and the section pointers were checked by repeated independent reads rather than by any instrument, and the per-entry marks record how far each verification reached. What was not checked at all: whether attrition differs systematically by tier, which §11's drop accounting cannot answer and which is the one selection channel capable of manufacturing the reported null.

The repository's `docs/` directory is public and holds the registrations, the run logs, the failed result, the reasoning that led to the stopping rule, and an internal adversarial referee report on this draft with the author's responses.

Appendix A · The framework the filter was built inside

*Three propositions about the composition of wealth, and the coupling they oblige. This material motivated the filter: it states the domain within which §2's two layers are the right two layers, and it carries the invariance evidence the ledger in §7 cites. It is an appendix rather than a section because **nothing in §§2–7 depends on it.** The identification result holds for any two-layer filter of the stated form, whatever one believes about the composition of wealth — and a result that needs a metaphysics is weaker than one that does not.*

A.1 · Three propositions

A.1.1 · What a first principle is, and what it is not

An appeal to "first principles" is worthless without a definition of the term, and the gap is not a wording problem — it is a type error, and type errors do not respond to prose:

An axiom is a proposition — truth-apt, deniable, the kind of thing that can be false. **A model is a structure** — it has interpretations, not a truth value. A structure cannot be promoted to a proposition by describing it more emphatically. The tensor is not the axiom. **The axiom is the proposition that wealth has the structure the tensor formalises.**

The second correction is that **"undeniable" must go.** An axiom nobody can deny is a definition, and definitions generate no empirical content. The useful notion is the one from computing: a **invariant** — never proved undeniable, proved *preserved*, within a **stated domain**. "Sound within a stated domain" is a defensible claim. "Undeniable" is a self-defeating one.

The test that separates a first principle from a result: **denying a first principle produces a different science; denying a result produces a wrong number.** Deny $r > g$ and you have a different empirical claim within the same science.

The example on the other side needs stating carefully, because the obvious version of it is a strawman and an economist referee would stop on it. **Neoclassical economics does not deny physical depreciation** — δK appears in every growth model from Solow forward, and claiming otherwise would be an error of exactly the kind §6.2 is about. What P3 puts at issue is different and narrower: whether an aggregate can be treated as a primitive carrying its own laws. Deny P3 — assert that an aggregate production function is a fundamental object rather than a fold over units whose validity requires conditions on the units — and the disagreement is not about a coefficient. It is about what kind of thing the object of study is, and it changes which questions are well-formed. That is what “a different science” is meant to pick out, and P3 rather than P2 is where this framework's commitment actually bites. The three propositions below are offered as principles on that test.

A.1.2 · The propositions

> **P1 · Composition.** Every unit of wealth is a compound of a physical component and a claim component, obeying different laws — thermodynamic and arithmetic respectively.
>
> *Domain:* units of wealth having any physical referent. Silent on purely contractual objects whose referent is another claim.
>
> **P2 · Decay.** The physical component degrades absent maintenance. No store is inert.
>
> *Domain:* physical referents over horizons long relative to their maintenance cycle. Silent on the short run, where degradation is negligible against measurement noise.
>
> **P3 · Atomism.** Measured aggregates are folds over units. No aggregate is more fundamental than its constituents.
>
> *Domain:* any measurement presented as a property of an economy rather than of a population. This is the proposition an aggregate-production-function economist denies, and denies knowingly.

Three, not ten. Each is stated so that a competent economist can say *no* to it and mean something specific by the refusal.

A.1.3 · The propositions are deniable, and this repository proves it

The claim that these are empirical rather than definitional is cheap to make and is usually made by assertion. Here it is demonstrated for **one** of the three, because the regime in which P2 fails is committed, tested code rather than a thought experiment. P1 and P3 are argued deniable in §A.1.1 and §A.1.4 and are not demonstrated in code, and the second item below is the mechanism's own switch-off rather than a proposition:

- **P2 fails at complete maintenance.** Effective decay is the entropy rate *net of maintenance*, so a fully maintained asset has no dynamics at all and the model collapses to an identity. This is the regime in which "no store is inert" is simply not true, and it is reachable by setting one parameter.
- **The framework's own central mechanism switches off at φ = 1**, which is a separate and equally important point. Perfect observability annihilates the entire phenomenon: recognition lag 0, deferred information 0.0, coupling identically 1, zero recognition events. Note carefully what this does *not* say — P2 still holds at φ = 1, and the physical layer still decays (from E_0 = 100 to 0.031 over 400 periods). What vanishes is the *gap*, and therefore everything this paper is about. A framework whose subject matter can be dialled to zero by an observability parameter is a framework making a claim about the world rather than a definition of it. These regimes are not embarrassments to be hidden behind a stronger word. **They are the evidence that the model has degenerate limits reachable by setting a parameter** — which is weaker than proving a proposition about the world false, and is stated at that strength deliberately. What a switch-off regime demonstrates is that the framework's subject matter is contingent on a quantity that could take another value, which is the minimum a claim must satisfy to be empirical rather than definitional. It is not itself a refutation, and no refutation is offered here.

A companion result on the same theme is cited rather than reproduced: ***Paper II*** *of this programme reports that a levy whose base cannot observe an accrual is inert regardless of its rate, and the evidence for it belongs to that paper. The identification of the two mechanisms that the shared theme invites is withdrawn. Put to a cross-scale check it does not hold: what this paper's filter fails to recognise is deferred, held in the gap and released at rate α, while what a levy's base fails to recognise is never assessed at all, and a* *levy has no parameter that plays α's part. The two results share the question and not the operator. The check, and the fact that the withdrawal was written down before it was run, are recorded in* `docs/RESULT-END-TO-END-001-E1.md`.

A.1.4 · Independence

P1 concerns composition and is silent about time. P2 concerns time and presupposes only that a physical component exists — it is not derivable from P1, since a compound whose physical component were inert would satisfy P1 and violate P2. P3 concerns the relation between measurements at different scales and is independent of both: an economy of inert single-component units would satisfy P3 while violating P1 and P2. No proposition is derivable from the others, and each is denied by an identifiable school.

A.2 · The coupling

This section defends Λ, at full strength and on three independent legs; the rest of the paper then uses it without re-arguing it.

A.2.1 · Λ is obliged by P1, not introduced

Notation, stated before the argument because two different objects are easily conflated under one symbol. Write **C** for the claim component and **E** for the physical component. Then:

- **λ = C/E** is **dimensionless** — a ratio of the claim measure to the physical measure once both are expressed in the same numeraire. This is the object §A.2.4 reports as a sawtooth, and it is the one with dynamics.
- **Λ = η·C/E** is **dimensional**, carrying units of currency per joule, where η is the numeraire conversion. This is the object the standing dimensional objection is aimed at, and the object §A.2.3 sweeps.

Conflating them is easy. Everything below is explicit about which is meant.

The entailment argument, and its actual reach. If P1 holds, wealth is a compound of two components measured in different units, so *some* relation between them exists in any unit of wealth. That much is entailed and it does useful work: it establishes that the framework is not smuggling in an extra object, and that asking "why did you introduce a coupling?" mistakes a consequence for a choice.

It does not, however, entail that the relation is a scalar, and the standing objection is aimed at the scalar. The relation could be state-dependent, non-stationary, set-valued, or not a function at all — and P1's own wording, *obeying different laws*, is if anything a reason to expect that no single constant suffices. So the honest version of this leg is narrower than the version this programme has previously stated: **the existence of a coupling is entailed; its representability as a scalar is an additional modelling assumption, and it is one this paper makes and does not prove.** What follows in §A.2.3 is a demonstration that no conclusion here depends on the scalar's *value*, which is a different and weaker guarantee than showing the scalar is the right object — and the difference is exactly the kind §6.2 is about.

λ is not stable, and §A.2.4 shows what shape its instability takes.

A.2.2 · Λ^{-1} is an indicator the United Nations already publishes

Energy intensity of output — the World Bank series *Energy intensity level of primary energy* (`EG.EGY.PRIM.PP.KD`, MJ per unit of PPP GDP) — is formally **SDG indicator 7.3.1**, co-tracked by the International Energy Agency, with global coverage and a long time series.

That series has **the dimensions of Λ^{-1}**, and the claim made here is exactly that and nothing more. It is emphatically **not** that SDG 7.3.1 measures Λ^{-1}. The two differ in a way this paper is obliged to name, since naming it is the discipline §6.2 arrives at the hard way: **Λ is a ratio of two stocks** (a claim stock in currency over a physical stock in joules), while **SDG 7.3.1 is a ratio of two flows** (annual primary energy over annual PPP output). Two quantities can share dimensions and remain different quantities, and a paper that lost a pre-registered test to precisely that error is in no position to commit it a second time in its own defence.

What survives the qualification is narrow and is still worth stating: **currency-per-energy is not an exotic dimension and not this author's coinage.** An institution with no stake in this framework tracks a quantity of that dimension against a global target, which places the construction outside the position that proved fatal to Odum's emergy programme — transformity coefficients derived from the accounting system that consumed them, and therefore unmeasurable from outside it. That is a claim about *availability in principle*, not about measurement in fact.

This leg is weaker than §A.2.3's. If, as §A.2.3 demonstrates, no conclusion in this paper depends on the coupling's value, then anchoring that value to a published statistic cannot be load-bearing for any result here. The two legs answer different objections — §A.2.3 answers *"your findings are an artefact of a number you made up"*, and this section answers *"the dimension you are working in is invented"* — and only the first is doing work for the results. A reader who finds this section unconvincing loses nothing downstream.

A.2.3 · The numeraire cancels — measured, not argued

The dimensional objection can be answered by algebra, and algebra is exactly what a sceptical reviewer declines to take on trust. So the two-layer system of §2 is **dressed in units** — the physical layer in joules at scale E_0, the claim layer in currency, the coupling η between them — and the invariance is measured on the dressed system.

Sweeping η across twelve orders of magnitude, from 10^{-6} to 10^{+6} currency units per joule, at **φ = 0.3, recognition mechanism live, 400 periods** (the diagnostics below are the same statistics §3.2 reports, under **this paper's names**; the module and `scripts/wt002_lambda_report.py` call them `variance_suppression`, `variance_concentration` and `n_crises`):

diagnostic	value at every η	spread across the sweep
recognition lag	22	**0.0**
inter-event smoothing	0.6097	**0.0**
share of reported movement inside recognition events	0.9199	**0.0**
recognition events	16	**0.0**
relative event magnitude	0.20138	**0.0**
mean / min / terminal coupling ratio	—	**0.0**

Not "within tolerance." **Bit-identical**, because the coupling never enters the recursion; it is dressing applied afterwards.

That result alone would be worthless, and it is important to say why: it is trivially easy to build a module in which nothing depends on a parameter *because the parameter is never used.* So the positive half is what makes this a test rather than a tautology. The dimensional quantities **do** move, and they move exactly as a unit conversion must:

quantity	at $\eta = 10^{-6}$	at $\eta = 10^{+6}$	log-log slope
deferred information (currency)	6.323144×10^{6}	6.323144×10^{18}	**1.000000000000**
terminal Λ	1.0×10^{-6}	$1.0 \times 10^{+6}$	**1.000000000000**

η is used, the currency figures track it linearly to twelve decimal places, and no conclusion moves at all. Both directions are mutation-tested: leaking η into the dynamics fails four tests, and removing the scaling fails two.

Scaling collapse. Two systems differing in energy scale (1 J against 6.02×10^{23} J) *and* in coupling (10^{-6} against 42) lie on a single dimensionless curve — every diagnostic identical, pairwise difference **exactly 0**, at φ = 0.3 over 300 periods. (The shorter horizon is stated because it changes the values: at 300 periods the system has had 12 recognition events rather than 16, and inter-event smoothing reads 0.6100 rather than 0.6097. The *collapse* is horizon-independent; the numbers collapsed onto are not.)

The sentence this licenses, and the paper will not need to say it twice: *the conversion coefficient is a numeraire; every result reported here is invariant to it across twelve orders of magnitude, while every currency-denominated quantity scales with it exactly linearly.*

A.2.4 · λ is not a constant that wobbles; it is a sawtooth

Throughout this section the object is $\boldsymbol{\lambda = C/E}$*, the dimensionless ratio of §A.2.1 — not the dimensional* $\boldsymbol{\Lambda = \eta \cdot C/E}$ *swept in §A.2.3. The numeraire enters nothing below.*

A freely-varying λ that is never pinned would forbid nothing, and a quantity that forbids nothing is the free parameter this programme has refused three times in other costumes. So the claim is not that λ *varies.* It is that λ varies **in a specific parameterised shape**, and the shape is a prediction.

At φ = 0.3 over 400 periods, with the recognition mechanism live:

	value
mean λ	1.136838
minimum λ	1.000000
maximum λ	1.245384
recognition events	16
λ = 1 exactly at every recognition event	**yes, all 16**

λ equals its physical value only at the instants the claim layer snaps to the physical one, and overstates it by ~14%

across the run. Floor pinned at unity by construction of the recognition event; ceiling set by observability; mean determined by φ. That is a shaped variable, not a free one — and it is the picture of the assertion that λ's drift *is* the accumulated deferred information.

References

How to read this list. **The edition cited is the edition *consulted* — the copy in the author's possession — not the earliest printing a catalogue happens to list. Where the original's date does argumentative work, because the entry is a translation or because a claim about priority rests on it, the entry is** dual-dated** `original/consulted`. A reprint that changes no pagination is a *printing*, not an edition, and is not dual-dated. Where the copy read was a working paper or an accepted manuscript rather than the typeset article of record, the entry is dual-dated in the other direction — `consulted/published`.**

Each entry carries a mark recording how far verification reached, because a bibliographic record and a text are different objects: a work can exist exactly as cited and still not contain the sentence attributed to it. Bibliographic verification was carried out on 2026-08-10, and the crash-risk entries were added on 2026-08-11. Per-entry findings are in the note attached to the entry they describe.

✓ — checked against a publisher page, a library-catalogue record, a Crossref record or the issuing body's own documentation, not recalled. ✓✎ — additionally checked against **the author's own copy**, by reading that copy's title page and colophon. The ✓✎ entries are the ones where doing so changed the citation. ✓⧗ — bibliographically verified, but the **text** consulted is a pre-publication version; any quotation is attributed to the version read and may not appear in the article of record; three entries carry it. ⧗ *alone* — the bibliographic record is verified and the **text was not read**; the characterisation rests on named secondary sources, and the entry says so in its own note. Three entries carry it. Two entries carry **no mark at all**, each stating in its own note why it is unmarked, and those two are the only unmarked entries in the list.

Andreou, P. C., Lambertides, N., & Magidou, M. (2023). A critique of the agency theory viewpoint of stock price crash risk: the opacity and overinvestment channels. *British Journal of Management*, 34(4), 2158–2185. ✓ *(Open access. The copy consulted is the publisher's own typesetting, deposited by the authors' institution,* *whose EarlyView pagination runs 1–28 and therefore does not match the issue pagination given here; quotations from it are cited without page numbers for that reason. §10 takes both the 5.5%⟶27% figure and the universe split from it directly: the 27% is the CRSP–Compustat–Execucomp sample and the CRSP-wide figure is 23%, a distinction the abstract does not make and the body does.)*

Basu, S. (1997). The conservatism principle and the asymmetric timeliness of earnings. *Journal of Accounting and Economics*, 24(1), 3–37. ✓ *(Cited for the asymmetric-timeliness result its title carries verbatim —* the asymmetric timeliness of earnings. ***Read at source****; the volume, year and page range are confirmed against the typeset article. Nothing is quoted from it.)*

Ball, R., Kothari, S. P., & Nikolaev, V. V. (2013). Econometrics of the Basu asymmetric timeliness coefficient and accounting conservatism. *Journal of Accounting Research*, 51(5), 1071–1097. ✓ *(§4.6 cites it for its stated expectation that firms with shorter asset maturity exhibit lower timely loss recognition, and for reading that dependence as the measure behaving correctly.* ***Read at source****, and the characterisation held: the passage sits in their §4.4, headed "Other Determinants of Conditional Conservatism," and the expectation is stated of "companies with short operating cycles, short investment cycles, or short asset maturity." The determinant reading is theirs and is explicit — they conclude that the measure "is unbiased under the null hypothesis of zero asymmetry, and that under the alternative hypothesis it captures conditional conservatism," in direct rebuttal of the invalidity critiques. Asset maturity is one of several examples they give of a comparative static, not their headline; §4.6 is worded accordingly.* ***No page is cited****: the text consulted was a full-text copy reporting itself as the published article, not the typeset original, and the MIT deposit (handle 1721.1/87767 —* not* 87766, which is the different 2013 paper in *The Accounting Review*) refused every retrieval route attempted, so nothing is quoted beyond the two phrases above and no absence is claimed of the typeset article.)*

Bateman, H. (1910). The solution of a system of differential equations occurring in the theory of radioactive transformations. *Proceedings of the Cambridge Philosophical Society*, 15(V), 423–427. ✓ *(Cited for the function that bears its name and for nothing else; §4.2 characterises only the functional form, which is standard. The bibliographic record is from catalogue listings rather than the author's own copy, and no text is quoted.)*

Beaver, W. H., & Ryan, S. G. (2000). Biases and lags in book value and their effects on the ability of the book-to-market ratio to predict book return on equity. *Journal of Accounting Research*, 38(1), 127–148. ✓ *(Cited for the bias/lag decomposition its title carries verbatim —* biases and lags in book value. *§10 identifies* *this as the closest prior art to §4's filter, so the entry is load-bearing against this paper rather than for it.* ***Read at source****; §4.7 quotes their method — regressing the ratio "on the current and six lagged security returns with fixed firm and time effects" — from* ***p. 128****. The sentence recurs at p. 135 as "six lagged* annual *security returns", which is not the wording quoted. That design is this paper's returns repair carried out twenty-six years earlier, and the decomposition is theirs: Ryan (1995) supplies the regression and assumes conservatism away.)*

Beaver, W. H., & Ryan, S. G. (2005). Conditional and unconditional conservatism: concepts and modeling. *Review of Accounting Studies*, 10(2–3), 269–309. ✓ *(****Read at source.*** *Cited in §4.6 as the nearest accounting-native ancestor of the present confound: their preemption mechanism runs a depreciation schedule against measured conditional conservatism explicitly. One sentence is quoted, from their development of the tangible-asset case. Theirs is a signed comparative static and not an identification claim, and §4.6 says so rather than recruiting it.)*

Bellman, R., & Åström, K. J. (1970). On structural identifiability. *Mathematical Biosciences*, 7(3–4), 329–339. ✓ *(Cited in §4.2 for the definition of structural identifiability and the transfer-function criterion, which is what the source is characterised on. Characterised at abstract level; the pole-set consequence drawn in §4.2 is this paper's statement of the mechanism, not theirs.)*

Barlow, R. E., Marshall, A. W., & Proschan, F. (1963). Properties of probability distributions with monotone hazard rate. *The Annals of Mathematical Statistics*, 34(2), 375–389. ✓ *(§4.9 cites Theorem 6.3, which is quoted in the paper in the form used here: the moment generating function is finite below the limit inferior of the hazard rate and infinite above its limit superior. That theorem is what turns this model's $\alpha > \delta$ into a statement about the recognition lag's tail. The paper's equation (6.2) — a hazard bounded between two constants bounds the survival function between the corresponding exponentials — is the same result in the form a reader may find more familiar.)*

Bleck, A., & Liu, X. (2007). Market transparency and the accounting regime. *Journal of Accounting Research*, 45(2), 229–256. ✓ *(Read in full text; the copy consulted carries the journal's own title page — vol. 45 no. 2, May 2007, DOI 10.1111/j.1475-679X.2007.00231.x — so it is the typeset article and not a pre-publication version. §3.2 and §10 both cite it against this paper: it states §3.2's volatility result nineteen years earlier.)*

Bushman, R. M., & Williams, C. D. (2015). Delayed expected loss recognition and the risk profile of banks. *Journal of Accounting Research*, 53(3), 511–553. ✓

Elbers, C., & Ridder, G. (1982). True and spurious duration dependence: the identifiability of the proportional hazard model. *The Review of Economic Studies*, 49(3), 403–409. ✓ *(§4.10 cites it for what identification of a hazard's shape costs elsewhere in the literature: a regressor with variation, a proportionality restriction and a moment condition on the mixing distribution.* ***The text was not consulted.*** *The bibliographic record is verified; the characterisation is taken from two independent secondary sources that state it identically, and the entry claims nothing beyond it. The point §4.10 draws is that a single reported series supplies none of the three, which is why that section measures rather than cites.)*

Fama, E. F. (1970). Efficient capital markets: a review of theory and empirical work. *Journal of Finance*, 25(2), 383–417. ✓

Financial Accounting Standards Board. *Accounting Standards Codification*, Topic 350 — *Intangibles — Goodwill and Other*; Topic 360 — *Property, Plant, and Equipment*; Topic 280 — *Segment Reporting*. ✓

Garrett, E. R. (1994). The Bateman function revisited: a critical reevaluation of the quantitative expressions to characterize concentrations in the one compartment body model as a function of time with first-order invasion and first-order elimination. *Journal of Pharmacokinetics and Biopharmaceutics*, 22(2), 103–128. ✓ *(Cited in §4.2 as the bridge between the Bateman function and the flip-flop phenomenon, which its own title and abstract establish. Characterised at abstract level; nothing is quoted.)*

Georgescu-Roegen, N. (1971). *The Entropy Law and the Economic Process*. Harvard University Press. ✓✎ *(The copy consulted is the Harvard Paperback second printing, 1974, ISBN 0-674-25781-2; a printing is not an edition, so no dual date.)*

Godley, W., & Lavoie, M. (2007). *Monetary Economics: An Integrated Approach to Credit, Money, Income, Production and Wealth.* Palgrave Macmillan. ✓✎ *(Copy consulted confirms first published 2007, ISBN 978-0-230-50055-6.)*

Hayek, F. A. (1945). The use of knowledge in society. *American Economic Review*, 35(4), 519–530. ✓

Khan, M., & Watts, R. L. (2009). Estimation and empirical properties of a firm-year measure of accounting conservatism. *Journal of Accounting and Economics*, 48(2–3), 132–150. ✓ *(§4.6 cites it both for C_Score and for its reported association between longer investment cycles and higher measured conservatism. Characterised at abstract level.)*

Kuan, I. H. S., Wright, D. F. B., & Duffull, S. B. (2023). The influence of flip-flop in population pharmacokinetic analyses. *CPT: Pharmacometrics & Systems Pharmacology*, 12(3), 285–287. ✓ *(Cited in §4.2 for the classification of flip-flop as a failure of global* *rather than local identifiability.* ***Read at source*** *(PubMed Central PMC10014047). They write of* local *identifiability rather than of a failure of global identifiability, and §4.2 uses their adjective. Their concluding sentence qualifies the "finite set" formulation as "not just a finite set of parameter values but a partial permutation of the set"; §4.2 carries that qualification too. Two short phrases are quoted; no page is cited, the article running to three pages without internal pagination in the deposit consulted.)*

Dutta, S., & Patatoukas, P. N. (2017). Identifying conditional conservatism in financial accounting data: Theory and evidence. *The Accounting Review*, 92(4), 191–216. ✓⧗ *(Cited in §4.6 as the nearest existing claim and the one most needing separation. The* ***text*** *consulted is the open UCLA Anderson working-paper version, read in full for the decomposition, the three named confounders and the accrual-variance-spread repair; the displayed algebra rendered unreliably in that copy, so nothing interior to their coefficient B is asserted here and no page is cited. Pagination of the published article is verified bibliographically and has* ***not*** *been checked against the working paper's.)*

Fisher, F. M., & McGowan, J. J. (1983). On the misuse of accounting rates of return to infer monopoly profits. *American Economic Review*, 73(1), 82–97. ⧗ *(Cited in §4.4 for the shape of its confound — a reporting-rule parameter against an asset-life parameter inside a published ratio — and for the fate of the inference drawn from it.* ***Not read****; the record is verified and the characterisation rests on the Long and Ravenscraft comment below, whose working-paper version was read in full, and on secondary accounts. No quotation is taken from it.)*

Hayn, C., & Hughes, P. J. (2006). Leading indicators of goodwill impairment. *Journal of Accounting, Auditing & Finance*, 21(3), 223–265. ✓ *(§4.9 cites it for the two figures its abstract states: goodwill write-offs lag the economic impairment of goodwill by an average of three to four years, and for a third of the companies examined the delay extends up to ten. Cited for the existence and length of the delay, which it establishes, and not for its shape, which it does not model. The page range is the publisher's landing-page range and was not checked against the typeset issue.)*

Jorgenson, D. W. (1966). Rational distributed lag functions. *Econometrica*, 34(1), 135–149. ✓ *(§4.10 cites it for the density result — that an arbitrary distributed lag may be approximated to any desired accuracy by a rational lag function, of which a constant hazard is the lowest-order member — which is the reason REG-005 predicted the shape would be invisible. Verified at abstract level against the Econometric Society's own record, from which the approximation claim is taken verbatim; the body was not read and nothing else is* *attributed to it.)*

Lanczos, C. (1956). *Applied Analysis.* Englewood Cliffs, NJ: Prentice-Hall. *(§4.10 cites the exponential-decomposition example at pp. 272–280 for the classical ill-conditioning of exponential-sum fitting.* ***Not read****, and the entry carries no verification mark for that reason: every copy located was lending-restricted and no full text was obtained. The page range is from the NIST Statistical Reference Datasets documentation of the same example, and everything §4.10 draws from it is drawn through Varah (1982), which quotes it with page citations. Nothing here rests on it alone.)*

Little, J. D. C. (1961). A proof for the queuing formula: L = λW. *Operations Research*, 9(3), 383–387. ✓ *(§4.9 cites it for the distribution-free stationary identity — average number in system equals arrival rate times average time in system — and, more to the point, for what that identity requires: a constant arrival rate. This model never has one, which is why its deferral measure is a transform of the lag rather than its mean. Characterised from the result its title carries —* L = λW*; nothing is quoted.)*

Marshall, A. W., & Proschan, F. (1972). Classes of distributions applicable in replacement with renewal theory implications. In *Proceedings of the Sixth Berkeley Symposium on Mathematical Statistics and Probability*, Volume I. Berkeley: University of California Press. *(§4.9 cites it for the new-better-than-used-in-expectation bound that signs the correction: an NBUE distribution's moment generating function is dominated by that of the exponential with the same mean.* ***Deliberately unmarked.*** *The Berkeley Symposium volume was not consulted and the page range is omitted rather than guessed; the attribution is taken from the reliability literature that rests on it. §4.9 does not depend on the citation, since the direction it predicts is also measured directly — but the reasoning that produced the prediction is this lemma's and is credited as such.)*

Nakagawa, T., & Osaki, S. (1975). The discrete Weibull distribution. *IEEE Transactions on Reliability*, R-24(5), 300–301. ✓ *(§5.4's fitted lag distribution is this one, in the survival parameterisation* $P(T \geq t) = q^{(t}k)$ *the source defines, whose discrete hazard is increasing exactly when* $k \geq 1$ *and which nests the geometric at* $k = 1$*. Named here because a distribution fitted and reported ought to say whose it is.)*

Potepa, J., & Thomas, J. (2023). Goodwill impairment after M&A: acquisition-level evidence. *Journal of Financial Reporting*, 8(2), from p. 131. ✓⧗ *(§4.9 cites it as the closest existing treatment of impairment* timing*: their design tracks each acquisition to its first impairment within a ten-year window and stops, which their own text describes as effectively a hazard model. It is cited for what it is and for what it is not — a covariate model with no baseline shape* *estimated — which is the gap §5.4's fit occupies. The text consulted is the authors' working paper rather than the typeset article, hence* ✓⧗*; the end page could not be confirmed from an open source and is omitted rather than guessed.)*

Kay, J. A. (1976). Accountants, too, could be happy in a golden age: The accountant's rate of profit and the internal rate of return. *Oxford Economic Papers*, 28(3), 447–460. ⧗ *(Cited in §4.4 for the analytical result that precedes Fisher and McGowan's numerical demonstration.* ***Not read****; record verified, characterisation from secondary sources.)*

Long, W. F., & Ravenscraft, D. J. (1984). The misuse of accounting rates of return: Comment. *American Economic Review*, 74(3), 494–500. ✓⧗ *(Cited in §4.4 for the rebuttal. The* ***text*** *consulted is the open FTC Bureau of Economics Working Paper No. 94, June 1983, read in full; the published comment is verified bibliographically and has not been read. Nothing is quoted.)*

Ryan, S. G. (1995). A model of accrual measurement with implications for the evolution of the book-to-market ratio. *Journal of Accounting Research*, 33(1), 95–112. ✓ *(Cited in §4.7 for the regression Beaver and Ryan (2000) adopt.* ***Read at source****, with the Autumn 1995 erratum at 33(2), 417, which corrects two typesetting errors in equation (5) — a "+" printed for the "=", and ΔMV_{i,t−10} printed for BV_{i,t−10} — and changes no coefficient, hypothesis or result. The erratum's γ term is absent from the equation §4.7 relies on, which is Beaver and Ryan's (4). Ryan's assumption (A8) "eliminates the possibility of conservative accounting," and his firm effects are a control for what that leaves unmodelled; the bias/lag reading is Beaver and Ryan's. In 1995 the journal carried the name given here.)*

Ryan, S. G. (2006). Identifying conditional conservatism. *European Accounting Review*, 15(4), 511–525. ✓ *(Cited in §4.6 solely to distinguish a near-identical title.* ***Read at source.*** *The characterisation holds and is now supported by the body rather than the abstract: the word "econometric" does not occur in the article, and "identify" and its cognates are used throughout in the empirical sense of detecting conservatism in data. Nothing is quoted.)*

Sims, C. A. (1971). Distributed lag estimation when the parameter space is explicitly infinite-dimensional. *The Annals of Mathematical Statistics*, 42(5), 1622–1636. ✓ *(§4.10 cites it for the conclusion that finite-dimensional approximations to a lag space are meagre in it and that their approximation error "cannot, in other words, be made asymptotically negligible" — quoted from §5, p. 1634 — and for his naming of "the finite-dimensional parameter spaces of rational lag distributions (see Jorgenson (1966))" as exactly the approximating class the result covers, p. 1628. The text consulted is* *an optically-recognised scan rather than the typeset original; the volume, issue and page range are confirmed against the journal's own table of contents.* ***The journal is the* Annals of Mathematical Statistics*, not* Econometrica*, and the title word is "explicitly", not "essentially" — this entry is commonly miscited on both counts.****)*

Varah, J. M. (1982). *On fitting exponentials by nonlinear least squares.* Technical Report TR-82-02, Department of Computer Science, University of British Columbia. ✓ *(§4.10 cites it for the quantified form of Lanczos's example, and it is the route by which Lanczos is cited at all.* ***Read in full.*** *It attributes the observation to "Lanczos (1956, pg. 279)" and locates the data at p. 273, and reports the Hessian's smallest eigenvalue falling by roughly three orders of magnitude per additional exponential term. A later journal version is believed to exist and was* ***not*** *verified, so the technical report is what is cited and what was read.)*

Nerlove, M. (1958). *The Dynamics of Supply: Estimation of Farmers' Response to Price.* Baltimore: Johns Hopkins Press. ⧗ *(Cited in §4.2 for the combined adaptive-expectations/partial-adjustment model, whose reduced form is the closest economics-native analogue of this paper's exchange symmetry.* ***Not read.*** *The bibliographic record is verified; the reduced-form algebra attributed here — symmetry of every systematic coefficient in $\beta \leftrightarrow \gamma$, asymmetry of the disturbance — was derived and checked in this repository (*`scripts/wt085_returns_conditioning.py`*, E7) rather than taken from Nerlove's text, and the entry claims nothing beyond the model's structure.)*

Hutton, A. P., Marcus, A. J., & Tehranian, H. (2009). Opaque financial reports, R^2, and crash risk. *Journal of Financial Economics*, 94(1), 67–86. ✓ *(Nothing is quoted from the body, which was read as full text rather than as the typeset article. The post-SOX dissipation §10 attributes to them is from the published abstract, checked at source.)*

International Energy Agency & United Nations Statistics Division. *SDG Indicator 7.3.1 — Energy intensity measured in terms of primary energy and GDP.* Reported as World Bank series `EG.EGY.PRIM.PP.KD`, *Energy intensity level of primary energy*, compiled for *Tracking SDG 7: The Energy Progress Report* by the IEA, IRENA, UNSD, the World Bank and the WHO. ✓

Jin, L., & Myers, S. C. (2006). R^2 around the world: New theory and new tests. *Journal of Financial Economics*, 79(2), 257–292. ✓ *(****Read at source****, typeset article. The sentence §10 quotes for the no-physical-layer reading — "For simplicity, we ignore depreciation and reinvestment" — is verified in the published text at* ***p. 262****, character for character, and unchanged from NBER Working Paper 10453. §10* *quotes the sentence because it is what establishes that the model has no physical layer, and a reader entitled to doubt that on a paraphrase should be able to see the words. Its footnote 3 is worth reading beside §4: the authors set aside depreciation "according to a pre-defined schedule" as an easy extension, and it is the interaction of exactly that schedule with recognition timeliness that §4.2 shows a reported series cannot resolve.)*

Jonckheere, A. R. (1954). A distribution-free k-sample test against ordered alternatives. *Biometrika*, 41(1–2), 133–145. ✓

Mann, H. B., & Whitney, D. R. (1947). On a test of whether one of two random variables is stochastically larger than the other. *Annals of Mathematical Statistics*, 18(1), 50–60. ✓

Mayo, D. G. (1996). *Error and the Growth of Experimental Knowledge*. University of Chicago Press. ✓✎ *(The copy consulted is the University of Chicago Press edition of 1996, which uses* severity *374 times and* severe test *232 times and is where the severity requirement is introduced. Mayo (2018),* Statistical Inference as Severe Testing *— a later restatement — has not been read. Both the edition-consulted rule and the first-appearance rule select 1996, and they agree here because the book he read is also the origin.)*

Mayo, D. G., & Spanos, A. (Eds.). (2010). *Error and Inference: Recent Exchanges on Experimental Reasoning, Reliability, and the Objectivity and Rationality of Science*. Cambridge University Press. ✓✎ *(Copy consulted gives © Cambridge University Press 2010, first published in print 2009.)*

Mises, L. von (1949/1998). *Human Action: A Treatise on Economics* (Scholar's ed.). Ludwig von Mises Institute. ✓✎ *(The copy consulted is the Scholar's Edition, ISBN 0-945466-24-2, whose own front matter states that it reissues* the first edition *— load-bearing, because the 1963 and 1966 editions differ from 1949. Original work published by Yale University Press, 1949.)*

Nosek, B. A., Ebersole, C. R., DeHaven, A. C., & Mellor, D. T. (2018). The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11), 2600–2606. ✓

Odum, H. T. (1996). *Environmental Accounting: Emergy and Environmental Decision Making*. John Wiley & Sons. ✓✎ *(Copy consulted gives © 1996 John Wiley & Sons, Inc., New York, ISBN 0-471-11442-1. This entry was re-pointed away to* Environment, Power, and Society *(Columbia, 2007) when the first library sweep did not find the 1996 book, then restored when it did. The sweep, not the citation, was wrong.)*

Piketty, T. (2013/2014). *Capital in the Twenty-First Century* (A. Goldhammer, Trans.). Belknap Press of Harvard University Press. ✓ *(Original work published as* Le Capital au XXIe siècle*, Éditions du Seuil, 2013. §10's relocation argument is about work that existed* *a year before the English text cited here.)*

Popper, K. R. (1935/2002). *The Logic of Scientific Discovery.* Routledge Classics. ✓✎ *(The copy consulted is the Routledge Classics edition of 2002. Its own colophon gives the chain:* Logik der Forschung *first published 1935, Vienna — its preface dated 1934 — first English edition Hutchinson & Co., 1959, Routledge from 1992. §10 cites Popper for the demarcation criterion, which is 1935's, not 1959's, so the original date is load-bearing and the entry is dual-dated.)*

Quine, W. V. O. (1951). Two dogmas of empiricism. *Philosophical Review*, 60(1), 20–43. ✓

Soddy, F. (1926/1961). *Wealth, Virtual Wealth and Debt: The Solution of the Economic Paradox* (3rd ed.). Omni Publications. ✓✎ *(The copy consulted is the third edition, LCCN 60-53331, printed in the United States under a Britons Publishing Company, London, title page, and described on that page as a reprint of the second edition of 1933 "containing new material and Foreword to the American Nation". The first edition is George Allen & Unwin, London, 1926; the copy's own* Preface to the First Edition *is dated January 1926 and its Addition to the Second Edition refers to "the book, which first appeared in 1926". The term* virtual wealth *is in the 1926 title, so 1926 is the earliest appearance this paper can support; Soddy's own footnote points back to* Cartesian Economics *(Hendersons, 1922) as prior work on the subject, and whether the term itself originates there has NOT been checked — the 1922 pamphlet is not in the author's library. No claim of priority is made in the text, so none is made here.)*

Terpstra, T. J. (1952). The asymptotic normality and consistency of Kendall's test against trend, when ties are present in one ranking. *Indagationes Mathematicae*, 14, 327–333. ✓

Zhu, W. (2016). Accruals and price crashes. *Review of Accounting Studies*, 21(2), 349–399. ✓ *(Cited in §10 against §2. **Read at source**, typeset article. High accruals predict firm-level weekly price crashes; the relation concentrates in the components Richardson, Sloan, Soliman and Tuna rank least reliable, including non-current operating asset accruals, and strengthens with CFO option incentives and weaker monitoring. The negative loading on current operating liability accruals is reported by Zhu as unexplained by his own mechanism, and is noted here for the same reason. Nothing is quoted.)*

The tensor composes, the behaviour does not: one atomic unit at the household, firm and sovereign scales

STOOD DOWN — do not cite, do not edit. This paper is not carried forward.

Decided and executed in the `paper-rebuild` pass of 2026-08-26. This manuscript's thesis is that the three scales are one structure — its own phrase for it is *"a chain rather than three analogies"*. The corpus's own end-to-end check, `docs/RESULT-END-TO-END-001-E1.md`, put that assertion under test and **failed it at leg `E1a`**: *"ρ and φ are not the same kind of object, so the join between Papers II and III is vocabulary at the sovereign scale."* The paper is left here intact, at v0.2, as the record of an argument that was made and withdrawn — which `docs/ADR-001-paper-decomposition.md` and `docs/METHOD-001-the-phantom-tag.md` are the house precedent for keeping in public.

Its three surviving parts went here:

part	was	is now
The SMD-as-boundary argument	§4 (4.1–4.4)	**Paper III, Appendix A.4** — *"The boundary on P3, and it is a fifty-year-old theorem."* Landed on P3 (atomism), where the objection actually bites. Limit 2 is remapped to Paper III's own section 5; limit 3 to its section 9.5. The six aggregation references travelled with it and were re-verified against Crossref on 2026-08-26.
The crossing-height instance	§5	**`docs/RESULT-CROSSING-HEIGHT-001.md`** — retired from the papers, kept whole. It was *not* folded into Paper III section 5.6: the two crossings share the word and no operator, and section 0 of that file argues the refusal.
The pre-registered citation-whitespace instrument	§6	**`docs/METHOD-002-citation-whitespace.md`** — a standalone method note beside `METHOD-001`. Not Paper III section 13, which enumerates only the registrations producing Paper III's numbers; REG-013 produces none of them.

Nothing else here was moved. `test_excess_demand_is_monotone_here_so_this_is_not_an_SMD_result` outlives this manuscript and Paper II section 7 says so. Sections numbered below are **this paper's own** and do not correspond to Paper III's post-rebuild numbering.

Jason C. Braatz *Independent researcher* jason@braatzresearch.com

Draft — not yet submitted. Version 0.2, 2026-08-24.

Declaration of interest. The author is employed by a company building accounting software for very small businesses. This work was conducted independently, on personal time, and without company funding, data or direction.

Use of AI assistance. Anthropic Claude Opus 5, at high reasoning effort, was used throughout as a research and drafting assistant: literature retrieval, adversarial review, code review and prose drafting. All claims, results and final text are the author's, and every computational result is produced by committed code in the repository named in §10.

Revision note: ***v0.2****, this draft — independent review passes; results, numbers, claims and citations were corrected where those passes found them wrong.*

Abstract

Three literatures describe wealth and do not read each other: biophysical economics, stock-flow consistent macroeconomics, and kinetic-exchange econophysics. This paper joins them on one claim — the same atomic **state** has one type at the household, firm and sovereign scales — and on one limit its own end-to-end test imposed: those scales share **one question, not one structure**.

The unit is an **extensive state**, and extensive states add. **Behavioural maps do not**, and the theorem saying so is Sonnenschein–Mantel–Debreu. SMD is therefore not an objection to composition across scales; it is its boundary. **Aggregation preserves the extensive state and destroys the behavioural map** — so a discipline that aggregates in order to recover behaviour is estimating the half that does not survive, while the half that does, the physical stock and the claim recorded against it, is largely unmeasured.

A worked instance at the smallest scale: for a single indivisible good, 25 allocations of one population yield 25 demand schedules, 25 supply schedules and **one** excess-demand schedule, and the crossing height is not a behavioural aggregate but the allocation mismatch.

The motivating whitespace is measured, not asserted. A preregistered co-citation instrument returns a split-half ceiling of

0.477, a floor of exactly **0.000** against a literature unrelated by construction, and target overlaps of **0.020, 0.011 and 0.005** — one pair undecided under a stricter ceiling, and **six works in the world cite both a stock-flow-consistent and a kinetic-exchange seed.**

Keywords: aggregation · Sonnenschein–Mantel–Debreu · biophysical economics · stock-flow consistency · econophysics · composition · citation graph · pre-registration

JEL classification: B41, D50, E01, Q57, C63

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1 · Introduction

Three literatures describe wealth and do not cite each other. **Biophysical economics** treats production as a transformation of energy and materials subject to thermodynamic constraint. **Stock-flow-consistent macroeconomics** insists that every flow leaves a stock somewhere and that the accounting must close. **Kinetic-exchange econophysics** treats a wealth distribution as the stationary state of a stochastic exchange process. Each is a serious research programme with its own journals, and each is describing a different layer of the same object.

This paper is the claim that they are layers of one stack, and that the object at the bottom of it has one type wherever it appears: **a household's holding, a firm's balance sheet and a sovereign's accounts are the same kind of object, and summing holdings does not change the kind.** That claim is worth stating carefully, because a nearby and stronger version of it is false, and the difference between the two is this paper's central technical content.

1.1 · Why now, and it is not because the idea is new

The models this framework declines to use were not errors. They were **force-fits, not form-fits** — rational responses to constraints that have since expired.

The distinction is worth stating exactly, because the soft version of the concession gives away more than it needs to. A *benign* simplification drops detail and preserves structure: refine it and the numbers move. A *structural* one collapses a distinction the theory needs: refine it and the conclusions move. Capital as a malleable scalar is the second kind, and the Cambridge controversy — opened by Robinson (1953), and given the reswitching apparatus by Sraffa (1960) — was fought over precisely that: Samuelson

(1966) conceded that with reswitching available the marginal-productivity account cannot be recovered from an aggregate capital measure. The scalar survived anyway, and it survived for a reason that had nothing to do with the argument being settled.

The reason was that the scalar was the only object anyone could populate. There was no firm-level panel in machine-readable form, no national input-output energy table at usable granularity, and no standing international series for energy intensity. A theory that requires a per-entity state vector was, in 1956, not a theory anyone could take data to. **Two of those three constraints are datable and have lapsed:** machine-readable structured filings became mandatory in the United States over a phase-in ending in 2011, and energy intensity of GDP is now published annually as a United Nations Sustainable Development Goal indicator (7.3.1). **The input-output energy table has no lapse to report.** The SDG series is a national aggregate; there is no counterpart giving energy input at the granularity the filings give claims, which is one reason §4.3 finds the composed state largely unmeasured rather than merely unassembled — and why no claim that firm-level panels of *both* are public and free is made here.

A parallel claim about computation — that models of that era had to be analytically solvable because numerical solution was unavailable — is true and is left out, because a claim about hardware dates a paper and a claim about what a model can be populated with does not. The data constraint is the stronger half of the argument and it is the half with an expiry date one can cite.

§7 states this as a method: the disagreement is never that a predecessor was wrong. It is that a predecessor was solving a differently-constrained problem, and the constraints moved.

Contributions.

1. **A composition result and its exact boundary** (§3, §4). The atomic unit is an extensive state, and extensive states compose by addition across scales. **Behaviour does not compose, and the theorem that says so is Sonnenschein–Mantel–Debreu.** SMD is therefore not an objection to this framework; it is the framework's own statement of where composition stops. A paper claiming clean composition across scales while citing SMD as a shield would be contradicting itself.
2. **The consequence, which is the corpus's thesis in one sentence** (§4.3): aggregation destroys the behavioural information macroeconomics believes it is measuring, and preserves the thermodynamic structure nobody is looking

at, because it is not what an aggregate is usually built for.

3. **A worked instance at the smallest possible scale** (§5): in a market for a single indivisible good, the quantity coordinate of the supply-and-demand diagram is not a behavioural aggregate at all. It is a state variable — the allocation mismatch — and the two curves' difference is invariant to the allocation identically, at every price.
4. **The whitespace, measured rather than asserted** (§6). The claim that these three literatures do not meet is normally made by an author reporting that he looked. Here it is a pre-registered citation-graph measurement (`REG-013`) with both ends of its scale established in the same run: a split-half control that cannot be tuned, and an unrelated-field floor.
5. **A stated method** (§7). The relocation move — *not wrong, differently constrained* — is used deliberately and named.

What this paper does not contain.

No new model code and no new simulation. Papers II and III carry the computational results and their test suites; this paper cites them as established and adds one measurement of its own, on the literature rather than on the model. Numbers appearing below without a citation to II or III have their record named in §10, one for each: §6's are `REG-013`'s, and its record is the committed output of that run rather than a command, because the instrument re-queries a live database; §5's and §8's results are the output of a fourth paper on price formation — written, refereed against itself and not published, for the reasons §8 gives — whose apparatus is still in this repository and still runs; and §8's word count for that paper is `wc -w` on its superseded draft.

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2 · The atomic unit

2.1 · The three propositions, cited not restated

Paper III states the framework's propositions with their domains (§A.1.2) and defends the coupling once. They are named here because §3 composes them and §4 bounds them, and the argument for each is in Paper III.

- **P1 · Composition.** A holding's value is a composite of a physical component and a claim component, and the two obey different laws. *(Domain, carried rather than cited because it is the one that bites here: units having a physical referent — silent on purely contractual objects whose referent is another claim.)*

- **P2 · Decay.** The physical component degrades absent maintenance. No store is inert.
- **P3 · Atomism.** Measured aggregates are folds over units, and no aggregate is more fundamental than its constituents.

P3 is the one this paper is about, and it is worth being precise about which P3. It is the weak form: aggregates are folds. It is **not** the strong form — that some aggregates carry information no fold contains — which §8 reports as tested and rejected.

2.2 · What kind of object composes

An **extensive** quantity is one whose value for a composite system is the sum of its values for the parts: mass, energy, a stock of steel, a balance of claims. An **intensive** one is not: temperature, a price, a rate of return. A **behavioural map** is neither — it is a function from states to actions, and the question of whether it composes is the question of whether the map of the whole is the map built from the maps of the parts.

The atomic unit of this framework is an extensive state. It composes by addition, and the statement that it does so is a definition rather than a discovery. The content of this paper is not that extensive states add; it is **what stops adding at the same moment**, and what that implies about which measurements survive aggregation.

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3 · Composition, at three scales

The claim to be checked is not "the framework applies at every scale" — a framework can be applied anywhere, which is why applicability is not evidence. The claim is that **the same object appears at each scale and the operator that moves between them is addition**.

Household. The unit is a holding: a dwelling, a vehicle, a claim on a pension. It has a physical component that degrades at some rate and a claim component recorded at some other rate. Nothing here is metaphorical; a roof has a service life and a mortgage has an amortisation schedule, and they are different numbers. **Paper III §2 is that holding**, before anything is summed: its physical layer degrades at an effective 0.02 per period and its claim layer recognises at $\alpha = 0.05$ — α being the rate at which a change already incurred but not yet recorded is released into the accounts, and 0.05 the model's swept calibration, which Paper III §5.4 goes on to measure at 0.408 per year on its registered sample and reports as low by an order of magnitude — and Paper III §3.1 reports what deferring a change at those rates costs. The household scale's quantitative answer is Paper III's, read before that paper's §4 indexes the holding by asset class.

Firm. A balance sheet is holdings of the household's kind, summed and reported. Paper III's result is that this reporting is a filter: a share φ of each true change passes through at once and the remainder is released at rate α from an unrecognised gap. The filter is a *per-class* object — Paper III indexes classes *i* and writes the recursion with a Hadamard product,

$$\mathbf{C}(t+1) = \mathbf{C}(t) + \boldsymbol{\varphi} \odot \Delta\mathbf{E} + \boldsymbol{\alpha} \odot \mathbf{gap}(t),$$

and the elementwise product is not notation. It is the claim that the reporting layer is **diagonal in class space**, which is exactly the statement that the firm's reporting composes from its classes' reporting without cross-terms.

Sovereign. National accounts are the same holdings summed across every institutional sector, households and firms alike, and Paper II's parameter space is what happens when a levy is assessed on that sum. Its central result is a composition result wearing different clothes: the base of a levy — stock or flow — is the question of *which component of the composed state the assessing layer can see*, and at zero realisation a confiscatory levy on flow leaves the wealth vector exactly unchanged, agent by agent.

Note what this is: three instances of one question, asked at three scales. At each step the same two components appear, the same question is asked of them — what does the measuring layer observe? — and the answer at each scale is a *quantitative* one that the paper for that scale reports. Paper II's κ, the share of aggregate wealth actually moved per assessment, is a composition quantity: it is defined at the sovereign scale and it is a fold over household-scale liabilities. Paper III's $\varphi \odot \delta$ — observability times the physical decay rate — is a composition quantity: it is defined at the firm scale and is *written* as diagonal over asset classes — a form Paper III §5.4 rejects, which changes what the link carries and not whether there is one.

And **the place where the firm-scale link could break was named, was tested, and the test rejected it.** Diagonality is an assumption, not a theorem: if recognition events cluster within firm-quarters rather than occurring independently across classes, the Hadamard form is wrong and the firm's reporting does *not* compose from its classes' without cross-terms. Paper III registered that test (`REG-003`) and ran it, and **independence is rejected in both universes in the same direction — 4.12× and 2.02×**

the independence expectation, both p = 0.0002, in a design that detects an injected excess of five per cent of events with probability 1.00 (Paper III §5.4; the headline survives that section's tag-list repair at 4.01× and 2.10×).

Three things follow. First, the consequence is *bounded rather than open*: the Hadamard form is now an approximation whose error is measured, not an assumption whose status is unknown, and a measured approximation is a stronger object to build on than an untested premise. Second, **what the rejection costs is the reporting layer's clean composition, not the state's.** Diagonality is a property of the *filter* — of how recognition in one class relates to recognition in another — and never of the extensive state, which adds by §2.2 whatever the recording practice does. The firm-scale link is therefore degraded and not severed: the firm's *state* still composes from its classes' states, and it is the firm's *reporting* of that state that carries cross-terms the diagonal form omits. Third, **Paper III's design cannot separate an economic coupling from the sequencing the standards impose** (ASC 350-20-35-31 and 35-32 order the tests), so this paper may not read the rejection as evidence that the underlying degradations are coupled. It inherits a measured departure with an unidentified cause, and §9 says so. **A stronger reading — that the three scales make *a chain rather than three analogies* — was registered, tested, and does not hold.** `END-TO-END-001` leg `E1` asked whether the sovereign and firm scales stand in the relation the word *chain* asserts — whether Paper II's realisation share ρ and Paper III's observability share φ are the same object seen twice — and they are not. What Paper III's filter does not recognise is **deferred**, held in an unrecognised gap and released at rate α; what Paper II's base does not recognise is **never assessed**. A lag and a loss are different operators, and Paper II has no parameter that plays α's part. So what joins the scales is the question, the fact that each scale answers it quantitatively, and the addition of §2.2 — and no more. The demotion was written into that document's §2 **before** the leg was run; `docs/RESULT-END-TO-END-001-E1.md` records the run and the reasoning.

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4 · The tension, resolved explicitly

4.1 · The objection, stated at full strength

Here is the objection this paper would deserve if §3 were the whole of it.

You claim a unit that keeps one type at the household, firm *and sovereign scales and survives being summed. But the best-established result about aggregation in economics says the opposite. Sonnenschein (1972, 1973), Mantel (1974) and Debreu (1974) proved that aggregate excess demand inherits from individually rational agents only continuity, homogeneity of degree zero and Walras's Law — not downward slope, not uniqueness, not stability. Aggregate demand can take essentially arbitrary shape. Worse: the price-formation paper of your own corpus — the fourth, which §8 records as written and withdrawn — cites SMD approvingly, as evidence that doubting inherited aggregation is inside the mainstream. You cannot cite the theorem that aggregation destroys structure and then claim structure survives aggregation.*

That is correct as far as it goes. It is also answerable in one distinction, and the answer is not a hedge.

4.2 · The distinction

SMD is a theorem about maps, and the atomic unit is a state.

What SMD constrains is the aggregate *excess demand function* — an object that takes prices and returns quantities, assembled from individual demand functions. The theorem says that essentially nothing about the individual functions survives the assembly. It is a statement about the **behavioural map**.

What P1–P3 assert composes is the **state**: how much physical stock is held, how much claim is recorded against it, and at what rates each moves. Summing steel is not summing preferences.

The two claims are therefore not in tension; they are complementary halves of one statement, and the statement is sharper than either half:

> **Aggregation preserves the extensive state and destroys the behavioural map.**

SMD is the second clause, proved, fifty years ago, inside the mainstream. This framework is the first clause, and the reason it is worth having is entirely the *conjunction*: if aggregation destroys the map, then a discipline that measures aggregates in order to recover behaviour is measuring the thing that does not survive, and one that measures aggregates in order to recover stocks is measuring the thing that does.

4.3 · The consequence

Macroeconomics aggregates in order to do behavioural inference. Estimate an aggregate production function, recover a technology; estimate an aggregate consumption function, recover a propensity; estimate an aggregate demand curve, recover an elasticity. SMD says the object being estimated need not have inherited any of the structure the inference requires. The response to SMD in practice has been to impose distributional restrictions on the population until the aggregate is well-behaved — Hildenbrand and Grandmont, where it is sufficient *dispersion* of household characteristics that does the work, and the representative agent as the degenerate case at the other end — which is a legitimate research strategy and is also an admission that the structure is imposed rather than inherited.

Meanwhile the extensive state — energy embodied, stock degraded, claim recorded against it, and the rate of each — **does** survive the sum, and it is very largely not being measured, because it is not what an aggregate is usually built for.

That is the thesis of this corpus in one sentence, and it is a *positive* claim about which measurements are informative, not a complaint about anyone's research programme.

4.4 · The limits of the resolution

Three.

1. **A state that composes is not thereby a state anyone can observe.** Composition is a property of the object; observability is a property of the measuring layer, and Papers II and III are both, in the end, about the measuring layer failing to see something. The framework's own results are the reason to be sceptical that the composed state is available.
2. **"Extensive" is doing real work and it excludes things one might want.** A rate is not extensive. δ, φ, α and ρ do not compose by addition; they compose, where they compose at all, as *weighted* combinations whose weights are themselves state. Paper III's asset-class ladder is what happens when one forgets this: a cross-class ranking computed from reported numbers is a ranking of the *product* $\varphi \odot \delta$, so it recovers a ranking of φ only where the decay rate δ is constant across the classes being ranked — and the classes accounting standards distinguish are very nearly defined by differing in δ. Ranking by the parameter can therefore invert the ordering rather than blur it.
3. **Diagonality is assumed at the firm scale, and the assumption is measurably wrong.** §3 names this; Paper III registered the test, ran it, and rejected independence across classes within a firm-quarter in both universes. The composition claim therefore has a *degraded* link at exactly the scale where the accounting is done — degraded rather than severed, because what was rejected is a property of the

reporting filter and not of the extensive state, and because the departure is now a measured quantity rather than an open exposure. What is not available is its cause: the design cannot say whether the coupling is economic or an artefact of the order the standards impose on the tests.

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5 · The smallest instance: the crossing height is the volume

This section carries the surviving result of a paper that was written and then not published; §8 records why.

Take a market for a single indivisible good. N agents each hold a reservation price m_i; S units exist; H is the set of current holders and T the set of the top-S valuers. The textbook reads two schedules off this population — a demand curve and a supply curve — and finds the price where they cross.

The two schedules are not independent equations. Measured over 25 allocations of the same population at 399 interior grid points: 25 distinct demand schedules, 25 distinct supply schedules, and **one** distinct excess-demand schedule, equal to $\#\{i : m_i > p\} - S$ at every point. The reason is a partition. At any price that is not itself a reservation price the S holders divide into those valuing above it and those valuing below, so the allocation enters the two counts with opposite signs and cancels — at every price, not merely at the zero.

So the *price* coordinate of the cross reads the population and nothing else. The **quantity** coordinate — the volume traded at the crossing, and the height of the cross in the diagram — reads something quite different. At a clearing price strictly inside the interval,

$$D(p^*) = |T \quad H|,\ S(p^*) = |H \quad T|, \text{ and the two are equal.}$$

The crossing height is the allocation mismatch — the number of units in the wrong hands — which is precisely the quantity excess demand cannot deliver. Verified for all 25 allocations.

Read as a composition statement, the diagram is doing two jobs in one picture: the price coordinate is a fold over the population, and the height is a *coupling* between the population and the current state. It is not a behavioural aggregate that has smuggled in state information; it is a state measurement that has been read as behaviour for a century and a half of teaching.

What this instance is not. It is not an SMD result and must not be sold as one. Excess demand here is monotone and single-crossing — zero monotonicity violations across 500 grid points, running from +249 to −150 with one sign change — because each agent demands at most one unit and there is no wealth channel from the endowment back into demand. Remove that restriction and income effects return, and income effects are exactly what SMD requires. **The two results sit at opposite ends of one axis**: at zero income effects the allocation cancels identically and the state is all there is; with income effects the behavioural map stops composing and SMD is what one gets. The test suite asserts the monotonicity deliberately, under the name `test_excess_demand_is_monotone_here_so_this_is_not_an_SMD_result`, as a standing limit on the claim rather than a property being celebrated.

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6 · The whitespace, measured

A paper that joins three literatures usually motivates itself by reporting that they do not meet, and the evidence offered is that the author looked. That is the first thing a referee attacks, and correctly: an absence found by searching is a property of the search.

So it was measured instead, and pre-registered first (`REG-013`, committed before the instrument existed). The design's whole content is that a low co-citation rate between two specialties is **uninformative on its own** — sociology and semiconductor lithography do not cite each other either. What makes a number here mean something is a scale with both ends fixed in the same run.

The instrument. Each literature is defined by seed works named in `REG-013`, and its *audience* is the set of works citing at least one seed, taken from OpenAlex. For two literatures the statistic is the overlap coefficient — the share of the smaller audience that also reads the other. Audiences are truncated at the 4 000 most-cited works before the overlap is taken; the second qualification below states what each truncation costs. Twenty-five of twenty-five seeds resolved.

The ceiling is each literature split in half by seed index and measured against itself. It cannot be tuned: if the instrument cannot see that half of econophysics is joined to the other half, it cannot see anything, and the registration voids the run below 0.20. It came back at **0.477** pooled.

The floor is each literature against six highly-cited CRISPR papers — unrelated to economics by construction. It came back at **exactly zero**: of the 4 000 most-cited works citing a CRISPR seed, not one is in any of the three economics audiences.

The three pairs, positioned on that floor-to-ceiling scale. The position z places a pair's overlap on that scale — (overlap − floor) ÷ (ceiling − floor), so the floor reads 0 and the ceiling 1 — and `REG-013` fixed the bar in both directions before the run: $z \le 0.10$ is whitespace, $z \ge 0.25$ is joined, and the band between them is undecided.

pair	works citing both	audience (smaller)	overlap	position z
biophysical × stock-flow	**23**	1 139	0.0202	0.042
biophysical × kinetic exchange	**15**	1 383	0.0108	0.023
stock-flow × kinetic exchange	**6**	1 139	0.0053	0.011

Against split-half intersections of 134, 155 and 380 *within* the same three literatures. **Six works in the world cite both a stock-flow-consistent seed and a kinetic-exchange seed.** All three pairs sit below the whitespace bar; the whitespace is where it was claimed to be.

One qualification. Biophysical economics is on this instrument a *loose federation*: its own split-half overlap is 0.168, well below stock-flow's 0.520 and kinetic exchange's 0.744, because its seeds are monographs spanning four decades and three sub-traditions that do not much cite each other. Scoring the biophysical pairs against the pooled ceiling is therefore generous to them. Under a stricter per-literature ceiling, biophysical × kinetic exchange and stock-flow × kinetic exchange remain whitespace, and **biophysical × stock-flow becomes undecided** ($z = 0.120$ against a 0.10 bar). The registered rule is the one that governs — re-choosing a ceiling after seeing which verdict it yields is not available — but the result is stated as it is: *two pairs are whitespace under every reading tried, and one is whitespace under the registered rule and undecided under a stricter one.*

A second qualification runs the same way. The biophysical audience was capped at 4 000 of 7 801 by descending citation count, so a bridging work in its tail is invisible here, which suppresses both biophysical overlaps. The floor's cap is tighter still — 4 000 of 43 048, 9.3 per cent — and it does the opposite and costs nothing: a cap can only remove intersections, so the measured floor is a lower bound on the true one, and a floor of exactly zero is the strictest value available. Both true-audience sizes are `RESULT-REG-013` §2's. The instrument stops retrieving at the cap and does not print them, which makes them the two numbers in this section that §10's command does not regenerate.

And what the measurement does not establish, stated in `REG-013` before the numbers existed: an unoccupied intersection is not thereby a fertile one. This section establishes that the three literatures do not read each other. That there is something worth finding where they meet is what §§3–5 have to earn on their own.

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7 · Relation to existing work, and the method used to state it

Everything in this section is one move, applied deliberately and named.

The move: relocation. The disagreement is never *you are wrong*. It is *you are a differently constrained case*. Piketty is not measuring the wrong thing; he is measuring a different layer. Solow's scalar was not an error; it was the only object anyone could populate. SMD is not an opposing result; it is this framework's own boundary, proved by the mainstream fifty years before the framework existed.

For a claim that spans three literatures, relocation is a **structural** requirement: a paper that must be read by biophysical economists, stock-flow theorists and econophysicists cannot afford to make an enemy in any of the three, and more to the point, in each case the relocation is *true*. A predecessor working under a binding constraint that has since lapsed has not made an error. Recording which constraint, and when it lapsed, is a stronger and more checkable statement than "the received view is mistaken," and it is falsifiable in a way that the received-view complaint is not.

Biophysical economics — Georgescu-Roegen, Daly, Odum, Ayres and Warr, Hall and Klitgaard — establishes that production is constrained by energy and materials and that the constraint is not a detail. What it has generally not had is an accounting-shaped object: a per-entity state that a balance sheet could carry. Paper III's decomposition is that object, and the coupling Λ it defines — the claim measure carried per unit of physical measure, in currency per joule — has the dimensions of a quantity the

United Nations already reports the inverse of (SDG 7.3.1). Paper III §A.2.2 makes that claim and bounds it in the same breath — the correspondence is dimensional, and *"emphatically not"* that the indicator measures Λ^{-1}. It is therefore evidence that the dimension is not this framework's coinage, and not evidence that the coupling has been measured.

Stock-flow-consistent macroeconomics — Godley and Lavoie, and the surveys of Caverzasi and Godin and of Nikiforos and Zezza — insists that the accounting close, and builds models in which it does. Its stocks are financial. The claim here is that the closure is incomplete in a specific and correctable way: a physical stock degrades on a schedule the financial accounting does not record, so a model whose accounting closes in claims can still be open in the thing the claims are against. That is a friendly amendment with a testable edge, and Paper III is where the edge is tested.

Kinetic-exchange econophysics — Drăgulescu and Yakovenko, Chakraborti and Chakrabarti, Bouchaud and Mézard, Chatterjee and Chakrabarti — has the stochastic machinery and the stationary distributions. What it has mostly not had is a *base*: its exchanges are of an undifferentiated scalar. Paper II's result is that the base is the decisive coordinate and the rate is not, which is a statement kinetic exchange can absorb directly, and κ — its compressive budget rather than its mechanism, in Paper II's own words — is the sort of closed-form quantity that literature likes.

And on the aggregation literature specifically. SMD is treated in §4 at length and is not re-argued here; its textbook statement is Mas-Colell, Whinston and Green (1995). The relation is complementarity: SMD is the theorem for the map, this framework is the claim for the state, and neither implies the other.

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8 · Abandoned approaches

The test applied to every entry is: had this route worked, which sentence in this paper would be different?

"A chain rather than three analogies." Had it survived, §3 would assert a structural correspondence between the sovereign scale's realisation share and the firm scale's observability share, and that correspondence — not the three separate results — would have been this paper's central contribution. `END-TO-END-001` leg `E1` shows the two shares are not the same kind of object: the unrecognised remainder is deferred in one and discarded in the other, and the firm scale carries a release rate α for which the sovereign scale has no counterpart. What is left is weaker and is still worth publishing: one question, asked at three scales, answered quantitatively at each. The surviving resemblance is not nothing and is not a structure — and the test that separated those two readings was designed, with its response to every outcome fixed in advance, before anybody knew which one it would return.

A fourth paper, on price formation, that was written and is not being published. The largest entry, and it cost the most. A complete draft existed — roughly 7,500 words, references verified — arguing that supply and demand are not independent equations. It is not published and is not one of the papers this corpus joins; the draft itself is in the repository, superseded and marked so, at `docs/papers/paper-I-price-formation/paper-I.md`, and §10 names the command and the locale that count it. Had the route worked, this paper would be the fourth in a series rather than the third, and §5 above would be a citation rather than a section.

Its first framing: attacking the diagram. The original claim was that partitioning agents into fixed buyers and sellers is mathematically invalid, presented as a critique of general equilibrium. It is not one. Walrasian excess demand already has that property and Arrow–Debreu never used the Marshallian cross, so the available referee reply — *the author is attacking the introductory diagram rather than the theory* — was fatal and unanswerable. **The cost was the most rhetorically satisfying claim the project had.** Had the route worked, §5 would indict the Marshallian cross rather than explain what its two curves are for, and §4 would be defending a critique of general equilibrium instead of using SMD once, as a boundary.

Its second framing: the strong form of P3. The replacement was to claim the diagram catches atomism in the act: excess demand is a fold over units, the two schedules are folds over units *and* the allocation, so the decomposition manufactures two objects presenting as aggregates while carrying information no fold contains. Three supporting results were built and all three arithmetics were correct. The framing died anyway, for three separate reasons, each established by running it rather than by argument:

1. **The conclusion was inverted.** $D(p^*) = S(p^*) = |H \quad T|$ — the crossing height *is* the allocation mismatch. The diagram is not concealing the coupling; it is displaying it. What survived is §5, which explains what the two curves are *for* rather than indicting them.
2. **The headline number was noise.** The reported effect — an allocation-indexed perturbation producing clearing-interval spread 26× the baseline width — has as its denom-

inator the gap between two consecutive order statistics, a single random draw with enormous relative variance. Repeating at $N = 400$, 1 000, 4 000 and 10 000 gave 26.1×, 8.3×, 113.5× and 47.2×: a 13.6-fold non-monotone swing. It is not a statistic.

3. **The load-bearing sentence was false in the formalism the paper cited.** *"H is not a property of the population"* fails under the standard unit (m_i, h_i), which makes the two schedules additive folds in exactly the sense excess demand is. This is Arrow–Debreu, Aumann and Hildenbrand (1994).

And a control that controlled for the wrong thing. The 26× contrast was measured against a rank-preserving perturbation. Against a *rank-scrambling* perturbation that never names the allocation at all, the effect was 0.89 against 0.96 — essentially identical. The comparison had been between scrambling and preserving, not between allocation-indexed and population-defined, and the allocation contributed almost nothing.

A registered test of the generality, which returned no verdict. With the P3 framing under pressure, a pre-registration (`REG-001`) was written to test whether the identity did work in a second setting where the incumbent apparatus could not follow. The instrument was mis-specified in four ways and the run produced no usable answer in either direction. It is recorded here as an instrument dead end — which is what it is — and not as a result, because it never became one. Under this section's own test: had it returned, §5's identity would be reported as holding beyond the single-good case, and §9's fifth limitation — that the zero-income-effect restriction is the load-bearing one — would be a narrower claim than it now is.

A superposition framing for agent role. Proposed to soften the critique: the agent is "in superposition" between buyer and seller until the price is observed. Rejected on technical grounds before it reached a draft. Superposition denotes genuine indeterminacy prior to measurement, and these agents hold a definite reservation price at all times; the role is a sign function. The cost of using it would have been a reviewer noting that quantum mechanics had been invoked to describe a piecewise-constant function, in a research area already carrying reputational damage from loose physics metaphor — and the metaphor undersells a claim that is exactly stateable without it. Had it been sound, §5's account of what an agent's role *is* would read differently, which is what earns it a place under this section's test despite its never having reached a draft.

Reporting a partial invariance that was a tie conven-

tion. §5's identity was first measured on a 12-point grid spanning the full range of reservation prices and returned **four** distinct excess-demand schedules rather than one. That reads as a partial invariance and was very nearly written up as one. Both grid endpoints coincide with data points, the strict inequalities in the two counts then disagree about a single agent whose holding status varies by allocation, and two endpoints × two holding states = four. Had it gone in, the central claim would have been published one full step weaker than it is true, and the identity would never have been looked for.

Centring SMD as the defence. Considered and refused. Making SMD the centrepiece hands a referee a ten-minute rejection — a multi-good general-equilibrium theorem applied to a single-good partial model — and worse, makes the paper's centrepiece a fifty-year-old theorem the author did not prove. §4 uses SMD as a *boundary*, once, which is the only load it can carry here.

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9 · Limitations

1. **The composition claim is about a state, and the state is largely unobserved.** §4.2's resolution buys its consistency by narrowing the claim to the extensive state, and Papers II and III are both, in substance, demonstrations that measuring layers do not see the thing that matters. A composed state nobody can read is a weaker asset than the argument's confidence might suggest.
2. **The three scales share one question, not one structure.** The corpus's own end-to-end test (`END-TO-END-001` leg `E1`) asked whether the sovereign scale's realisation share and the firm scale's observability share are one object seen twice, and rejected it: what the firm scale's filter does not recognise is deferred and released at rate α, what the sovereign scale's base does not recognise is never assessed, and there is no parameter on one side playing α's part on the other. What the framework offers across scales is therefore one type, one question answered quantitatively at each, and addition within a scale — not a correspondence between the scales' parameters. §3 records the demotion and §8 the route.
3. **Diagonality at the firm scale is assumed, its test is closed, and it went against the assumption.** Recognition events do cluster within firm-quarters — 4.12× and 2.02× the independence expectation, both universes, both $p = 0.0002$ (Paper III §5.4) — so the Hadamard form in §3 is

an approximation and not an identity, and the composition claim's link at the scale where accounting happens is degraded. It is degraded rather than broken for the reason §4.2 gives: diagonality is a claim about the *reporting filter*, and this paper's composition claim is about the *extensive state*, which adds regardless of how it is recorded. The honest cost is that **the firm's reported object no longer composes from its classes' reported objects without cross-terms**, which is the scale at which anyone would actually read it. And the cause is unidentified: the standards sequence the tests, so an accounting artefact and an economic coupling are not separated by the design that found the departure.

4. **This paper contributes no new computation.** Its claims are joins over results established elsewhere, plus one measurement on the literature. A reader who rejects Paper II or Paper III should reject the corresponding link here.
5. **§5's instance is the zero-income-effect case, which is the load-bearing restriction.** The allocation cancels identically because each agent's demand does not depend on their own endowment. That is the special case; the general one is where SMD lives.
6. **The three-literature framing is a choice, not a partition.** Ecological economics, industrial ecology, national accounting theory and the aggregation literature proper all have claims on this territory. The three named here are the three this paper *joins*. The aggregation literature is used, and heavily — §4 takes SMD as the boundary of the composition claim — but it is used as a limit rather than joined, which is why §7 treats it under a heading of its own.
7. **Nothing here is normative.** The corpus characterises what is measurable and what composes. It does not say what should be measured, taxed, recorded or done.
8. **The whitespace measurement is about occupancy, not fertility.** `REG-013` can establish that an intersection is unoccupied. That it is worth occupying is what the argument has to earn, and the measurement cannot help with it.
9. **One of this paper's absences is measured and three others named here are asserted.** §6 exists because *an absence found by searching is a property of the search*, and it measures the absence that motivates the paper. Three others, named here, are not measured: §1.1's *the input-output energy table has no lapse to report*, which is load-bearing for §1.1's reading of §4.3 as *largely unmeasured rather than merely unassembled*; and §7's two within-literature absences, *has generally not had* an accounting-shaped object and *has mostly not had*

a base. None of the three is known to be false. The point is that they stand on the evidentiary footing §6 was built to escape.

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10 · Data and code availability

This paper's own contribution is one measurement, on the citation graph rather than on a model. Numbers reported without a citation to Paper II or Paper III have their record named here, which is what §1 says: §6's are `REG-013`'s; §5's and §8's results are the surviving apparatus of the fourth paper §8 describes; and §8's word count for that paper is the `wc -w` named below, on the superseded draft §8 names. Everything else is cited from Paper II or Paper III and is regenerated by those papers' scripts.

- **Repository:** `https://github.com/jasoncbraatz/wealth-tensor` (public)
- **Pre-registration:** `docs/preregistration/REG-013-citation-graph-whitespace.md`, committed in `fff7063`, before the instrument existed
- **Instrument:** `scripts/reg013_citation_whitespace.py`
- **The record of §6:** `docs/preregistration/RESULT-REG-013-run.log` and `docs/preregistration/RESULT-REG-013-run.json`, the committed output of the 2026-08-16 run, with `docs/preregistration/RESULT-REG-013.md` reading the verdict off them. **Those files, not a command, are the record of §6.**
- **Re-run the instrument:** `python3 scripts/reg013_citation_whitespace.py`. It reproduces the instrument and not the pull. Each cluster's audience is the set of works citing its seeds, retrieved live from OpenAlex, so a run depends on that graph on the day it is made and on the API's rate limit: one attempt on 2026-08-18 exited non-zero on `HTTP 429` inside the ceiling control, and a second attempt the same day exited zero and returned every figure §6 reports unchanged — the three intersections, the three overlaps, the three split-half intersections, the pooled ceiling and the zero floor — while the seed citation counts the run prints had already moved. That second run is committed at `docs/preregistration/RESULT-REG-013-rerun-2026-08-18.json`, and it replicates §6 on the live graph rather than regenerating it from anything held here. Nothing in this repository re-derives §6's figures from committed data.

- **Test suite:** `python3 -m pytest tests/ -q`
- **Papers cited as established results:** Paper II (`src/wealth_tensor/redistribution.py`, regenerated by `scripts/wt030_report.py`); Paper III (`src/wealth_tensor/lag.py`, regenerated by `scripts/wt027_report.py`, and `src/wealth_tensor/lambda_sensitivity.py`, regenerated by `scripts/wt002_lambda_report.py`). The diagonality rejection cited in §3, §4.4 and §9 comes from neither: it is Paper III §5.4's, and its command is `python3 scripts/wt089_recognition_and_offdiagonal.py`, which runs both universes in one pass and prints the 4.12× and 2.02× lifts, their $p = 0.0002$ and the power curve §3 quotes. `python3 scripts/wt026_severe_test.py --universe pilot --onset peak` is Paper III §5.3's command, it answers a different question, and it prints none of those numbers.
- **Regenerate §5 and §8:** `python3 scripts/wt018_report.py` prints §5's table — the 399 interior grid points, the 25 demand, 25 supply and one excess-demand schedules, and the 500-point monotonicity sweep with **0** violations, endpoints **+249** and **−150**, and one sign change. `python3 -m pytest tests/test_excess_demand.py -q` asserts those schedule counts and the twelve-point tie convention §8 records; `python3 scripts/wt071_refuter.py` gives §5's crossing-height identity across all 25 allocations and §8's N-dependence table and its control. The module under all three is `src/wealth_tensor/excess_demand.py` — this paper's only `src/` dependency that is not a sibling's. This is the surviving apparatus of the fourth paper §8 describes. It is in this repository, and every number §5 and §8 report was re-run from it. `tests/test_excess_demand.py` *asserts* §5's **399** and §8's twelve-point **four** rather than printing them, so its output is a verdict and not a table; `wt018_report.py` is the table.
- **Regenerate §8's word count:** `LC_ALL=C.UTF-8 wc -w docs/papers/paper-I-price-formation/paper-I.md` returns **7,527**, the *roughly 7,500* §8 reports. **The locale is part of the command:** the same bytes return **7,367** under GNU `wc` in a non-UTF-8 locale, so a word count quoted without a locale is checkable only by accident.
- **Commit for the results reported here: 5efe626** — the last commit touching `scripts/reg013_citation_whitespace.py`, and therefore the state of the instrument that produced every number in §6. It is also the commit that first added this manuscript: the instrument and the paper

entered the repository together, so the pin is exact rather than approximate. What is pinned is the state of the code, which is what a replicator needs and is verifiable today.

Two of the suite's overclaim-forbidding tests are named here, and the suite holds more than two: `test_the_forbidden_claim_is_red` appears in two registration modules and Paper III names a third, `test_pre001_constants_are_what_was_registered`. `test_excess_demand_is_monotone_here_so_this_is_not_an_SMD_result` forbids §5's instance being sold as an SMD result, which is this paper's own claim. `test_a_flat_gini_does_not_mean_a_bounded_one` forbids a saturating statistic being read as convergence, which is **Paper II's** claim and not this one's — this paper reports no Gini — and it is named here because §3's sovereign scale rests on the result it guards.

Known limitations of this review. This paper contributes one measurement of its own, and it is pre-registered, run against a floor and a ceiling fixed in the same run, and replicated on the live graph a day later; §6 states both truncations and what each costs. Everything else here is cited from Papers II and III, so it inherits their checks and no more — a reader who rejects either should reject the corresponding link. The joins between the three scales, the cross-paper references and the relocation claims of §7 were checked by reading rather than by any instrument. What was not checked at all: three of the four absences this paper asserts, which §9's ninth limitation names one by one and declines to defend on the footing §6 was built to escape.

The repository's `docs/` directory is deliberately public and is the project's working notebook, including the pre-registration whose prediction failed and the result document recording that a fourth paper died to its own referees.

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Bibliographic details are to be verified against live sources before submission, per `docs/papers/PREPRINT-CHECKLIST.md` *and* `docs/REFERENCE-POLICY.md`. *Entries marked ✓ were verified against the sources the mark table in* `docs/REFERENCE-POLICY.md` *requires.*

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